

UNIVERSITY OF BENIN
Faculty of Engineering - Department of Production Engineering

PRODUCTION ENGINEERING LABORATORY AND WORKSHOP PRACTICE

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HSE 201: Health and Safety in the Workplace

1. Aims

To study the Health and Safety Management System (HSMS) of a workplace, and to understand the responsibilities of the people who work in it, the nature of workplace hazards, and the general safety rules that must be observed in the engineering workshop.

2. Objectives

- To expand our knowledge of the elements that make up a workplace.
- To know the role of the various health and safety Acts and regulations.
- To understand how the physical environment is organised so that hazards can be controlled.
- To be able to recognise, evaluate and control hazards using the three-step approach to safety.

3. Apparatus / Materials

- Departmental Laboratory and Workshop Practice manual (Section: Health and Safety).
- The workshop environment itself (machines, benches, electrical outlets, floors and storage areas) used for the hazard-recognition walk-through.
- Personal Protective Equipment (PPE): safety glasses or goggles, closed shoes, overalls, gloves, ear protection.
- First aid box, fire extinguisher and the Material Safety Data Sheet (MSDS) book kept at the safety cabinet.

4. Theory / Diagram

A Health and Safety Management System (HSMS) is an organised process for reducing the number of injuries, illnesses and even deaths that occur at work. Many safety elements have been identified, but four have been found to be critical to the success of any system. These are: the quality of leadership shown by management (the employer, supervisors and foremen) on safety and on the involvement of employees; analysis of the worksite for potential hazard spots; hazard prevention and control; and the quality and frequency of safety and health training programmes. Put simply, a safe workplace is one where the people in charge care, the dangers are known, the dangers are controlled, and everyone has been taught how to stay safe.

What is a workplace?

A workplace is anywhere that work is done, or is to be done, by a worker, a person or a group of persons carrying out an undertaking or business. By this definition a workshop, a business shop, an office, a farm, a home or even a playground can be a workplace. The components of a workplace are the physical objects (machinery, furniture, structures), the human beings (operators, workers, inspectors, visitors) and the environment in which the work is carried out.

The environment

An environment is the totality of all the factors that make up a workplace, and two are always identifiable. The physical environment is the actual place or site where the work is done: factories and plants, workshops, farms, roadways, vehicles and machinery, and buildings. The operating environment is the set of physical and atmospheric conditions that influence job delivery, such as seasonal variation, weather, temperature and relative humidity. Other

environment-dependent factors include exposure to harmful substances, working in confined spaces, poorly ventilated or poorly lit enclosures, and attacks by insects or wild animals. Each of these carries a hazard if it is not properly controlled.

Hazard and hazard points

A hazard is any situation, condition or extreme event (natural or caused) that carries a certain probability of producing adverse consequences for the safety or health of workers, property or the environment. The specific places where hazards occur are called hazard points: moving machine parts, work at heights, slippery floors, live electrical equipment, excessive noise, toxic substances and the lifting of heavy objects.

Types of hazard

Workplace hazards fall into four categories. (1) Physical hazards cause traumatic injury through lifting heavy loads, repetitive motion, slips, trips and falls, contact with moving parts of machinery such as unguarded belts and gears, electric shock and fire, excessive noise, and extreme temperatures or radiation. (2) Chemical hazards are injuries caused by exposure to substances such as battery acid, solvents, cleaners and pesticides, to dusts from grinding or sawmills, and to fumes from welding or gas emissions. (3) Biological hazards come from exposure to pathogens such as viruses, fungi, bacteria and mould, which can attack through the blood and body fluids. (4) Psychological hazards arise from unfavourable working conditions, inadequate or inappropriate tools, and working with old or worn-out equipment, which may lead to stress, fatigue, exhaustion and job depression.

Accident, injury and safety

An accident is an unforeseen occurrence with negative consequences, resulting from sudden impact, system malfunction, human error, carelessness or unsafe acts. It results in pain, loss of a body part or even death, and also in waste of time, money and materials and damage to equipment. An injury, in public health terms, is physical harm to a person's body caused by contact between people and harmful objects, substances or other things in their surroundings: broken bones, cuts, bruises, brain damage, poisoning and burns. Safety, in its simplest form, is a state of being at little or no risk of injury from a harmful external impact, inhalation or contact. It is a holistic approach in which people feel free from harm in addition to actually being safe, and it requires thinking ahead about the hazards associated with the job.

The three-step approach to safety

Step 1, Recognition: identify the hazards to which a person at the workplace is likely to be exposed. Step 2, Evaluation: assess the risk of injury or harm from each hazard identified. Step 3, Control: consider the means by which the risk may be reduced, whether by elimination, substitution, guarding, safe procedures or protective equipment. If hazards are not recognised, evaluated and controlled, a worker may be injured or killed by machinery, electricity, fires or falls.

General shop safety rules

The manual lists 29 rules for all students using the workshop. In summary: wear safety glasses or goggles in any shop area; wear shoes that cover the whole foot (no sandals); operate only equipment you are familiar with and have been authorised to use; never use power tools unless at least two people are in the shop; avoid blowing compressed air at people or at chips; report every injury, however slight, to the instructor; wash chemical splashes in the eye for 15 minutes before seeking medical help; shut machines off before cleaning, repairing or oiling; do not wear ties, loose clothing, jewellery or gloves around rotating machinery; tie back long hair; use a brush or hook, never the hands, to remove chips;

keep floors clean and dry; do not work when tired or in a hurry; never indulge in horseplay; keep all machine guards in place; store oily rags in an approved metal container; check power cords for damage; consult the MSDS before using unfamiliar materials; and do not drink alcohol or bring food into the machine shop.

5. Procedure

Procedure followed in this exercise

1. We were briefed on the meaning of a workplace, its components and its environment, with the workshop used as the working example.
2. We were taken through the types of hazard (physical, chemical, biological and psychological) and shown where each could be found in the workshop.
3. We carried out a hazard-recognition walk-through of the workshop, identifying hazard points such as rotating machines, electrical outlets, floors, and storage areas for oily rags and chemicals.
4. For each hazard, we applied the three-step approach: recognition, evaluation of the risk, and consideration of control measures.
5. The 29 general shop safety rules were read out and explained, and the location of the first aid box, extinguisher and MSDS book were pointed out.
6. The correct use of personal protective equipment was demonstrated, and the instructor answered questions.

6. Precautions

- Always wear the correct PPE before entering any shop area.
- Never run, play or use a phone carelessly in the workshop.
- Listen to the instructor and do not touch any machine without permission.
- Know where the exits, first aid box, extinguisher and emergency switches are located.
- Report every injury or unsafe condition immediately.

7. Discussion

This exercise showed us that safety in the workshop is not a matter of luck; it is the result of deliberate planning. When we walked through the workshop, hazards that we had earlier ignored became visible: a loose power cable, a floor with oil on it, a machine belt with its guard removed. The three-step approach (recognise, evaluate, control) gave us a simple habit we can apply to any job, in any shop, and even outside engineering.

We also observed that most of the 29 rules are common sense that is easily forgotten under pressure. Rules such as not working when tired or in a hurry, keeping the floor clean, and tying back long hair are rarely broken on purpose; they are broken through carelessness. It is therefore important that leadership (the instructor, the technologist, the supervisor) sets a good example, which agrees with the manual's point that management leadership is the first of the four critical safety elements.

It is also worth noting that hazards are not only physical. Psychological hazards such as stress, fatigue and working with worn-out tools reduce concentration and indirectly cause accidents. A safe workplace therefore deals with the person as well as the machine.

8. Conclusion

We conclude that a health and safety management system, built on good leadership, hazard analysis, hazard control and regular training, is essential to preventing injury and loss in any

workplace. All the aims of the experiment were met: we understood the components of a workplace, the types of hazard, the three-step approach to safety and the general shop rules that govern our conduct in the workshop.

9. Questions Asked in the Material

What are the four critical elements of a successful safety and health management system?

Leadership qualities of management on safety and employee involvement; worksite analysis of potential hazard spots; hazard prevention and control; and the quality and frequency of safety and health training.

What is a hazard?

Any situation, condition or extreme event with some probability of causing adverse consequences to the safety or health of workers, property or the environment.

State the three-step approach to safety at the workplace.

Recognition of hazards, evaluation of the risk each hazard presents, and control, which means finding ways to reduce the risk.

10. What I Have Learned

- A workplace is any place where work is done, and it is made up of objects, people and the environment.
- Hazards can be physical, chemical, biological or psychological, and each needs its own controls.
- Accidents are caused mainly by unsafe acts and unsafe conditions, both of which can be prevented.
- The rules of the workshop exist to protect me and the people around me, so I must follow them every time.

GWP 201: Introduction to the Workshop

1. Aims

To become familiar with the various shops that make up an engineering workshop and to understand the importance of the workshop in engineering training.

2. Objectives

- To identify the common shops and the special shops of a workshop.
- To state the purpose and typical activities of each shop.
- To appreciate the role of the workshop in the practical training of an engineer.

3. Apparatus / Materials

- Departmental workshop (guided tour of the shops).
- Laboratory and Workshop Practice manual (Section: General Workshop Practice).
- Personal protective equipment (safety glasses, overall and covered shoes).

4. Theory / Diagram

The workshop is the centre of learning about engineering materials, manufacturing practices, equipment and tools, and the safety precautions to be observed in manufacturing operations. A mechanical or production engineer who has never touched a lathe, struck a weld or poured a mould will struggle to design components that can actually be made. For practical knowledge and skills, therefore, the workshop is indispensable. The workshop gives us the chance to link the theory taught in the lecture room to real materials, real tolerances and real mistakes.

The shops in a workshop are divided into two groups: the common shops and the special shops.

Common shops

The common shops are those that every engineering student passes through, because they cover the basic manufacturing processes. They are the following.

Carpentry (woodwork) and pattern-making shop: here timber is cut, planed and jointed. Besides furniture and doors, the shop makes the wooden patterns used in the foundry to form moulds.

Bench work and fitting shop: here metal parts are marked out, cut, filed and assembled to the required fit using hand tools such as the hacksaw, files, scribes and the bench vice.

Smithy and forging shop: here metal is heated and shaped by hammering or pressing while it is plastic, which also refines the grain structure of the metal.

Sheet metal shop: here thin metal sheets are marked, cut, bent, folded and joined to make trays, funnels, ducts and similar articles.

Electric arc and gas welding shop: here metals are joined permanently by fusion, using the heat of an electric arc or a gas flame.

Machine shop: here machine tools such as the lathe, shaper, milling and drilling machines remove metal to produce parts of accurate size and finish.

Foundry shop: here metal is melted and poured into moulds to produce castings.

Special shops

The special shops cover more specialised trades that support engineering work. They are: electrical wiring, soldering and electronics; painting; heat treatment; plumbing; automobile; refrigeration and air-conditioning; and material testing. Heat treatment, for example, changes the properties of steel (hardness, toughness) by controlled heating and cooling, while the material testing shop measures strength, hardness and impact resistance of specimens.

Importance of the workshop

The workshop develops manual skill, helps us to appreciate the limits and capabilities of the processes we will later specify in drawings, and teaches the discipline of working to dimensions and to safety rules. It also introduces us to the vocabulary and tools of each trade, which makes communication with technicians and craftsmen easier in industry.

5. Procedure

1. We assembled at the workshop with the necessary PPE and were briefed on the safety rules.
2. The instructor explained the division of the workshop into common shops and special shops.
3. We were taken round each of the seven common shops and shown the main machines, tools and products of each.
4. The special shops were then described, and the processes carried out in each were explained.
5. We noted down the functions of each shop and asked questions where necessary.

6. Precautions

- Remain with the group and do not touch any machine during the tour.
- Wear safety glasses, closed shoes and suitable clothing throughout.
- Keep out of the way of operators who are working.
- Do not run in the workshop and observe all warning signs.

7. Discussion

The tour made it clear that a workshop is not one room but a family of specialised areas, each with its own tools, hazards and skills. Moving from the carpentry shop to the foundry, for example, we saw how one object can pass through several shops: a wooden pattern made in the carpentry shop is used to produce a mould in the foundry, the casting is machined in the machine shop, and the finished part may be heat treated or welded to another component.

We also appreciated why the common shops are compulsory for engineering students. Each of them teaches a basic process that the others depend on. The special shops, although less general, show how engineering extends into electrical work, automobiles, refrigeration and material testing.

8. Conclusion

The workshop is made up of seven common shops and seven special shops, each dedicated to a particular type of manufacturing or support activity. The exercise achieved its aim of familiarising us with these shops and with the importance of practical training in producing competent engineers.

9. Questions Asked in the Material

List the common shops in a workshop.

Carpentry and pattern making, bench work and fitting, smithy and forging, sheet metal, electric arc and gas welding, machine shop and foundry shop.

List the special shops.

Electrical wiring, soldering and electronics; painting; heat treatment; plumbing; automobile; refrigeration and air conditioning; and material testing.

10. What I Have Learned

- The workshop is the place where engineering theory becomes practice.
- Each shop teaches a distinct process, and a single product often passes through several of them.
- Working safely and to dimensions is as important as the finished product.

EM 201: Study of Workshop Measuring and Marking Instruments (Linear Measuring Devices)

1. Aims

To study the various linear measuring devices used in the workshop, and to learn how to choose a suitable instrument according to the job and the precision required.

2. Objectives

- To become familiar with the different types of linear measuring instruments.
- To select a suitable measuring device according to the job and the accuracy required.
- To learn how to read the vernier scale and the micrometer scale correctly.

3. Apparatus / Materials

- Steel rule (150, 300, 500 and 1000 mm).
- Outside and inside spring callipers, divider and telescopic gauge.
- Depth gauge.
- Vernier calliper and vernier height gauge.
- Outside micrometer (and samples of inside, screw-thread, V-anvil, blade, groove and digital micrometers for demonstration).
- Sample work pieces of known dimensions.

4. Theory / Diagram

Linear measurement covers the measurement of lengths, diameters, heights and thickness. The basic principle of mechanical linear measurement is comparison with standard dimensions on a suitably engraved instrument. Linear measuring instruments are categorised by accuracy into two groups: non-precision instruments, which measure to the line graduations of a rule, and precision instruments, which use a vernier or screw principle to measure to a fraction of a millimetre. There are also electrical measuring devices, which are mainly transducers that convert displacement into a measurable voltage or current, such as strain gauges, linear variable differential transformers (LVDT) and potentiometers.

Non-precision instruments

Non-precision instruments are limited to the measurement of parts to a visible line graduation on the instrument. They are used where high accuracy is not required.

Steel rule. This is the simplest and most common measuring instrument. It works by comparing an unknown length with one that has been previously calibrated, so the rule must be graduated uniformly throughout its length. Rules are made in 150, 300, 500 and 1000 mm lengths and may be of folding type so that they can be kept in a pocket. The accuracy obtained depends on the quality of the rule and on the skill of the user in estimating part of a millimetre, which usually limits the reading to about 0.5 mm.

Callipers. Callipers are used to measure parts that cannot be measured directly with a scale. They consist of two legs hinged at the top; the ends of the legs span the part to be inspected, and this span is then transferred to a scale. Callipers are of two types, the spring type and the firm-joint type. In the spring type the two legs are attached with a spring, and the setting is adjusted by a knurled nut running on a finely threaded screw. The spring keeps the legs rigid at all points of the adjustment. Spring callipers are of two forms. The outside spring calliper measures outside dimensions such as the diameter of a shaft, and its legs are bowed inwards.

The inside spring calliper measures inside dimensions such as the diameter of a hole or the distance between shoulders, and its legs are bent outwards. In both cases accuracy depends on the user's sense of feel, and a steel rule must be used with them to obtain a reading.

Divider. A divider is similar in construction to a calliper except that both legs are straight with sharp hardened points at the ends. It is used for scribing arcs and circles and for general layout work. Its nominal size is the distance from the fulcrum roller centre to the end of a leg, and it is available in sizes such as 100, 200 and 300 mm.

Telescopic gauge. This is used to measure the internal diameter of a hole during machining. It consists of a handle and two plungers, one telescoping into the other, under spring tension. The plungers are compressed and locked, the plunger end is inserted into the hole and allowed to expand to the opposite edges, then locked again, withdrawn, and measured with an outside micrometer.

Depth gauge. This is used to measure the depth of blind holes, grooves, slots and the heights of shoulders. It is essentially a narrow steel rule clamped to a head that sits at right angles to the rule.

Precision instruments

Vernier calliper. The vernier calliper gives an accuracy of 0.1 mm down to 0.01 mm. The main scale carries the fixed graduations and one of the two measuring jaws, while a sliding head carries a vernier scale and the other jaw. The sliding head can be locked on the main scale by a knurled screw and moved finely by a fine-adjustment screw. The lower jaws measure external dimensions, the upper jaws measure internal dimensions, and the rectangular rod at the end measures depth. To take a reading, the main-scale reading up to the zero of the vernier scale is noted. Then the vernier division which coincides exactly with a main-scale division is found, and this number multiplied by the vernier constant (least count) gives the vernier reading, which is added to the main-scale reading.

The least count is the difference between one main-scale division and one vernier-scale division. As worked in the sample questions, a vernier with 25 divisions spanning 12 mm, on a main scale with 24 divisions per 12 mm, gives a main-scale division of $12/24 = 0.5$ mm, a vernier division of $12/25 = 0.48$ mm and therefore a least count of $0.5 - 0.48 = 0.02$ mm.

Vernier height gauge. It uses the same principle as the vernier calliper and is mainly used to measure height. It has a heavy base block, a graduated beam, a sliding jaw assembly and a removable clamp, and a scriber can replace the measuring jaw so that lines can be scribed at a given height above a surface plate.

Micrometer. The micrometer is one of the most widely used precision instruments, used mainly for external dimensions such as the diameter of shafts and the thickness of parts, to an accuracy of 0.01 mm. Its parts are the frame, anvil and spindle, screwed spindle, graduated sleeve or barrel, thimble, ratchet or friction stop, and spindle clamp. The spindle has a thread of 0.5 mm pitch, so one complete turn of the thimble advances the spindle by 0.5 mm. The thimble has 50 equal divisions around its circumference, so one thimble division corresponds to $0.5/50 = 0.01$ mm, which is the least count. The ratchet slips when the pressure on the screw exceeds a set amount, ensuring uniform pressure and preventing distortion of the work or the instrument.

To read a micrometer we note (1) the number of whole millimetres visible above the reference line, (2) whether a half-millimetre graduation below the line is visible, and (3) the thimble division in line with the reference line, multiplied by 0.01 mm. For example, 8 divisions above the line (8.00 mm), 1 division below the line (0.50 mm) and 28 thimble divisions (0.28 mm) give 8.78 mm. If the micrometer does not read zero when closed, the zero

error must be applied as a correction.

Special types of micrometer

The inside micrometer calliper has jaws with hardened and ground contact surfaces and no U-shaped frame, and it is used for small internal dimensions. The inside micrometer, for internal measurement, is used for holes over 50 mm in diameter and has a measuring unit with extension rods of 13, 25, 50, 100, 150, 200 and 600 mm. The screw-thread micrometer has a pointed spindle and a double V-anvil to contact the thread flanks. The V-anvil micrometer, with a 60 degree V, is used for checking out-of-roundness of centreless-ground parts and fluted tools such as taps and reamers. The blade-type micrometer has a non-rotating spindle and is used for narrow grooves and keyways. The groove micrometer measures grooves and shoulders inside a bore. The bench micrometer is a high-precision micrometer with an anvil retractor. The digital micrometer gives a direct reading up to 0.001 mm and has push-button zero reset and hold functions.

Diagrams

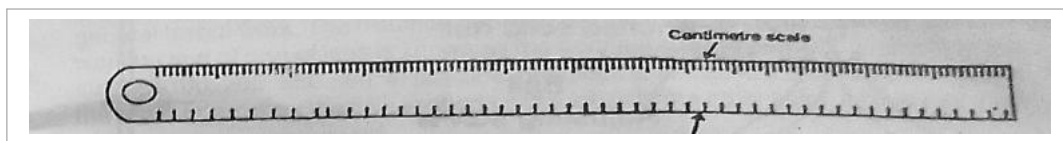


Figure 1: Steel rule graduated in centimetre and inch scales (Departmental Lab Manual, p. 11).

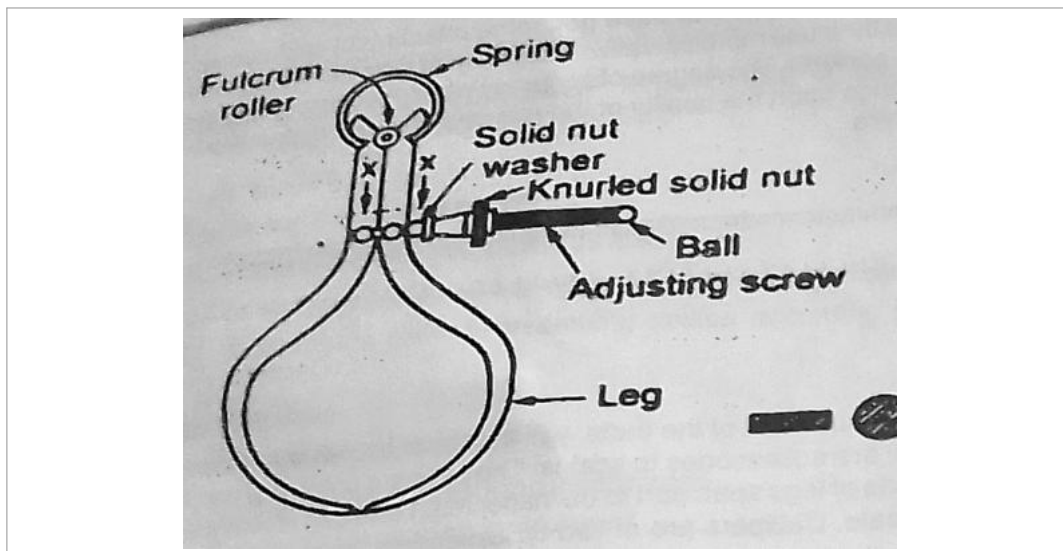


Figure 2: Outside spring calliper showing leg, spring, adjusting screw and knurled nut (Lab Manual, p. 12).

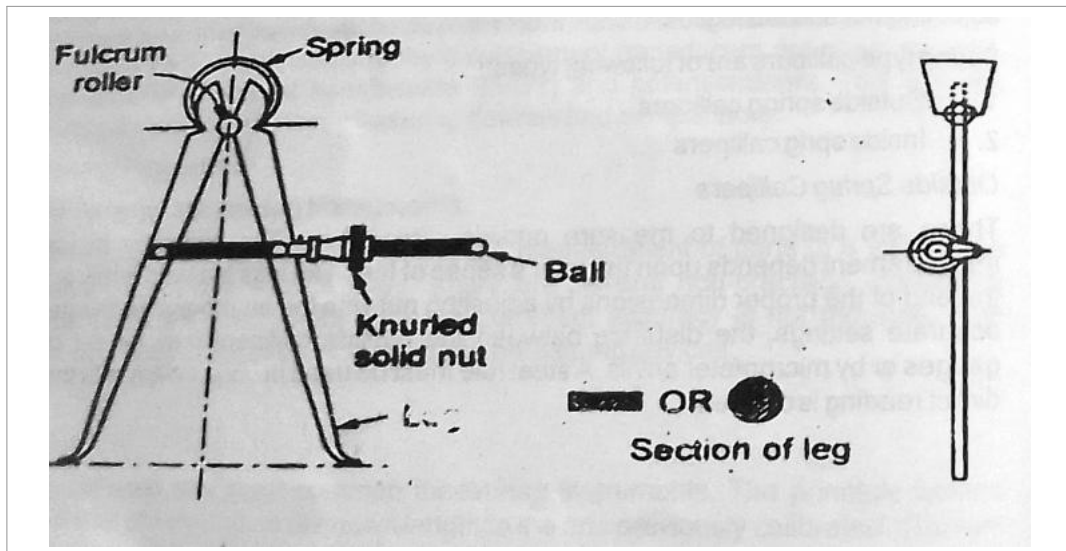


Figure 3: Inside spring calliper (Lab Manual, p. 12).

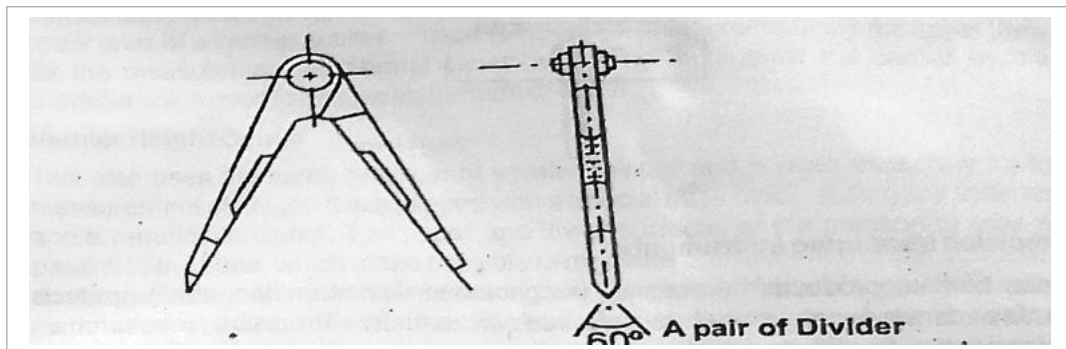


Figure 4: A pair of dividers (Lab Manual, p. 13).

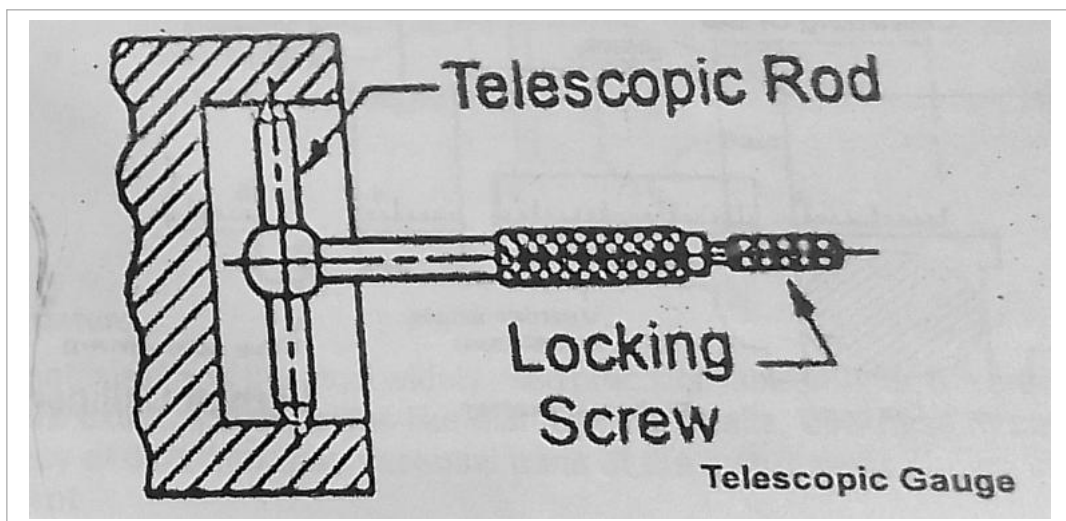


Figure 5: Telescopic gauge used to measure the internal diameter of a hole (Lab Manual, p. 13).

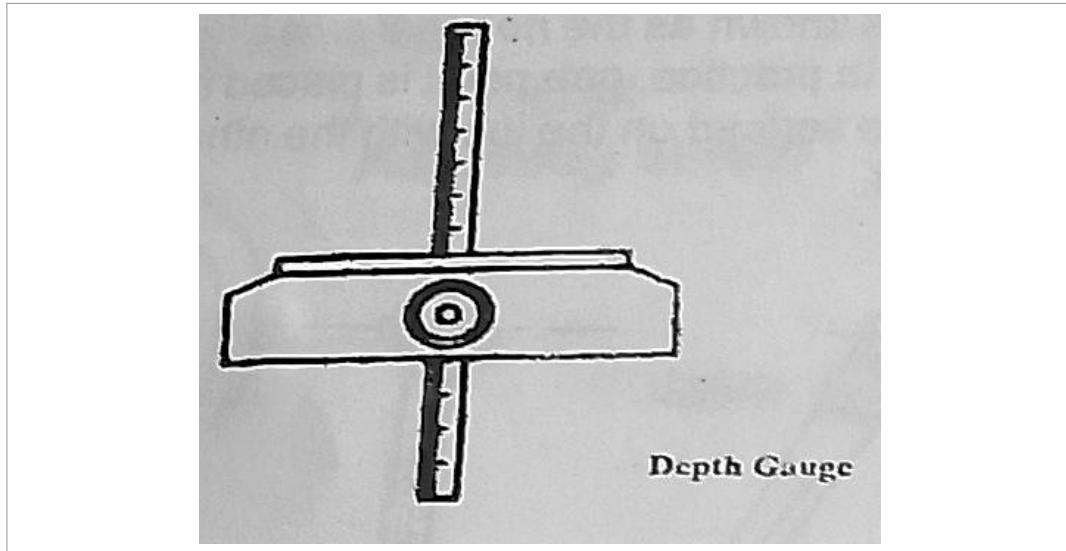


Figure 6: Depth gauge (Lab Manual, p. 14).

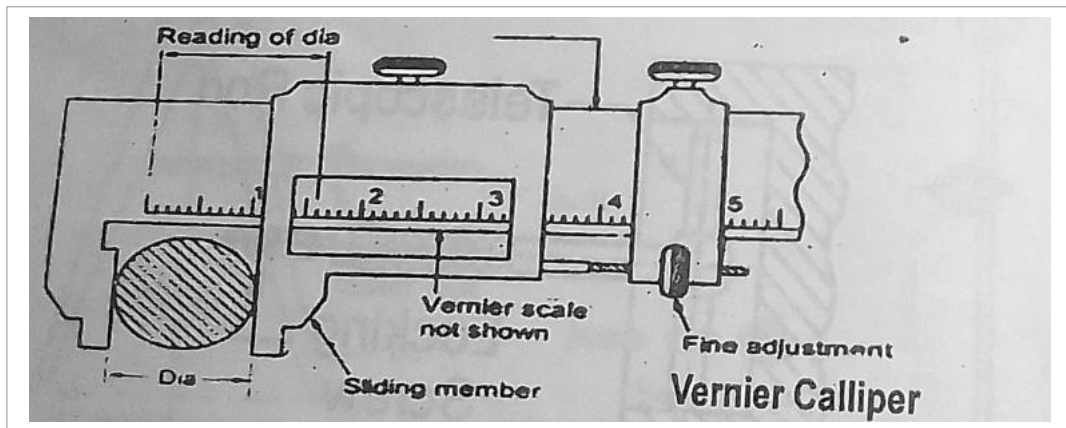


Figure 7: Vernier calliper measuring the diameter of a round bar (Lab Manual, p. 14).

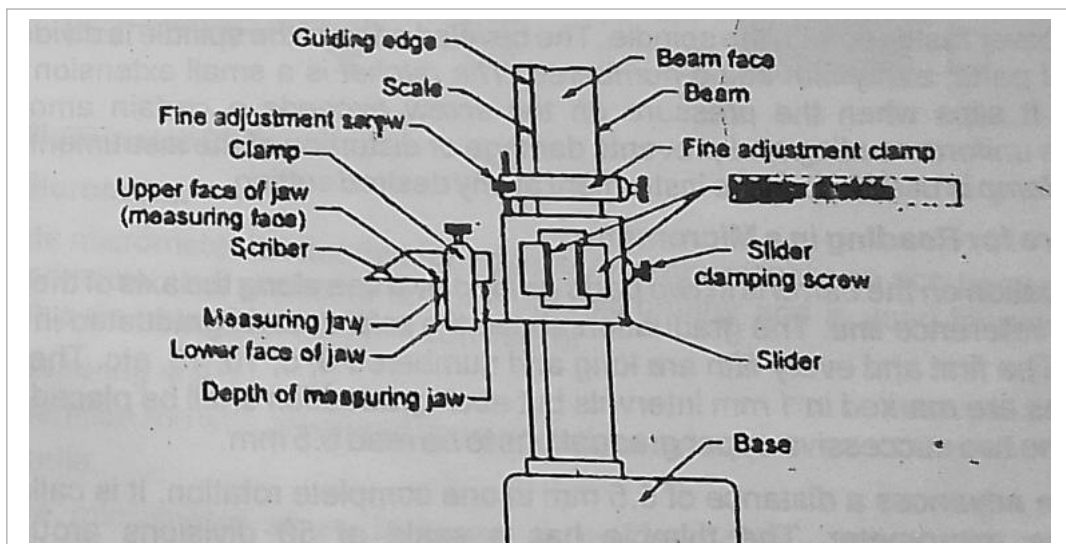


Figure 8: Vernier height gauge with its parts labelled (Lab Manual, p. 15).

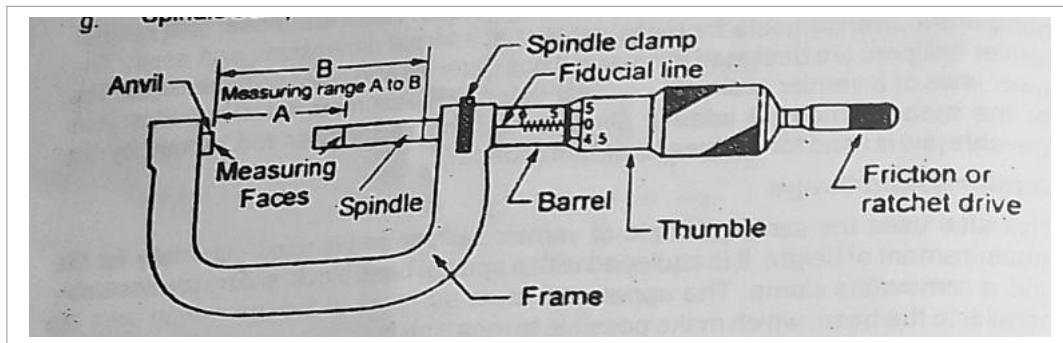


Figure 9: Outside micrometer: anvil, spindle, barrel, thimble, ratchet and frame (Lab Manual, p. 16).

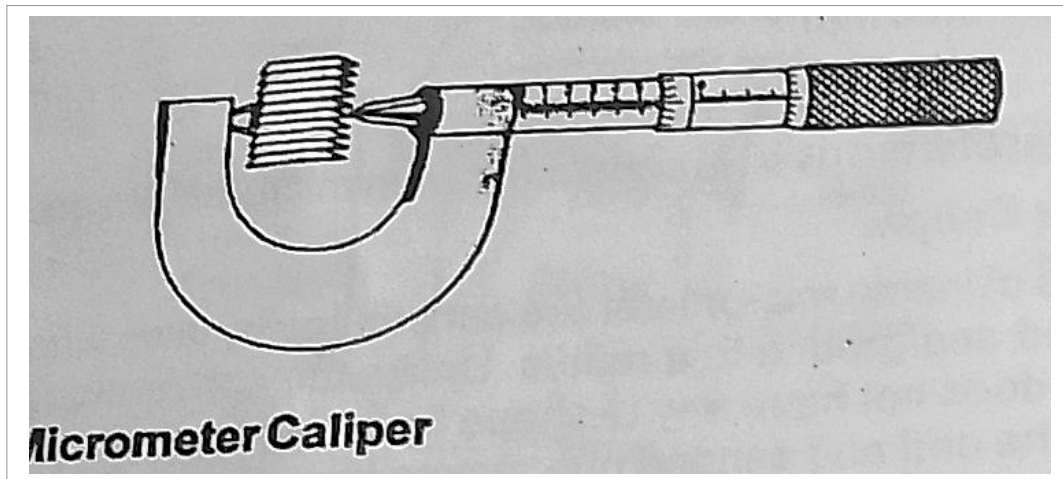


Figure 10: Screw-thread micrometer calliper (Lab Manual, p. 18).

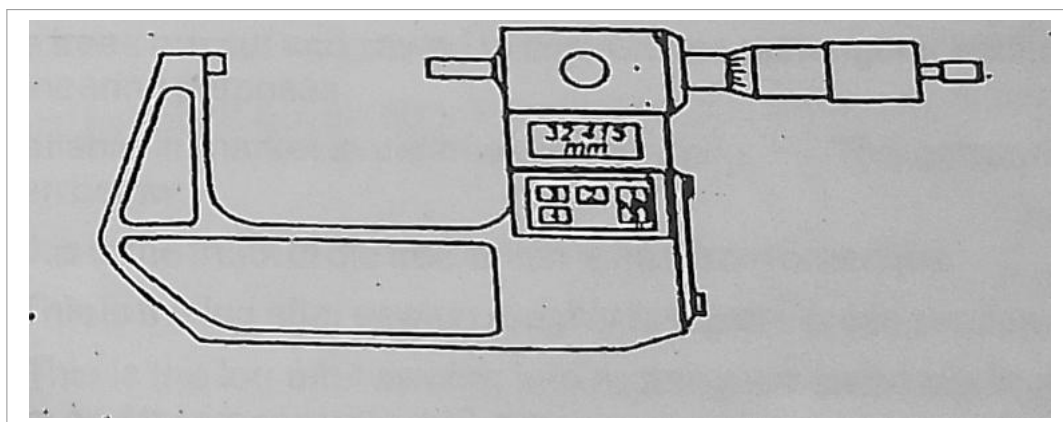


Figure 11: Digital micrometer (Lab Manual, p. 19).

5. Procedure

Non-precision measurement

1. A sample work piece of unknown length was selected, and the steel rule was placed along its edge with the zero graduation exactly at one end; the reading at the other end was recorded.
2. For the outside diameter of a round bar, the outside spring calliper was set by turning the knurled nut until the legs just touched the work gently. The setting was then transferred to the steel rule and read.
3. For a hole, the inside spring calliper was set to touch the walls lightly, then read against the steel rule. The telescopic gauge was also locked in the hole and measured with a micrometer.

4. The depth of a blind hole or slot was measured with the depth gauge by seating the head on the surface and extending the rule to the bottom.

Vernier calliper

5. The jaws were closed, and the instrument was checked for zero error.
6. The work piece was placed between the jaws and the fine adjustment screw turned until the jaws fitted snugly. The lock screw was tightened.
7. The main-scale reading just before the vernier zero was noted, then the vernier division coinciding with a main-scale division was found. The reading is main scale reading + (coinciding division x least count).

Micrometer

8. The measuring faces were cleaned and the micrometer was closed using the ratchet to check zero error.
9. The work was placed between the anvil and spindle, and the thimble turned using only the ratchet until it clicked.
10. The sleeve reading (mm and half-mm) was added to the thimble reading (thimble division x 0.01 mm), and the zero error corrected for.
11. The various types of micrometer and the vernier height gauge were examined, and their special uses were noted.

6. Precautions

- Clean the measuring faces and the work piece before measuring.
- Always check for zero error before taking readings.
- Do not force the callipers or the vernier; use gentle, uniform pressure.
- Use the ratchet when closing the micrometer to prevent over-tightening.
- Read the scale with the eye directly in line to avoid parallax error.
- Do not measure moving or hot work, and never use the instruments as hammers or levers.
- Store precision instruments with their faces slightly apart in their cases.

7. Discussion

The experiment showed that the choice of instrument must follow the accuracy that the job demands. The steel rule is fast and convenient but only reliable to about half a millimetre, so it is suitable for rough layout and general lengths. When the same bar was measured with a vernier calliper, the reading carried a precision of 0.02 mm, and with the micrometer, 0.01 mm. The difference between these readings, for the same object, illustrates why precision work such as fits and shafts requires precision instruments.

Spring callipers do not give a reading by themselves; they only transfer the dimension to a rule. This means that their accuracy is limited by both the feel of the user and the reading of the rule. They are, however, very useful for checking diameters on a lathe and for comparing a part with a mating part. The telescopic gauge solves a problem that the calliper cannot, which is measuring the diameter of a bore from inside, but it too depends on a micrometer for its final reading.

Common sources of error that we identified include zero error, parallax when reading the vernier, uneven pressure on the work (which the micrometer ratchet is designed to prevent), dirt on the faces and temperature differences. A good practice is to take more than one reading and to check the instrument against a known standard.

8. Conclusion

The workshop linear measuring instruments were studied and used. Non-precision instruments (steel rule, callipers, divider, telescopic and depth gauges) are quick and suit general work, while precision instruments (vernier calliper, vernier height gauge and micrometers) provide an accuracy of 0.02 mm down to 0.01 mm or better. A suitable instrument must be selected according to the precision required, and care must be taken to read it correctly.

9. Questions Asked in the Material

Describe the different types of calliper for measuring linear dimensions.

Callipers are of the spring type and the firm-joint type. The spring type includes the outside calliper (for outside diameters and thickness), the inside calliper (for hole diameters and internal distances), and the divider and odd-leg (hermaphrodite) calliper used for scribing. The vernier calliper is a precision calliper with a main scale and a vernier scale that measures external, internal and depth dimensions.

A vernier scale consists of 25 divisions on 12 mm spacing and the main scale has 24 divisions on 12 mm. What is the least count?

One main-scale division = $12/24 = 0.5$ mm. One vernier division = $12/25 = 0.48$ mm. Least count = $0.5 - 0.48 = 0.02$ mm.

Describe different types of micrometers for measuring linear dimensions.

The outside micrometer (external dimensions), inside micrometer calliper and inside micrometer (internal dimensions), screw-thread micrometer, V-anvil micrometer (out-of-roundness), blade-type micrometer (narrow grooves and keyways), groove micrometer, bench micrometer and digital micrometer (direct reading to 0.001 mm).

What is the degree of accuracy of a steel rule and a vernier calliper?

A steel rule can be read to about 0.5 mm (half a millimetre) depending on the skill of the user, while a vernier calliper reads to 0.1 mm, 0.05 mm or 0.02 mm depending on the vernier scale, and up to 0.01 mm for finer designs.

Select the most appropriate linear measuring tool for a given job.

Rough lengths: steel rule. Outside diameters for comparison: outside spring calliper. Bore diameters: inside calliper or telescopic gauge. Slot depth: depth gauge. Accurate diameters, lengths and depths: vernier calliper. High precision shaft diameter: micrometer. Heights and scribing: vernier height gauge.

10. What I Have Learned

- How to read a steel rule, a vernier calliper and a micrometer, and how to compute the least count.
- That non-precision instruments are for quick, general work and precision instruments are for close tolerances.
- That zero error, parallax and uneven pressure are the main sources of error.
- How to choose the right instrument for each type of measurement.

WW 201: Introduction to Woodwork Practice

1. Aims

To study the materials, tools and basic operations used in the wood workshop, and to learn how timber is marked, cut and planed.

2. Objectives

- To know the materials used in the woodwork shop.
- To become familiar with the tools used in woodwork.
- To know how to cut wood.
- To know how to plane wood.

3. Apparatus / Materials

- Timber (Afara) in the form of boards and scantlings.
- Marking and measuring tools: steel rule, steel tape, try square, marking gauge, marking knife, bevel square.
- Holding tools: work bench, bench vice, C-clamp, bar clamp.
- Cutting tools: hand saws, tenon saw, bow saw and chisels.
- Planing tools: wooden and metal jack planes, rasp.
- Striking and boring tools: mallet, claw hammer, bradawl, brace and bit, hand drill.

4. Theory / Diagram

Carpentry is the process of shaping timber using hand tools. The products include doors and windows, furniture, building construction work and the patterns used for moulding in foundries. Carpentry mainly involves joining wooden pieces together and finishing the surfaces after shaping them, and for this reason the word joining is often used for carpentry. A student of woodwork must therefore know about timber and other carpentry materials, the tools, the operations and the methods of making common joints.

Materials used in carpentry

The basic materials are timber and plywood. The auxiliary materials include nails, screws, adhesives, paints and varnishes. Timber is the name given to wood obtained from exogenous (outward-growing) trees. Such a tree grows outwards from the centre by adding concentric layers of fresh wood each year, called annual rings. After full growth the trees are felled and sawn into rectangular sections of various sizes for engineering use. The common shapes and sizes of timber on the market are: log (the trunk of the tree freed of its branches); balk (a log roughly sawn to a square section); deal (width about 225 mm and thickness up to 100 mm); plank (width above 275 mm and thickness 50 to 150 mm); board (thickness below 50 mm and width above 125 mm); batten (width below 175 mm and thickness 30 to 50 mm); and scantlings (assorted pieces of non-standard sizes).

Classification of wood

Timber for commercial use is divided into softwood and hardwood. Softwood is light in weight and light in colour; annual rings are distinct but medullary rays are not visible, and the sap wood is not distinct from the heart wood. Softwoods cannot resist stresses developed across their fibres and are therefore not well suited to woodworking. Hardwood has compact, thin annual rings and its medullary rays are usually visible. It is nearly equally strong along and across the fibres, which makes it the preferred material for woodworking. The cross-section of a tree trunk shows the pith or medulla at the centre, the medullary rays, the annual rings,

the heart wood, the sap wood and the bark.

Tools for woodworking

The principal hand tools of a carpentry workshop are classified as marking and measuring tools, cutting tools, planing tools, boring tools, striking tools and holding tools.

Marking and measuring tools. Steel rules (30 cm and 60 cm) and flexible measuring rules are used for dimensions. The straight edge and square is a flat strip of wood or metal with straight, parallel edges. The steel tape is used for long dimensions, such as marking on boards and checking overall dimensions. The marking gauge consists of a wooden stem sliding in a stock; the stem carries a steel point, and the stock can be fixed at any position with a thumb screw. It is used to mark lines parallel to an edge. The mortise gauge has two points so that both sides of a mortise can be marked together. The try square has a rectangular steel blade (150 to 300 mm) fixed rigidly to a cast iron stock, and it is used to mark and check right angles. The marking knife (scriber) turns pencil lines into deep scratch lines on the surface. The bevel square is an adjustable try square used to mark and measure angles between 0 and 180 degrees.

Holding tools. The work bench is a strong table of hard wood, about 50 to 80 cm in length and 90 cm in width (for the carpenter's type described), at which two to four carpenters can work. The carpenter's bench vice has one fixed jaw and one movable jaw operated by a screw and handle, and its jaws are lined with hard wood so that the work is not damaged. The C-clamp is used to hold short pieces together, for example after gluing, and is available in sizes from 70 to 800 mm. The bar or T-cramp has a steel bar with a threaded spindle and an adjustable shoe and is used to hold glued pieces tightly or to hold unglued pieces in an assembled position.

Cutting tools. Saws cut by a reciprocating motion of the toothed edge relative to the work. In a push-type saw the cut occurs during the forward stroke. The hand saw (rip saw and cross-cut saw) is used for straight cuts; its blade is 25 to 40 cm long and 6 to 10 cm wide, with 3 to 5 teeth per cm. The tenon (back) saw has a parallel blade with a stiffened back, with 5 to 8 teeth per cm, and is used for fine, accurate joint work. Other saws are the compass saw, key-hole saw, cross-cut saw and bow saw. Chisels are used to cut and shave wood. The firmer chisel is the most general-purpose chisel, with a flat blade of 15 to 50 mm width. The dovetail chisel has bevelled edges and is used for fine work and corners. The mortise chisel has a thick blade for heavy, deep cuts.

Planing tools. A plane smooths timber surfaces. The wooden jack plane is the most commonly used plane; its main part is a wooden block called the sole, in which a steel blade with a knife edge is held at about 45 degrees to the sole by a wooden wedge. The metal jack plane serves the same purpose, but its cast iron body is machined and ground to give smoother operation and a better finish, and it has an adjusting lever and a lever cap.

Boring tools. The bradawl bores small holes to start a screw or a nail. The carpenter's brace rotates auger bits and twist drills, often with a ratchet device. The auger bit is the most common tool for making holes in wood; its lead screw guides it into the wood and its helical flutes carry the chips out. The hand drill is used for small holes and has a chuck turned by a gear and pinion arrangement. The gimlet has cutting edges like a twist drill and is used for large holes with hand pressure.

Striking tools. The mallet is a wooden-headed hammer used to strike chisels without damaging their handles. The claw hammer has a steel head, with a flat face to drive nails and a claw to extract them. The pincer is used to pull out small nails and pins. The screwdriver drives wood screws, and its size is given by the length of the blade. The wood rasp is a finishing tool with sharp cutting teeth that is used to smooth surfaces and remove sharp

edges.

Diagrams

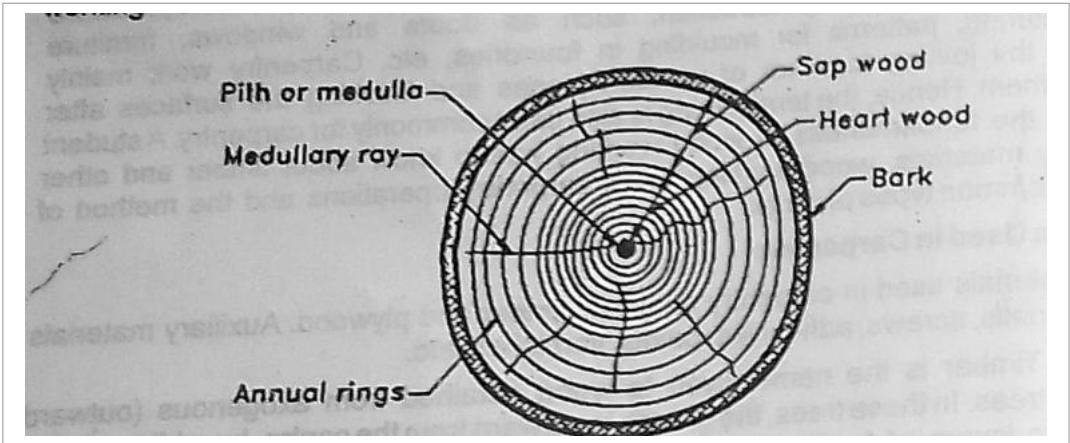


Figure 12: Cross-section of a tree trunk showing pith, medullary rays, annual rings, heart wood, sap wood and bark (Lab Manual, p. 22).

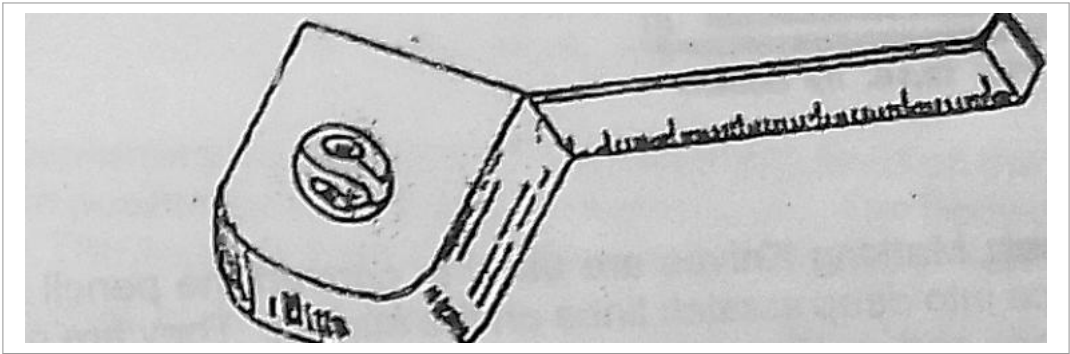


Figure 13: Measuring steel tape (Lab Manual, p. 23).

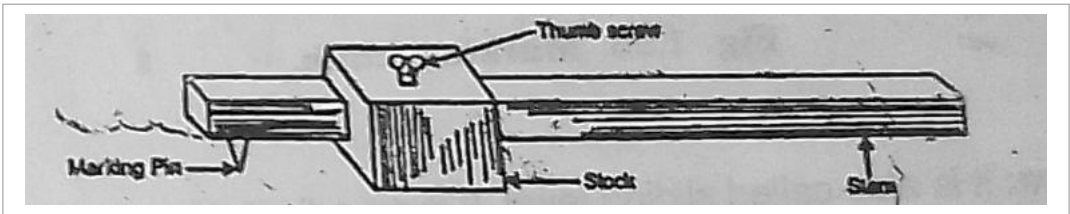


Figure 14: Marking gauge (Lab Manual, p. 23).

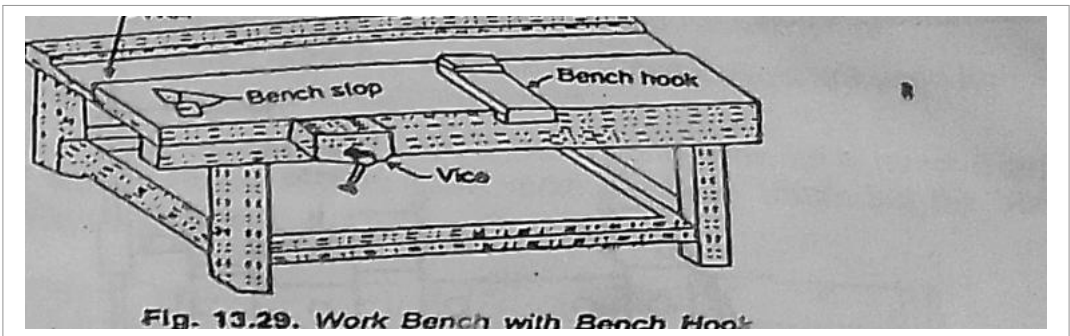


Figure 15: Work bench with bench hook, bench stop and vice (Lab Manual, p. 25).

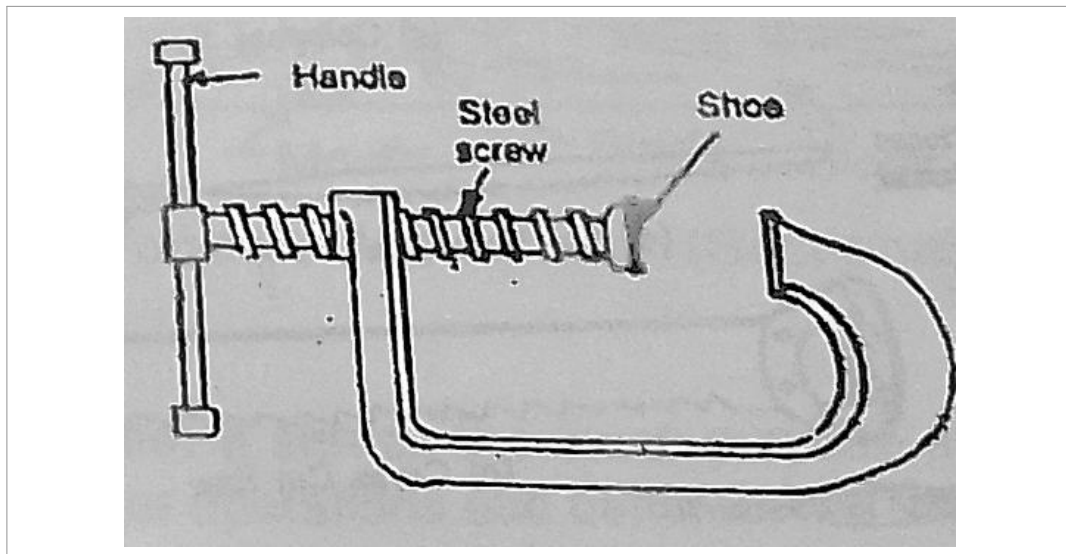


Figure 16: C-clamp (Lab Manual, p. 25).

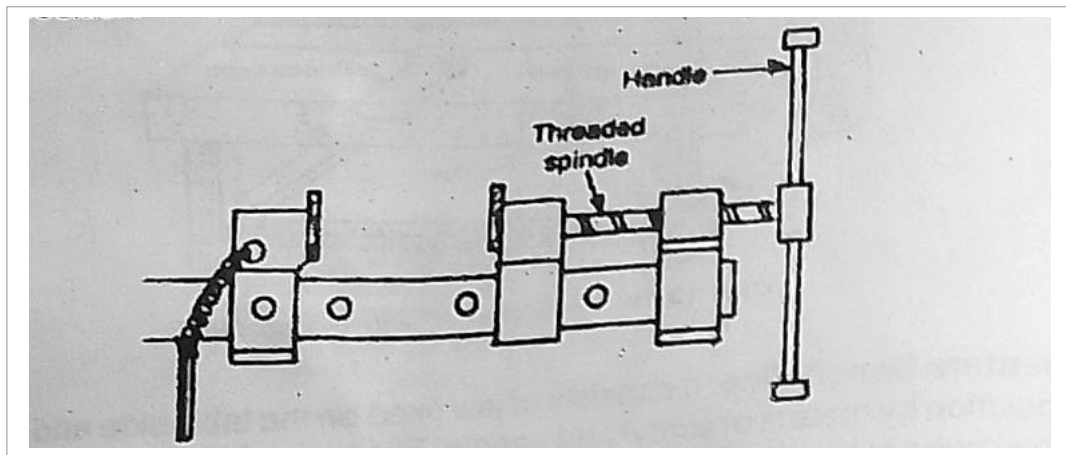


Figure 17: Bar or T-cramp (Lab Manual, p. 26).

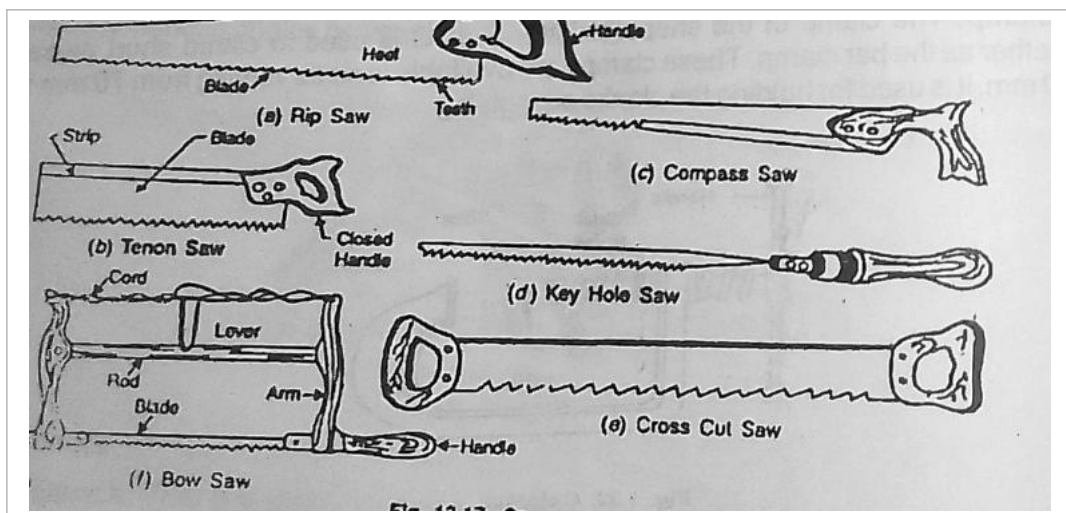


Figure 18: Types of saw (Lab Manual, p. 26).

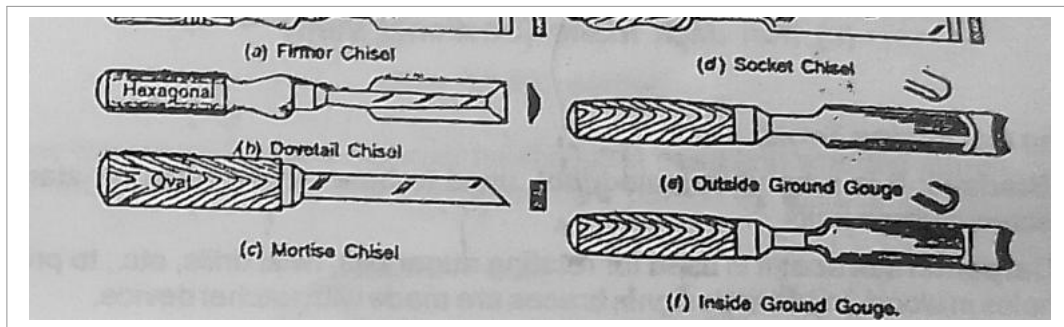


Figure 19: Types of chisel (Lab Manual, p. 27).

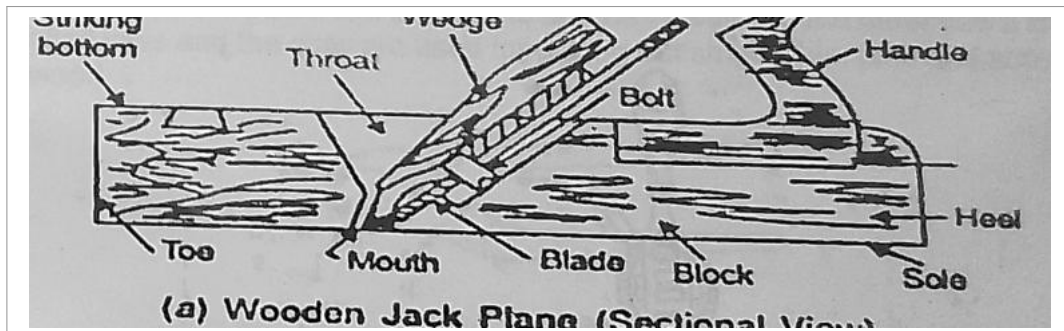


Figure 20: Wooden jack plane (sectional view) (Lab Manual, p. 27).

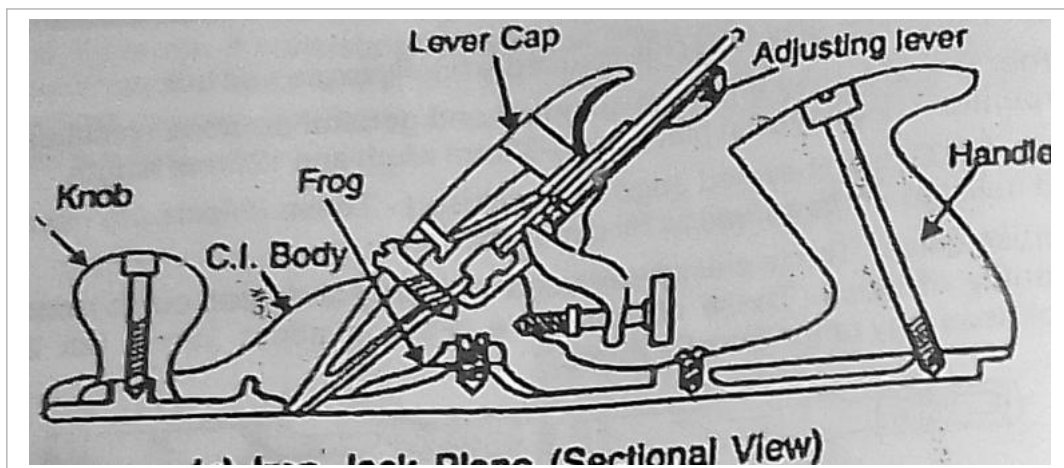


Figure 21: Iron jack plane (sectional view) (Lab Manual, p. 28).

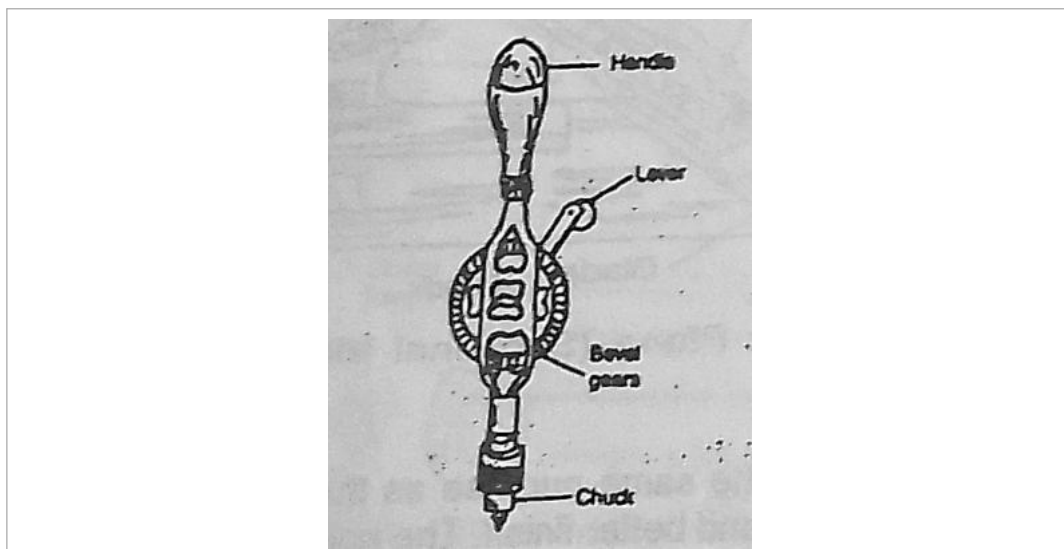


Figure 22: Hand drill (Lab Manual, p. 28).

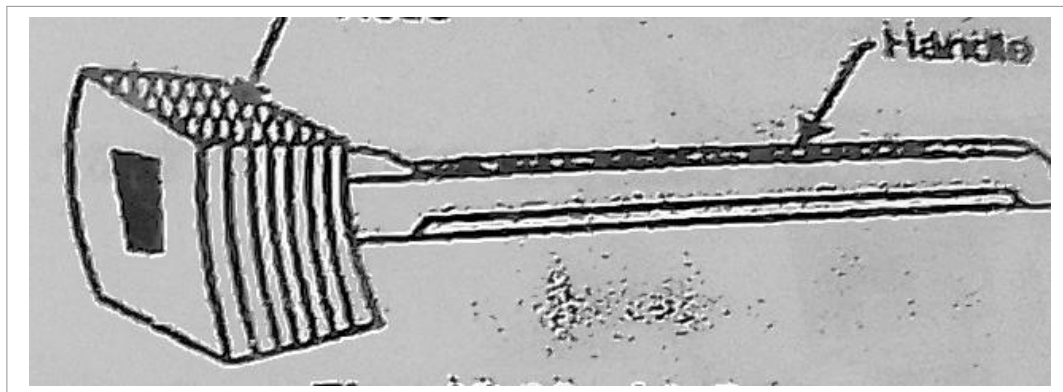


Figure 23: Mallet (Lab Manual, p. 29).

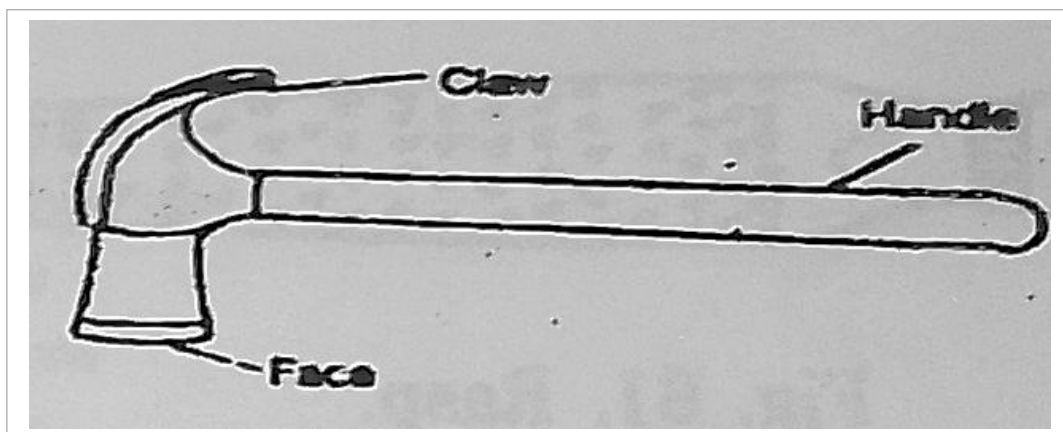


Figure 24: Claw hammer (Lab Manual, p. 29).

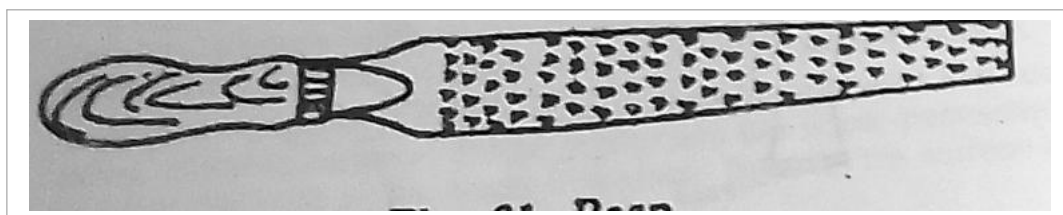


Figure 25: Wood rasp (Lab Manual, p. 30).

5. Procedure

1. We were introduced to timber, its classification (softwood and hardwood) and the standard shapes and sizes available.
2. The marking and measuring tools were demonstrated: steel rule, tape, try square, marking gauge, marking knife and bevel square.
3. We practised holding a piece of timber securely on the work bench using the bench vice, C-clamp and bar clamp.
4. A demonstration of cutting with the rip saw, cross-cut saw and tenon saw was given on a piece of timber.
5. We observed the use of the chisel and mallet and the jack plane for smoothing a surface.
6. Boring tools (bradawl, brace and bit, hand drill) and striking tools were demonstrated.

6. Precautions

- Always secure the work before cutting or chiselling, and keep both hands behind the cutting edge.
- Keep cutting tools sharp, because blunt tools slip and cause injury.

- Carry chisels and saws with the blade pointing downward, and never leave tools on the edge of the bench.
- Do not use a metal hammer to strike a chisel handle; use a mallet.
- Wear safety glasses and keep the floor free of shavings and sawdust.

7. Discussion

The introduction to woodwork showed us that the quality of a wooden product depends on three things: the right material, the right tool and the right method. Hardwoods, such as Afara used in our practicals, are preferred because they are strong in both directions and take a good finish. Softwood is lighter but weaker across the grain.

The range of tools might look confusing at first, but each has a clear purpose, and they are grouped logically: marking tools define what is to be done, holding tools secure the work, cutting and planing tools remove material, and striking tools assemble the parts. A rule of good woodwork is to mark accurately, since an error in marking cannot be corrected by sawing. The sharpness of the tool and the correct setting of the plane blade determine how clean the surface will be.

8. Conclusion

The materials, tools and basic operations of the woodwork shop were studied. We learned the shapes and sizes of timber, the difference between softwood and hardwood, and the purpose of the marking, holding, cutting, planing, boring and striking tools, which prepares us for the practical joints that follow.

9. Questions Asked in the Material

Mention the types of tools used in the woodwork shop.

Marking and measuring tools, cutting tools, planing tools, boring tools, striking tools and holding tools.

Differentiate between softwood and hardwood.

Softwood is light in weight and colour with indistinct sap and heart wood, and it is weak across its fibres, so it is not suitable for woodworking. Hardwood has compact annual rings, visible medullary rays and strength both along and across the fibres.

What is a batten, a plank and a board?

A batten is timber below 175 mm wide and 30 to 50 mm thick. A plank is wider than 275 mm and 50 to 150 mm thick. A board is below 50 mm thick and above 125 mm wide.

10. What I Have Learned

- The standard forms of timber and the difference between softwood and hardwood.
- The names, parts and uses of the main woodworking tools.
- That accurate marking is the foundation of accurate joints.

WW 201 (Practical 1): Cutting of Wood

1. Aims

To cut a piece of timber to a required length and to a diagonal (bevel) line using the correct saws.

2. Objectives

- To measure and mark out timber accurately using a tape, rule, pencil, try square and sliding bevel.
- To select and use the rip saw, cross-cut saw and tenon saw for their correct purposes.
- To cut along a marked line without drifting from it.

3. Apparatus / Materials

- Afara timber, 600 x 75 x 25 mm (the manual lists the stock as 12 x 1 x 12 in).
- Hand saws: rip saw, cross-cut saw (or framed saw), tenon saw and dovetail saw.
- Steel rule or measuring tape, pencil, try square and sliding bevel.
- Bench vice or bench hook for holding the work.

4. Theory / Diagram

Sawing is the first step in converting rough timber into usable pieces. A saw is a cutting tool with teeth on one edge. The cutting is done by a reciprocating motion between the toothed edge and the work. In a push saw the teeth cut on the forward stroke, while a pull saw cuts on the backward stroke. The saw teeth are set, meaning that alternate teeth are bent slightly to the left and right, so that the cut (the kerf) is wider than the blade. This stops the blade from binding in the wood.

The rip saw is used to cut along the grain. Its teeth are shaped like small chisels that pare the fibres away. The cross-cut saw cuts across the grain; its teeth are shaped like knives that sever the fibres first. The tenon saw (back saw) has a stiff steel strip along the back of a thin blade and has fine teeth (5 to 8 per cm), so it makes straight, accurate cuts for joints. The dovetail saw is a smaller version for very fine work.

Before sawing, the cut must be planned. A pencil line is drawn with the help of the rule and try square. A try square gives a line square to the edge, while a sliding bevel is set to the required angle (the manual uses a diagonal line) and then used to transfer the angle to the timber. Marking is done on the face side with a sharp pencil so that the line is thin and accurate. The saw is cut on the waste side of the line so that the finished piece keeps its full size.

Diagrams

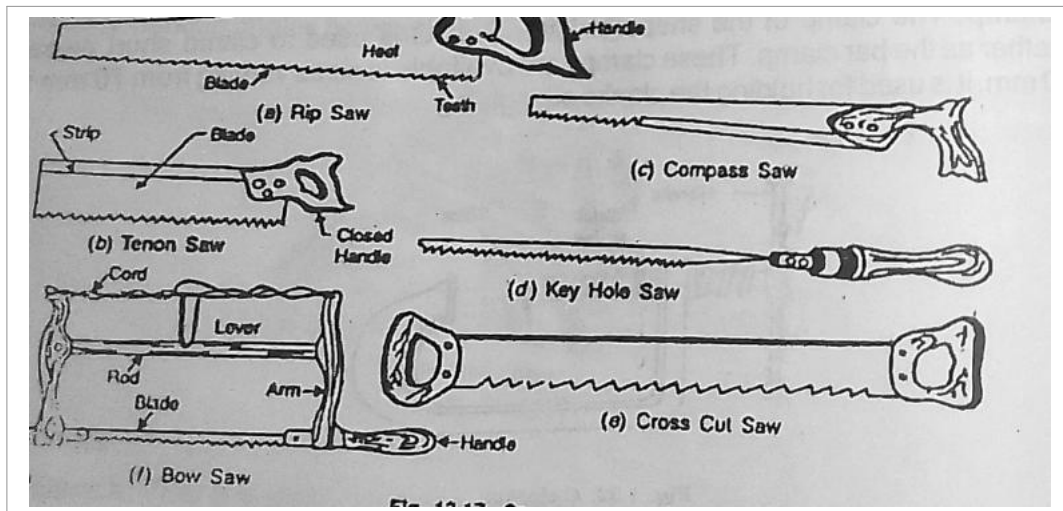


Figure 26: Types of saw used in woodwork (Lab Manual, p. 26).

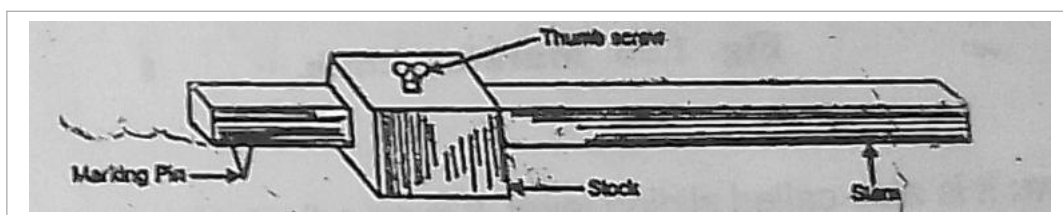


Figure 27: Marking gauge used to scribe lines parallel to an edge (Lab Manual, p. 23).

5. Procedure

1. Each student measured out a piece of wood 600 x 75 x 25 mm.
2. The piece was marked using the tape and pencil, and the lines were squared across the faces with the try square.
3. The timber was held firmly on the bench, and the rip saw was used to cut the piece to a length of 600 mm along the grain.
4. The cross-cut saw was used to cut to the required length across the grain.
5. A diagonal line was marked out using the sliding bevel set to the required angle.
6. The tenon saw was used to cut along the diagonal line, with the cut made on the waste side of the line.
7. The sawn edges were checked against the marked line and with the try square.

6. Precautions

- Hold the work firmly before cutting, and keep the free hand clear of the saw path.
- Start the cut with short, light strokes using the thumb as a guide.
- Cut on the waste side of the line.
- Keep the saw in line with the cut and do not force it, and use the full length of the blade.
- Wear safety glasses and keep the floor clear of sawdust.

7. Discussion

We observed that the quality of a cut depends first on the marking and second on the technique. When the line was clear and the saw was started carefully using the thumb as a guide, the cut followed the line with little effort. When the saw was forced or held at an angle, it drifted away from the line and the surface became rough. The rip saw cut quickly along the grain, while the cross-cut saw produced a smoother edge across the grain. The tenon saw, because of its fine teeth and stiff back, gave the most accurate diagonal cut.

It is also important to note that cutting on the waste side of the line leaves a small allowance that can be planed off to the exact size, whereas cutting on the wrong side makes the piece undersize and unusable.

8. Conclusion

The piece of timber was successfully cut to length and to the marked diagonal using the rip saw, cross-cut saw and tenon saw. The exercise showed the importance of accurate marking, correct choice of saw and careful sawing technique.

9. Questions Asked in the Material

Show a working drawing of the marked-out wood.

The working drawing shows a rectangular piece 600 mm long, 75 mm wide and 25 mm thick, with a diagonal line marked across one end using the sliding bevel (see the diagram of the saws and the dimensions above).

Using good sketches and short notes, state the operational steps.

Measure and mark the stock; hold in the vice; rip saw to width along the grain; cross-cut to length across the grain; mark the diagonal with the sliding bevel; cut the diagonal with the tenon saw; check.

State the uses of each tool used during the exercise.

Rule and tape: measuring. Pencil: marking. Try square: marking and checking right angles. Sliding bevel: transferring angles. Rip saw: cutting along the grain. Cross-cut saw: cutting across the grain. Tenon saw: fine, accurate cuts.

State the safety precautions observed.

See the precautions listed above.

10. What I Have Learned

- How to select the correct saw for cutting along or across the grain.
- How to use the sliding bevel to mark an angle.
- That cutting on the waste side of the line keeps the piece to size.

WW 201 (Practical 2): Planing of Wood

1. Aims

To reduce timber to the required thickness and to produce a flat, square and smooth surface by planing.

2. Objectives

- To surface, edge and thickness a piece of timber.
- To set and use a jack plane and a smoothing plane correctly.
- To use a try square and a marking gauge to test the work for flatness, squareness and thickness.

3. Apparatus / Materials

- Afara timber, 600 x 75 x 25 mm (manual stock 12 x 1 x 12 in).
- Jack plane and smoothing plane.
- Marking gauge, pencil and try square.
- Bar clamps or bench vice and bench stop.

4. Theory / Diagram

Planing is the process of reducing wood to usable sizes and smoothing it so that it is ready for finishing. The material to be dressed is planed to the required flatness and thickness. The main operations are surfacing the wood (producing one flat face), edging the wood (producing one flat edge square to the face), thicknessing to the required thickness, and marking out the width required.

A hand plane is basically a cutting blade (iron) held at a fixed angle, about 45 degrees, in a body. The blade projects slightly through a slot called the mouth, and as the plane is pushed along the timber the blade removes a thin shaving. In the wooden jack plane the blade is held in place by a wooden wedge, while in the iron jack plane the blade is held by a lever cap and is adjusted by an adjusting lever and a frog. The sole of the plane runs on the high spots of the wood, and so a long plane such as the jack plane levels the surface. The smoothing plane is shorter and is used for finishing.

The face side is the first surface prepared, and it is marked with a face mark to remember that it is the reference. The face edge is then planed square to the face and given an edge mark. All other dimensions are marked from these two reference surfaces using the marking gauge. This is the principle of working from datum surfaces, and it ensures that all faces are square to each other. A true surface is tested by looking along it, by laying a straight edge on it, and by testing with a try square.

Planing is done with the grain wherever possible. Planing against the grain tears the fibres and leaves a rough surface. Large planing and moulding machines are used in industry to do the same job in bulk, using a rotating cutter block, but hand planing teaches the principles.

Diagrams

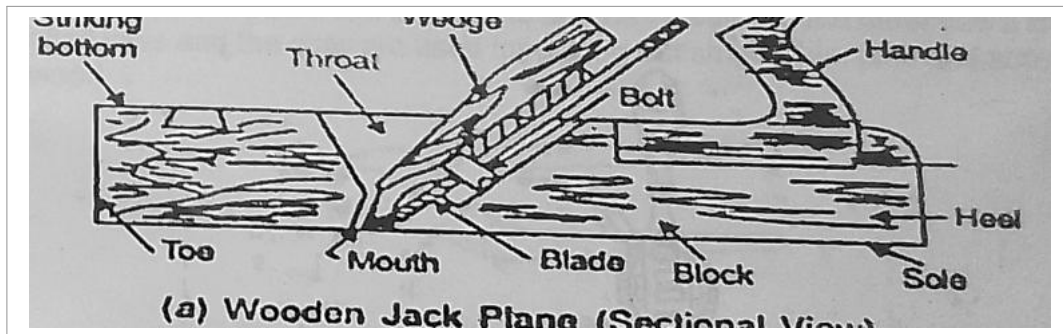


Figure 28: Wooden jack plane (sectional view) showing the blade, wedge, mouth, sole and handle (Lab Manual, p. 27).

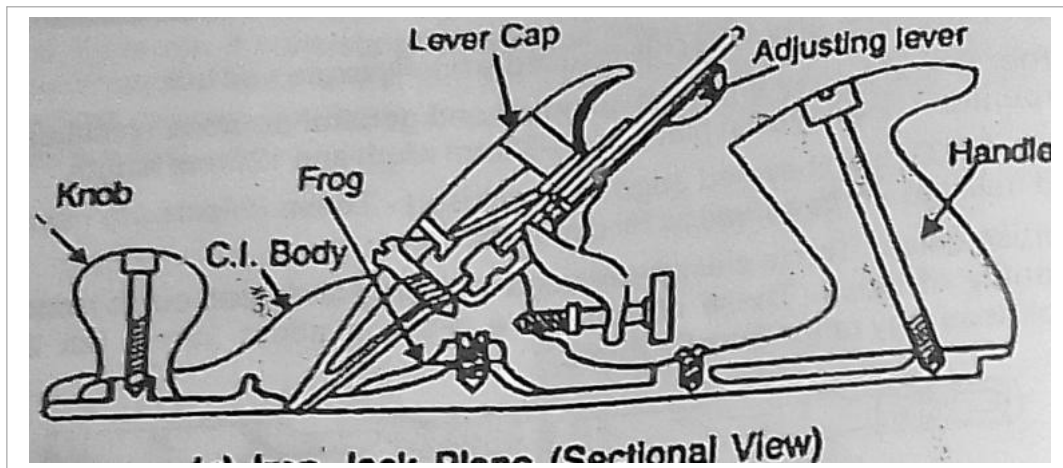


Figure 29: Iron jack plane (sectional view) (Lab Manual, p. 28).

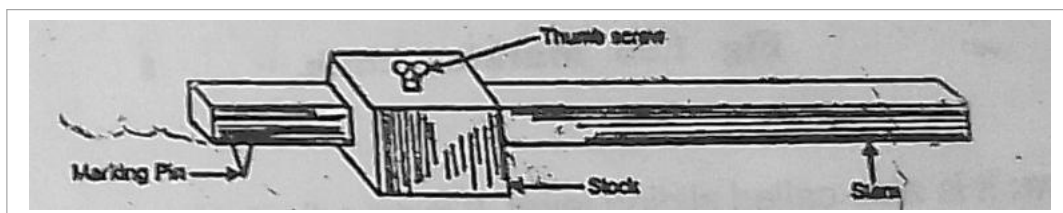


Figure 30: Marking gauge used for gauging the thickness (Lab Manual, p. 23).

5. Procedure

1. Each student was given a piece of timber 600 mm x 75 mm x 25 mm.
2. The timber was fixed using cramps or the bench vice.
3. The plane was used to smooth the surface until it was flat, and this was tested with the try square. The flat surface was marked with a face mark.
4. The edge was turned up and planed until it was square to the face; it was tested with the try square and given an edge mark.
5. The marking gauge was set to the required thickness and used to gauge a line all around from the face side.
6. The surplus was planed off down to the gauge line to give a uniform thickness.
7. The other edge was planed to give the required width.
8. The work was finally checked for flatness, squareness and dimension.

6. Precautions

- Keep the plane blade sharp and set properly, and do not set it too deep.
- Plane with the grain and keep the plane flat on the wood.

- Clamp the work firmly and keep fingers away from the blade.
- When the plane is not in use, lay it on its side so the blade does not touch the bench.
- Check the work often against the try square and gauge lines.
- Clear shavings from the floor to avoid slipping.

7. Discussion

Planing showed us how much patience precise woodwork requires. A first attempt at surfacing usually leaves hollow areas or ridges, and only repeated testing with the try square and by sighting along the work allowed us to correct them. We noticed that the plane cut best when the blade was sharp and set to take a thin shaving, and that planing against the grain gave a torn surface.

The use of a reference face and a reference edge was the key to success. Because every other measurement was taken from these two surfaces, the final piece had square corners and uniform thickness. Without them, small errors would accumulate and the piece would be out of square.

8. Conclusion

The timber was planed to a flat, square and smooth surface with the required thickness and width. The exercise demonstrated the use of the jack plane and smoothing plane, the importance of datum surfaces and the need to plane with the grain.

9. Questions Asked in the Material

Show a working drawing of the marked-out wood.

The drawing shows a rectangular block 600 x 75 x 25 mm with the face side and face edge marked.

State the operational steps.

Clamp; plane the face; test for flatness; mark face mark; plane the edge square to the face; mark the edge mark; gauge the thickness; plane off the surplus; plane the other edge to width.

State the uses of each tool used.

Jack plane: removing large amounts of waste and flattening. Smooth plane: final smoothing. Marking gauge: marking thickness and width parallel to an edge. Try square: testing squareness. Pencil: marking.

10. What I Have Learned

- The meaning of surfacing, edging and thicknessing.
- Why we work from a face side and a face edge.
- How to set and use a jack plane safely.

WW 201 (Practical 3): Tee Halving Joint

1. Aims

To make a tee halving joint from two pieces of timber using marking, cutting and shaping tools.

2. Objectives

- To mark out a socket and a pin accurately using a marking gauge, try square and pencil.
- To cut the joint with the tenon saw and to remove waste using a chisel and mallet.
- To fit the two members together so that the joint is tight and square.

3. Apparatus / Materials

- Two pieces of Afara, 300 x 60 x 42 mm.
- Pencil, measuring tape, try square, marking gauge.
- Tenon saw, chisel, mallet and rasp.

4. Theory / Diagram

A halving joint is a joint in which each of the two members is cut away to half its thickness, so that the two pieces fit together and are flush on both faces. The tee halving joint is a joint in which the end of one member meets the other at right angles, somewhere along its length, forming a T-shape. It is useful for making table legs, door frames and roof carcassing, where a quick and reasonably strong right-angled connection is needed.

One member is cut with a socket (a recess, sometimes called a housing) at the position where the other member joins. The other member is cut with a corresponding pin (the end of the member reduced to half thickness) which fits into the socket. The recess has a depth equal to half the thickness of the timber, which is marked with the marking gauge, so that when the joint is assembled the surfaces are flush.

The sequence is: marking out with marking tools, cutting with the tenon saw down to the gauge line, removing the waste with a chisel and mallet, cleaning up the bottom of the socket, and shaping with the rasp. The joint is then assembled by applying light blows from the mallet and tested for squareness with the try square. A good halving joint fits tightly without gaps, and it is usually fixed with glue, screws or nails.

Diagrams

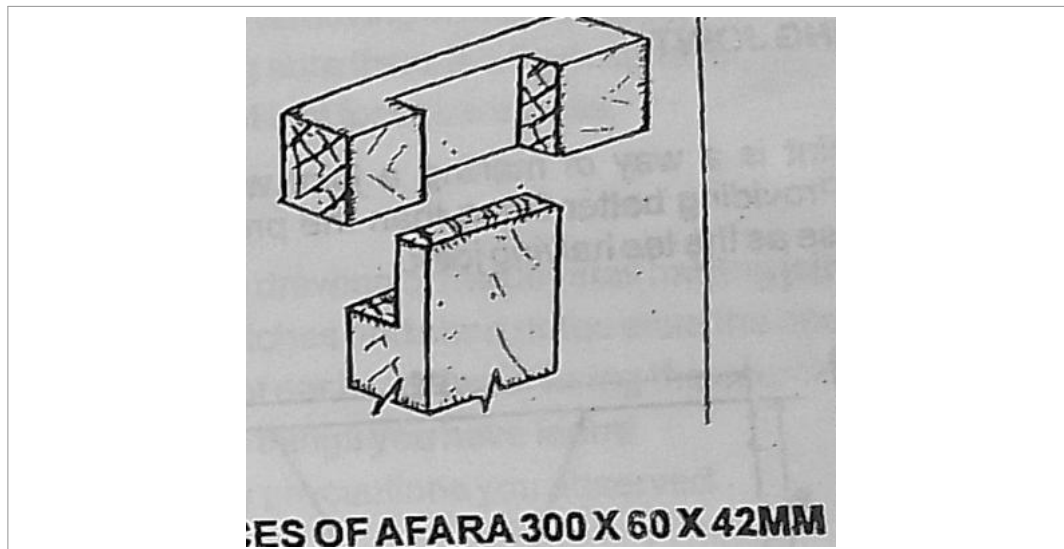


Figure 31: Tee halving joint showing the socket on one member and the pin on the other (Lab Manual, p. 33).

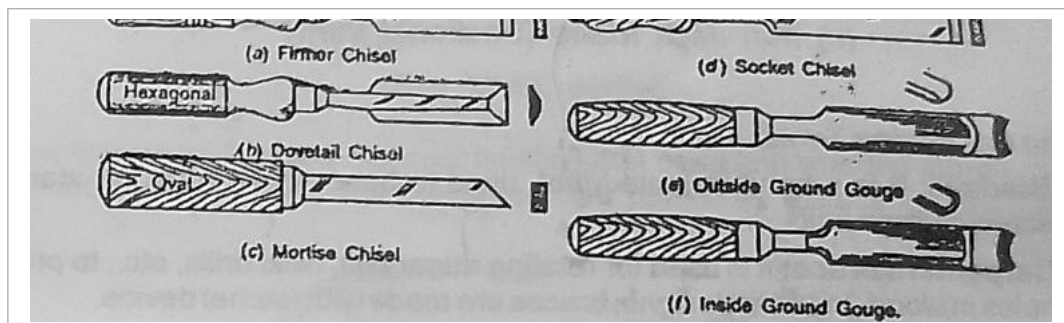


Figure 32: Chisels used to remove the waste (Lab Manual, p. 27).

5. Procedure

1. The pieces were marked out, one for the socket and the other for the pin, using the tape, marking gauge, pencil and try square.
2. The marking lines were cut as specified using the tenon saw, stopping at the gauge line.
3. The waste was removed using the chisel and mallet.
4. The bottom of the socket was cleaned using the rasp.
5. The pin was cut to size to fit the socket.
6. The joint was fixed by applying light blows with the mallet.
7. When properly fitted, the joint was tested for squareness using the try square.

6. Precautions

- Cut on the waste side of the marked lines.
- Cut only to the gauge line so that the depth of the recess is correct.
- Always chisel away from the body and clamp the work down.
- Do not strike a chisel with a hammer; use a mallet.
- Do not force the joint together; trim the pin gradually until a snug fit is reached.

7. Discussion

The tee halving joint looked simple, but it demanded accuracy. If the depth of the socket was too shallow, the surfaces of the two pieces would stand proud of each other after assembly, and if it was too deep there would be a gap. The marking gauge was essential in getting the

correct depth on both pieces. The quality of the fit was also dependent on cutting exactly to the lines, because it is much easier to remove a little more wood than to put it back.

We found that the chisel gave good results when taken in thin layers and when the bevel faced the correct direction, while heavy cuts split the wood along the grain. When the pin was trimmed in small steps the final joint fitted firmly and was square when tested.

8. Conclusion

A tee halving joint was successfully produced from two pieces of timber. The exercise strengthened our skills in marking, sawing, chiselling and fitting, and showed that accuracy in marking is the key to a well-fitting joint.

9. Questions Asked in the Material

Show a working drawing of the tee halving joint.

The drawing shows two members, 300 x 60 x 42 mm, with the socket on one equal to half the thickness and the matching pin on the other (see the figure).

State the uses of each tool used.

Tenon saw: cutting down to the gauge line. Chisel and mallet: removing waste. Rasp: cleaning the bottom of the socket. Try square: testing squareness. Marking gauge: marking depth.

List the new things you have learnt.

See the section below.

10. What I Have Learned

- The structure of a halving joint and where the tee halving joint is used.
- How to use a marking gauge to set the depth of a recess.
- How to remove waste with a chisel and mallet.
- How to test a joint for squareness.

WW 201 (Practical 4): Dovetail Halving Joint

1. Aims

To make a dovetail halving joint, a joint that keys the two wood members together and fits better than a simple tee halving joint.

2. Objectives

- To mark out a dovetail pin and socket using a dovetail marker, try square and marking gauge.
- To cut the dovetail with a dovetail saw and to remove waste with chisels.
- To fit and adjust the joint to give a flush surface.

3. Apparatus / Materials

- Two pieces of Afara, 300 x 60 x 42 mm.
- Pencil, measuring tape, try square, dovetail marker, marking gauge.
- Dovetail saw, chisel, mallet, paring chisel and jack plane.

4. Theory / Diagram

The dovetail halving joint is a way of making a halving joint in which the pin is shaped like a dovetail (a wedge that widens towards its end). The two wood members therefore key together properly, which means that the joint cannot be pulled apart in the direction of its length. This gives a better fit than the ordinary tee halving joint, and it serves the same purpose. The joint is used where tension is likely to try to pull the members apart, in frames, in table rails and in cabinet work.

The pin has sloping sides, usually in a ratio between 1 in 6 and 1 in 8, drawn with the dovetail marker, so the pin is wider at its end than at its neck. The socket is cut to the same shape in the other member. The depth is half of the thickness, as in the tee halving joint. In the dimensions of the manual the pin is 150 mm wide at the widest part and the pieces are 50 mm wide with 12 mm for the depth recess.

The socket is cut first, because it is easier to mark the pin from the socket than the socket from the pin. The pin outline is then transferred from the socket to the other member by a pencil or a scribe, and the pin is cut to fit. Surplus wood is scooped out with chisels and the pin is trimmed with a paring chisel. Any fine adjustment is made by removing small amounts of wood from the pin, since the joint must fit tightly without the use of excessive force. At the end the joint should lie flush on the surface, and the surface is levelled with the plane.

Diagrams

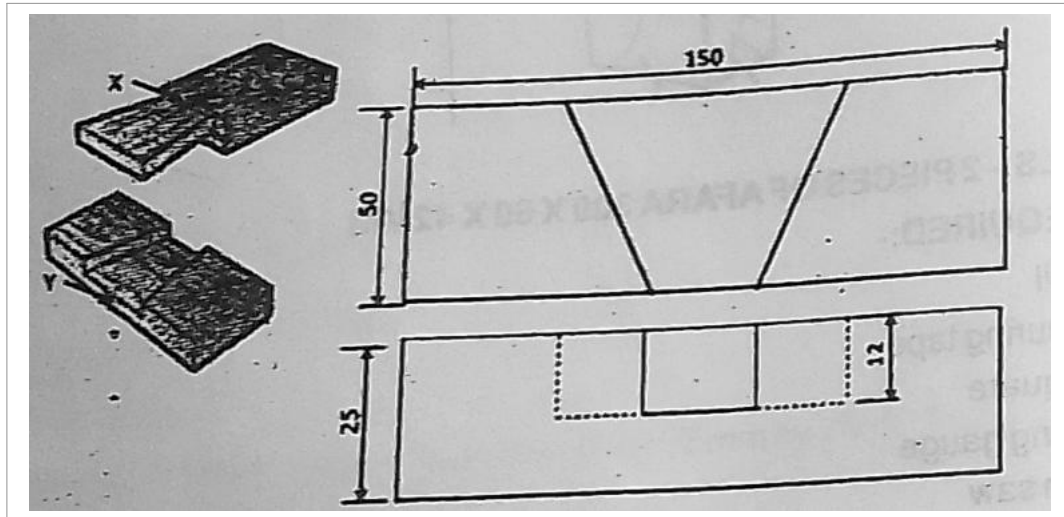


Figure 33: Dovetail halving joint with its pin and socket and main dimensions (Lab Manual, p. 34).

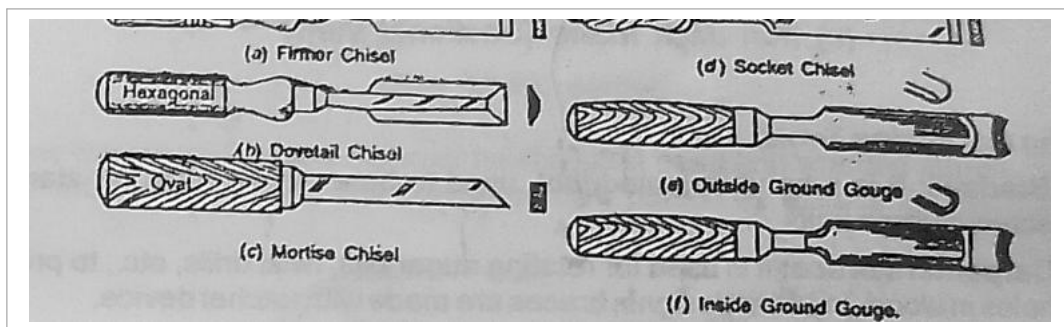


Figure 34: Chisels used to scoop out waste and to pare the pin (Lab Manual, p. 27).

5. Procedure

1. The two pieces were marked out, one for the socket and the other for the dovetail pin.
2. The try square was used to transfer the marks to the edges.
3. The socket was cut first using the dovetail saw.
4. The waste was scooped out and the socket cleaned up with chisels.
5. The pin and the dovetail were cut.
6. The pin was cleaned, including the waste portion, using the paring chisel.
7. The joint was fixed and adjusted by removing unwanted parts.
8. The joint was fitted to make a flush surface, and the jack plane was used to level it.
9. The joint was tested for squareness.

6. Precautions

- Cut on the waste side of the lines and never cut beyond the gauge lines.
- Use sharp chisels and keep both hands behind the cutting edge.
- Clamp the work during chiselling.
- Trim the pin in small cuts rather than forcing the joint together.
- Wear safety glasses and keep your bench clear of tools.

7. Discussion

The dovetail halving joint was more difficult than the tee halving joint because the sloping sides had to be marked and cut accurately. Where the angle was slightly off, the pin did not seat properly and a gap appeared. The best method was to cut the socket first and then mark

the pin from it, which meant that any small error in the socket was copied into the pin and the two still fitted.

The joint also demonstrated the benefit of the dovetail shape. Once assembled, the joint could not be pulled apart lengthwise, unlike the simple tee halving joint. Paring with a sharp chisel was the most useful technique for the final fit because it removes only a very thin layer at a time.

8. Conclusion

A dovetail halving joint was produced and tested. The exercise showed the value of careful marking and of fitting the pin to the socket, and it demonstrated that this joint is stronger and tighter than a tee halving joint.

9. Questions Asked in the Material

Show a working drawing of the dovetail halving joint.

The drawing shows two members with a dovetail pin of 150 mm and the socket with 12 mm depth recess (see the figure).

State the uses of each tool used.

Dovetail marker: marking the dovetail slope. Dovetail saw: fine cutting. Chisel and mallet: removing waste. Paring chisel: trimming the pin. Jack plane: levelling the surface. Marking gauge: marking depth.

10. What I Have Learned

- How a dovetail locks two members mechanically.
- How to cut a socket first and mark the pin from it.
- How to use a paring chisel for fine fitting.

SM 201: Introduction to Sheet Metal Work

1. Aims

To study the processes, metals, hand tools and operations involved in sheet metal practice, and to understand how a flat sheet is developed and formed into a useful article.

2. Objectives

- To know the processes involved in sheet metal practice.
- To know the metals used in sheet metal work.
- To become familiar with the hand tools used in sheet metal practice.
- To know the various sheet metal operations.
- To be able to produce a rectangular plate (tray).

3. Apparatus / Materials

- Galvanised plain (G.P.) sheet, 28 gauge.
- Measuring and marking tools: steel scale, steel square, scribe, divider, trammel, wire gauge.
- Cutting tools: straight snips, bent snips and circular snips.
- Punches: prick punch, centre punch and solid punch.
- Hammers and mallets, stakes, files, pliers and soldering iron.

4. Theory / Diagram

To make a sheet metal object, the flat sheet is first cut to a particular shape. It is then formed by laying its entire lateral surfaces next to each other, and after that it is shaped by bending, forming and folding as appropriate, and finally joined by soldering, welding or riveting. The sheet metal process therefore consists of five stages: study of the drawing, development, preparation of patterns or templates, operations on the sheet metal, and joining.

Development and pattern

The complete surface of an object, opened out onto one plane and drawn on paper with suitable allowances for seams and hems, is called a development. The sheet obtained by laying all the outer surfaces of the object together, with the necessary allowances, is called a pattern or template. All lines in the development must be drawn in true length. Four methods of development are used. (1) The parallel-line method is used for right prisms, cylinders, cubes and rectangular trays. (2) The radial-line method is used for the lateral surfaces of right pyramids and cones. (3) The triangulation method is used where the surface is made of a number of triangles laid out in true size, for example transition pieces such as square to round. (4) The approximate method is used for double-curved surfaces such as a sphere.

Sheet metal operations

Marking: the development is placed on the G.P. sheet, and the boundary line is marked with a scribe, with the bending lines punched. Cutting and shearing: the sheet is cut by chisel, hand snips or tin cutter, or by shearing machines. Circle cutting: curved contours are cut with curved chisels, hand snips or a circle-cutting machine. Hand forming: the sheet is shaped in three dimensions to its final shape and size. Hemming, edge stiffening and wiring: the edge of the sheet is folded to ensure safety, to give stiffness and to give a good appearance. Joint making or seaming: sheet parts are joined by folded or self-securing joints. Bending: the sheet is bent by hammering over a stake or by a bending or rolling machine. Notching: the opening left at the corners of seams and edges is notched to prevent bulging at those places.

Notches can be straight, square or V-shaped. Drawing: done with a die and a punch on a press. Hollowing: beating the sheet into a bowl shape on a hollowing block or sand bag. Sinking: the bottom is sunk to form a tray with a flat rim. Raising: dinting the metal down over a tool with a raising hammer. Planishing: improving the surface and giving the final shape and finish. Soldering: joining sheet metal parts using a low-melting-point alloy (solder) that is melted between the surfaces and allowed to solidify.

Metals used

A large variety of metals are used as sheet and plate. The metal sheet is specified by its gauge number, length and width. The gauge number corresponds to the thickness of the sheet as measured by a standard wire gauge. The higher the gauge number, the thinner the sheet. Common metal sheets include black plain (hot rolled) sheet, cold-rolled sheet (0.1 to 0.3 % carbon), galvanised plain sheet (coated with zinc), aluminium sheet, copper sheet, tin sheet, stainless steel sheet (about 18 % chromium and 8 % nickel with less than 1 % carbon) and brass sheet.

Hand tools

Measuring and marking tools are the steel scale, wire gauge, micrometer and vernier calliper, steel square, scribe, divider and trammel point. Cutting tools are hand snips (straight, bent, combination and circular snips). Punches include the prick punch, solid punch, centre punch and hollow punch. Others are chisels, stakes of different shapes, hammers and mallets, files, pliers and soldering iron. Sheet metal fabrication is applied in automobile, aircraft and ship building, packaging, air conditioning, ventilation, and in the manufacture of funnels, hoppers, bins, trays, buckets, storage tanks and chimney boilers.

Hems and seams

A hem is a folded edge. Single hems and double hems are used to stiffen and strengthen the edge and to remove the sharp edge. A seam is a joint between two sheet metal pieces. Common seams include the single lap seam, the grooved seam and the double seam. A wired edge is made by folding the sheet around a wire for rigidity.

Diagrams

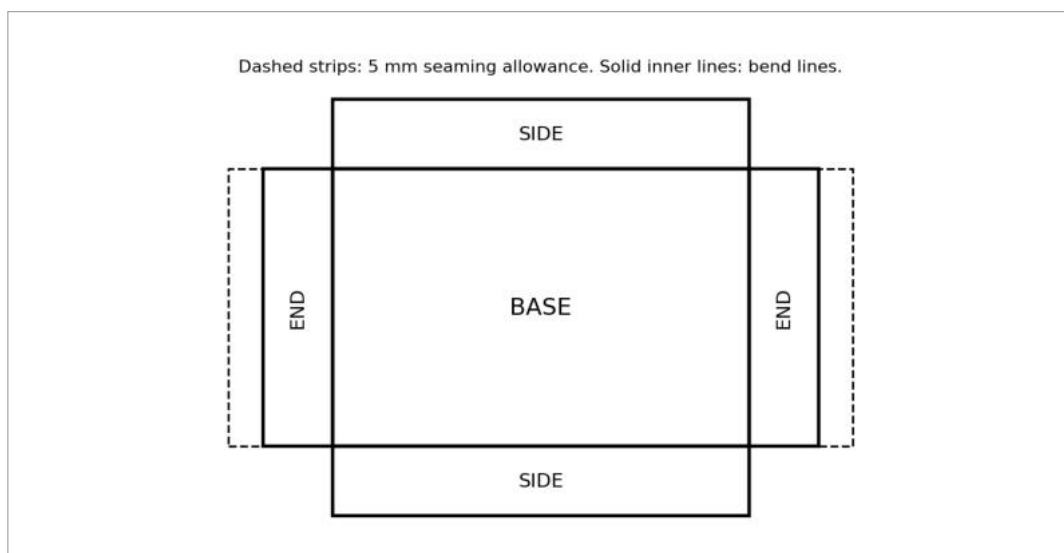


Figure 35: Development (pattern) of a rectangular tray showing the base, sides, ends and seaming allowance (redrawn for clarity from Lab Manual, p. 40).

5. Procedure

1. The drawing of the article was studied and the dimensions were noted.

2. The pattern was developed on paper, including allowances for seams and hems, and with all lines drawn in true length.
3. The pattern was cut out and placed on the G.P. sheet, and the boundary and bend lines were marked with a scribe.
4. The sheet was cut along the boundary lines using hand snips.
5. The corners were notched to prevent bulging at the seams.
6. The sheet was bent along the bend lines over a stake or bench edge.
7. The edges were hemmed or seamed and the joints were soldered.
8. The finished surface was cleaned and checked.

6. Precautions

- Wear gloves when handling sheet metal, since the edges are razor-sharp.
- Remove burrs after cutting.
- Do not use the snips to cut beyond their capacity.
- Keep fingers away from the cutting edge and from the stake while hammering.
- Use the soldering iron carefully and ventilate against fumes.

7. Discussion

Sheet metal work depends on correct development, since a mistake in the pattern appears as a gap, a misfit or a twisted article. The use of true lengths for all the lines in the development was emphasised, since the sloping sides of a tapered article look shorter in the plan than they really are. Allowances for seams and hems must be added to the pattern, or the finished article will be undersize.

The choice of sheet also matters: galvanised plain sheet can be soldered, bent and formed easily, while stainless steel is harder to work. The thin sheets are easily distorted, and so it is important to work carefully and hammer lightly.

8. Conclusion

The processes, metals, tools and operations of sheet metal work were studied. We learned the methods of development, the importance of true length and seam allowance, and the sequence from marking and cutting to bending and joining, which prepares us to make the tray, dustpan and tapered tray.

9. Questions Asked in the Material

What are the five processes involved in sheet metal practice?

Study of the drawing, development, patterns or templates, operations on sheet metal, and joining.

What are the methods of development?

Parallel line, radial line, triangulation and approximate methods.

What does gauge number mean?

It indicates the thickness of the sheet, measured with a standard wire gauge; the higher the number, the thinner the sheet.

10. What I Have Learned

- The five stages of a sheet metal job and the four methods of development.
- Why a development must be drawn in true length with allowances.

- The names and uses of the main sheet metal hand tools and operations.

SM 201 (Practical 1): Rectangular Plate (Tray)

1. Aims

To produce a rectangular tray from G.P. sheet, from working drawing, development and pattern to the finished soldered article.

2. Objectives

- To draw the 2D and 3D views and the full-scale development of the tray.
- To mark, cut, bend and seam a sheet metal tray.
- To solder the joint and finish the surface.

3. Apparatus / Materials

- G.P. (galvanised plain) sheet 28 gauge.
- Paper, pencil, scale and paper-cutting scissors.
- Steel rule, scribe, marking pen, dot or prick punch.
- Hand snips, mallet, flatter, stake, soldering iron, solder and flux.

4. Theory / Diagram

A tray is a typical example of an article developed by the parallel-line method. The base is a rectangle, and the four sides are rectangles that are folded up at 90 degrees from the edges of the base. Because the sides are perpendicular to the base, every line in the development appears in its true length, which makes the development simple. The sides and ends are attached to the base at the bend lines, and extra strips (allowances) are added to the free edges so that the corners can be joined by seaming and soldering.

The manual stipulates a 5 mm extra allowance on the side faces to be used for joining the corners by seaming. At the corners the metal would overlap and bulge if bent as it is, so the corner points are notched (a small piece removed) to prevent bulging. Bending is done along the seaming line first and then along the remaining lines. The seams are then soldered, using a low-melting-point alloy that flows between the surfaces by capillary action once the metal and flux are hot enough, and finally the surface is finished.

The development, made on paper first, serves as a template. It is cut out, fixed on the sheet with a sticker, and the bend lines are transferred by punching two points on each bending line, because a punched dot is more permanent than a pencil line. The boundary lines are then marked in with a marker pen and sheared.

Diagrams

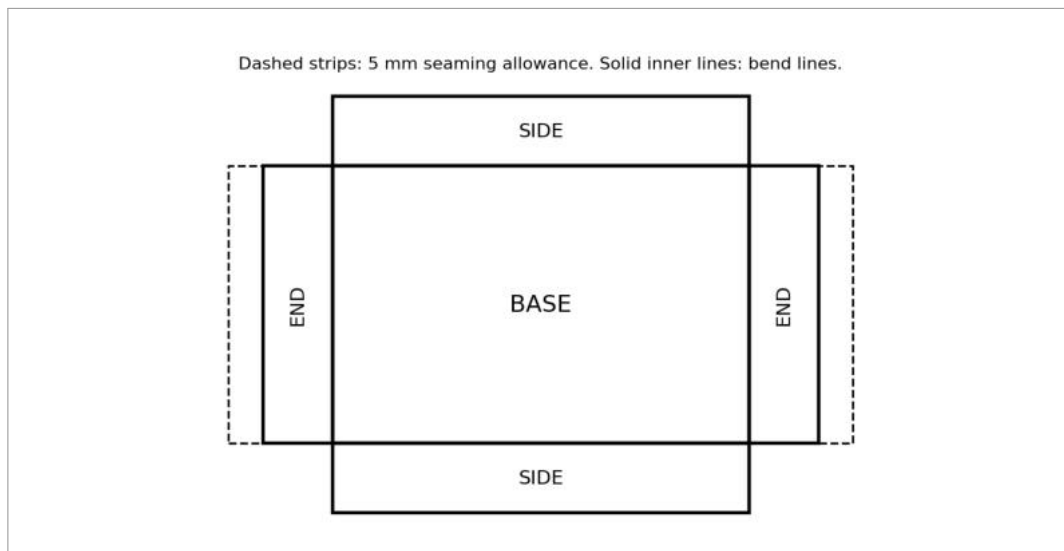


Figure 36: Development of the rectangular tray: a central base with four folded sides and 5 mm seaming allowances (redrawn for clarity from Lab Manual, p. 40).

5. Procedure

1. 2D and 3D sketches were drawn to scale.
2. The development was drawn, taking all lines in true length.
3. The base of the tray was constructed, followed by the two sides and the two ends.
4. A 5 mm extra allowance was set on the side faces for joining the corners by seaming.
5. The points were notched to prevent bulging on seaming.
6. The development shape was cut out along the lines using paper-cutting scissors.
7. The development was placed and fixed on the G.P. sheet using a sticker.
8. Two points were punched on each bending line.
9. The boundary line was marked with a marking pen.
10. The paper development was removed, and the two punch points were joined using a scribe or marking pen.
11. The boundary lines were sheared using the hand snip.
12. The sheet was flattened using a mallet on a flatter.
13. The seaming line was bent first, followed by the remaining lines, to get the desired shape by forming.
14. The surface was finished, keeping the work on a stake to shape it.
15. The joint was soldered.
16. Finally, the surface was cleaned and finished to complete the tray.

6. Precautions

- Wear gloves when handling the cut sheet and remove burrs from the edges.
- Mark carefully, as errors cannot be corrected after the sheet is cut.
- Bend the seaming line first to avoid distorting the other bends.
- Do not hammer too hard, since the sheet is thin and can stretch or crack.
- Handle the hot soldering iron with care, and work in a ventilated place.

7. Discussion

The tray was a good introduction to sheet metal because every line in its development is in true length. The principal difficulty was bending accurately along the bend lines. When the bend lines were punched clearly and the sheet was bent over a straight edge or stake, the sides came up square. Where the lines were not clear, the bend wandered and the sides did not close neatly.

The 5 mm seaming allowance and the corner notches both proved their worth. Without notching, the metal at the corners bunched up and prevented the sides from closing. With the notches cut, the corners met neatly and the soldering produced a tidy seam. Soldering required clean metal and flux, because otherwise the solder would not flow.

8. Conclusion

A rectangular tray was produced from G.P. sheet following the sequence of development, marking, cutting, bending, seaming and soldering. The objective was achieved, and we learned the importance of correct development, true lengths, allowances and notching.

9. Questions Asked in the Material

Produce the working drawing of the tray.

The tray is drawn in plan, front view and a 3D view, all dimensions in mm (see the 2D/3D drawing in the manual, p. 40).

Produce the development of the tray.

See the diagram above: a base with the sides and ends attached on the bend lines, and a 5 mm seam allowance.

Using simple sketches, state the operational sequence of how the tray was produced.

Sketch and develop, cut pattern, fix on sheet, punch bend lines, mark, shear, flatten, bend, form, solder, finish (see procedure).

State the safety precautions you observed.

See the precautions above.

10. What I Have Learned

- How to develop a tray by the parallel-line method.
- Why notching and seaming allowances are required.
- How to bend, form and solder a sheet metal joint.

SM 201 (Practical 2): Dustpan

1. Aims

To produce a dustpan from G.P. sheet, from working drawing and development to the finished article.

2. Objectives

- To draw the 2D and 3D views and the development of a dustpan.
- To mark, cut, bend and join the sheet to form the dustpan.
- To compare the dustpan with the simple tray and understand the effect of the tapered side.

3. Apparatus / Materials

- G.P. sheet, 28 gauge.
- Paper, pencil, steel rule, scribe and marking pen.
- Dot punch, hand snip, mallet and flatter, stake.
- Soldering iron, solder and flux.

4. Theory / Diagram

A dustpan is similar to the tray in its construction, except that it has a shallow body with a sloping (tapered) front lip and a raised back, so that dust can be swept onto it easily. It consists of a base, two sides and a back. The sides are not rectangles but taper from a low front to a higher back, so their development is a trapezoid. The slanting edge must be drawn in true length because the vertical height of the side is not the same as the length of the sloping edge.

A handle is usually attached at the back, riveted or soldered. The dustpan is formed by the same sequence as the tray: the development is drawn, the base is constructed, then the two sides, then the ends. A 5 mm allowance is provided on the side face for joining the corners by seaming, the points are notched to prevent bulging on seaming, and the sheet is bent, formed and soldered. The front edge is generally hemmed or kept low so that dust can slide on.

The design of a dustpan shows the practical use of sheet metal development in domestic products. It also shows that more than a single development method may be needed when edges are inclined; the sides are developed using the parallel-line method, but their true lengths are obtained from projection.

Diagrams

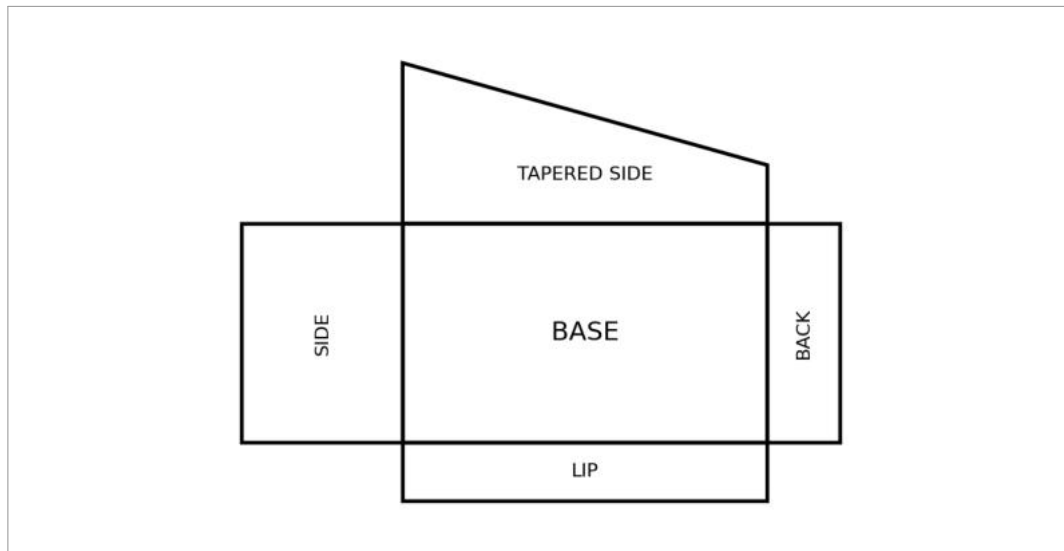


Figure 37: Development of a dustpan: base, tapered sides, lip and back (redrawn for clarity from Lab Manual, p. 41).

5. Procedure

1. 2D and 3D sketches were drawn to scale.
2. The development was drawn, considering all lines in true length.
3. The base of the dustpan was constructed, then the two sides and the two ends.
4. A 5 mm allowance was set on the side face for seaming.
5. The corners were notched to prevent bulging on seaming.
6. The development was cut from paper and fixed on the G.P. sheet with a sticker.
7. Two points were punched on each bending line, and the boundary lines were marked.
8. The paper was removed, the punch points were joined with a scribe, and the sheet was sheared along the boundary using the hand snip.
9. The sheet was flattened with a mallet on a flatter.
10. The seaming lines were bent first, then the remaining lines, to form the shape.
11. The surface was finished on a stake, the joints were soldered, and the dustpan was cleaned.

6. Precautions

- Take care with sharp edges and burrs after shearing.
- Do not over-stretch the sheet when hammering.
- Check the dimensions of the development before cutting.
- Use the soldering iron and flux carefully.
- Wear gloves and safety glasses.

7. Discussion

The dustpan was a little more demanding than the tray because the sides are tapered, and their slanting edges had to be set out in true length. If we had used the plan length, the finished sides would have been too short and the joints would not have met properly. The difference between plan view and true length is an important concept in sheet metal development.

The low front lip and the sloping sides also made the bending order important. We found that bending the seaming line first prevented distortion. The solder must be applied sparingly, to

give a neat joint without leaving lumps that would hinder the dust from sliding into the pan.

8. Conclusion

A dustpan was developed, cut, formed and soldered from G.P. sheet. The exercise reinforced the principles of development in true length, notching, seaming and soldering, and showed how the tray technique can be adapted to a different product.

9. Questions Asked in the Material

Produce the working drawing and development of the dustpan.

See the development above and the 2D/3D drawings in the manual (p. 41).

Using simple sketches, state the sequence of operations.

Sketch, develop, cut pattern, mark sheet, punch, shear, flatten, bend, form, solder, finish.

State the safety precautions.

See above.

10. What I Have Learned

- How to develop an article with tapered sides using true lengths.
- Why the order of bending matters.
- How to adapt a basic technique to a new product.

SM 201 (Practical 3): Tapered Tray

1. Aims

To produce a tapered tray from G.P. sheet, in which the sides flare outwards from the base.

2. Objectives

- To draw the working drawing and the development of a tapered tray.
- To determine the true lengths of the slanting sides and ends.
- To cut, bend, seam and solder the sheet to form the tray.

3. Apparatus / Materials

- G.P. sheet, 28 gauge.
- Paper, pencil, scale, scribe and marking pen.
- Dot punch, hand snip, mallet and flatter, stake.
- Soldering iron, solder and flux.

4. Theory / Diagram

A tapered tray is a tray whose sides flare outwards from the base, so the top opening is larger than the base. It is typical of many household and industrial articles such as bowls, trays, hoppers and chutes. Its sides are trapezoids, and each side is inclined to the base at an angle larger than 90 degrees.

Because the sides are inclined, the true length of the slant edges cannot be taken directly from the plan or the elevation. The true length is found by projecting the inclined edge into a plane in which it appears in its real length, or by calculation using Pythagoras' theorem from the vertical height and the horizontal run of the slant edge. This is the reason why the manual repeatedly insists that the development must be drawn with all lines in true length.

The development is made by the parallel-line (or by an approximate triangulation) method. The base is drawn in the middle, and the sloping sides and ends are attached to the edges of the base at the bend lines. At the corners, adjacent sides meet at angles that are slightly different from 90 degrees, so the corner flaps and notches have to be drawn carefully. Seaming allowance of 5 mm is provided on the side faces, and the corners are notched to prevent bulging.

After cutting, bending starts from the seaming line and then the other bend lines are bent to give the desired shape by forming. The tray is finished on a stake, soldered at the joints, and cleaned.

Diagrams

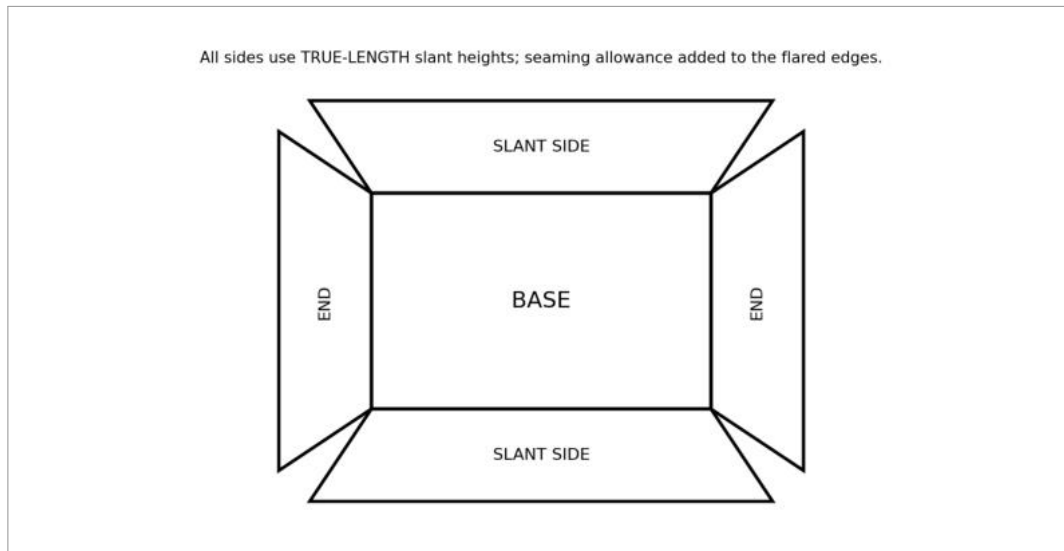


Figure 38: Development of a tapered tray: the sides and ends are trapezoids attached to the base and flare outwards (redrawn for clarity from Lab Manual, p. 42).

5. Procedure

1. 2D and 3D sketches were drawn to scale.
2. The development was drawn with all lines in true length.
3. The base was constructed, then the two sloping sides and the two sloping ends.
4. A 5 mm seam allowance was set on the side faces to join the corners.
5. The points were notched to prevent bulging.
6. The development was cut out with paper-cutting scissors and fixed on the G.P. sheet with a sticker.
7. Two points were punched on each bend line, and the boundary line was marked.
8. The paper development was removed, and the punch points were joined by a scribe.
9. The boundary lines were sheared with the hand snip, and the sheet was flattened with a mallet on a flatter.
10. The seaming line was bent first, followed by the other lines, to get the desired shape.
11. The surface was finished on the stake, the joint was soldered, and the tray was cleaned and completed.

6. Precautions

- Wear gloves and avoid contact with sharp edges.
- Check the true lengths before cutting.
- Make small, even hammer blows when forming.
- Notch the corners carefully, and do not cut beyond the notch lines.
- Ventilate the area when soldering.

7. Discussion

The tapered tray was the most challenging of the three trays. The slanted sides meant that the corners of the development did not meet at right angles, and a small error in the true length of the edges caused the adjacent sides to be of different lengths so that the seams did not line up. By working out the true lengths before cutting, we avoided this error.

The tapered tray also showed the value of the stake. The flared shape could be formed gradually by hammering the sheet over the stake, and this gave a smooth finish. Once

soldered, the tray was rigid, even though the sheet itself was thin, which shows how forming and seaming add stiffness to thin metal.

8. Conclusion

A tapered tray was produced from G.P. sheet. The experiment showed how true lengths and notching are applied to articles with inclined sides, and it consolidated our skills of cutting, bending, seaming and soldering.

9. Questions Asked in the Material

Produce the working drawing and the development of the tapered tray.

See the diagram above and the 2D/3D drawings in the manual (p. 42).

State the operational sequence using simple sketches.

Sketch, find true lengths, develop, cut the pattern, mark the sheet, punch, shear, flatten, bend, form, solder, finish.

State the safety precautions you observed.

See above.

10. What I Have Learned

- How to find the true length of an inclined edge.
- How the shape of the development changes for a tapered article.
- How to form a flared shape on a stake.

BF 201: Introduction to the Fitting Shop (Bench Work)

1. Aims

To study the tools used in the fitting shop, the methods of marking out, cutting and filing metal, and the preparation needed to produce a square fit.

2. Objectives

- To know the tools used in the fitting shop.
- To know how to mark out.
- To know the methods of filing.
- To produce a square fit.

3. Apparatus / Materials

- Work bench and bench vice, V-block, angle plate and surface plate.
- Steel rule, callipers, try square, scribe, surface gauge, vernier height gauge, dot and centre punches.
- Hacksaw frame and blades, files (flat, hand, square, round, half-round, triangular, knife-edge).
- Ball peen hammer.

4. Theory / Diagram

The term fitting is related to the assembly of parts, after bringing their dimensions or shapes to the required size or form, in order to secure the necessary fit. The operations required for this are usually carried out on a work bench, hence the term bench work is also added to the name fitting. Bench work and fitting play an important role in engineering. In today's industries most of the work is done by automatic machines that produce jobs with good accuracy, but some hand operations, called fitting, are still required to complete the job. The person who works in the fitting shop is called a fitter.

Fitting tools are grouped into work-holding or clamping tools, measuring and marking tools, cutting tools, striking tools, drilling tools and threading tools.

Work-holding and clamping tools

The work bench is a strong, heavy and rigid table of hard wood, about 150 to 180 cm long, nearly 90 cm wide and 76 to 84 cm high. The bench vice is fixed firmly to the bench with nuts and bolts. It has a cast iron body, two jaws with hardened carbon steel plates that are knurled for gripping, a square-threaded main screw and a handle; one jaw is fixed and the other slides. The V-block has V-grooves to hold round work longitudinally, and a clamp with a screw applies the holding pressure.

Measuring tools

The steel rule is made of stainless steel in lengths from half a foot to two feet and marked in inches or millimetres. Callipers measure inside and outside diameters. Outside callipers measure outside dimensions and inside callipers measure inside dimensions; the spring calliper has a spring to apply pressure and a nut to lock any desired position. The hermaphrodite (jenny or odd-leg) calliper has one leg bent inwards at the tip and the other straight and pointed, and it is used to scribe lines parallel to straight edges. The vernier calliper measures the outer dimension of round and flat components, the inner size of holes

and bores, and with its blade the depth of slots, with a reading accuracy of 0.02 mm (or 0.001 in).

Marking tools

The surface plate is made of cast iron or graphite and has a very smooth, planed upper face. It is used to test the flatness of surfaces and in scribing work, and it must be protected with a cover when not in use. The angle plate is a cast iron L-shaped block with two planed faces at right angles, with slots to take clamps and bolts to hold the work. It is never hammered to tighten the nuts. The scribe is made of high carbon steel with a hardened point, and the surface gauge carries a scribe on a stand to mark lines at a given height. Punches are made of high carbon or high speed steel with one end sharpened and the other struck by the hammer. The dot punch (60 degree point) marks dotted lines and the centre punch (90 degree point) marks the centre of a hole before drilling. The try square checks squareness of two surfaces and is also used to mark right angles and to measure straightness; it should never be used as a hammer. The vernier height gauge is similar to a large vernier calliper but has a heavy base so that it stands upright on the surface plate and carries a scribe on its jaw.

Method of marking

Marking means setting out the dimensions with the help of a working drawing, or transferring them directly from a similar part. First the surface to be marked is coated with a paste of chalk or red lead and allowed to dry, so the lines will show up. Then the work is held in a holding device suitable for its shape (surface plate for flat work, V-block with a clamp for round work, angle plate otherwise). Lines at right angles are scribed using the surface gauge, and lines in the other direction by turning the work through 90 degrees. The centre of a round bar can be found using an odd-leg calliper or a surface gauge. Circles and arcs are marked using a divider. Finally, the lines are made permanent by dotting along them with the dot punch and a light hammer so that they remain visible during machining.

Cutting tools: the hacksaw

The hacksaw cuts rods, bars, pipes and flats. It consists of a mild steel frame (fixed or adjustable), a handle and a blade of high carbon steel or high speed steel held by a flange nut. The teeth are set in a zig-zag fashion so that the cut is wider than the blade and the blade does not bind. The teeth of the blade cut in the forward direction, so the pressure is applied forward only. Blades are classified by pitch as coarse (8 to 14 teeth per inch), medium (16 to 20 teeth per inch) and fine (24 to 32 teeth per inch). Thick material takes a coarse blade, and thin sheets and tubes take a fine blade so that at least two or three teeth are in contact with the work at all times.

Files

Files are multi-point cutting tools used to remove material by rubbing them on the metal. They differ in length (4, 6, 8, 12 inches), grade, cut and shape. By grade, files are: rough (about 20 teeth per inch), bastard (30), second cut (40), smooth (50) and dead smooth (100). Rough and bastard files remove a lot of metal, but leave a rough surface, whereas smooth and dead smooth files remove little metal and give a fine finish. By the number of cuts, files are single cut (parallel rows of teeth at 60 degrees), double cut (a second row of teeth cut in the opposite direction, which removes material faster) and rasp cut (separate pointed teeth). By shape, they are flat, hand, square, round (or rat-tail), half-round, triangular, knife-edge, diamond and needle files. A flat file is double cut on its faces and single cut on its edges. A hand file has one safe (uncut) edge for filing a right-angled corner. A square file is used for square holes and slots, a triangular file for square corners and sharpening saws, a round file for opening out holes and producing round corners, a half-round file for curved surfaces, and the knife-edge file for sharp corners of grooves and slots.

Methods of filing

Cross-filing is used to remove the maximum amount of metal in the shortest possible time. The file is held horizontally throughout the stroke (long, slow and steady) with pressure only on the forward stroke. Draw filing is used to remove file marks and for finishing; the file is gripped close to the work between two hands and drawn across the work, using a fine cut file with a flat face.

Diagrams

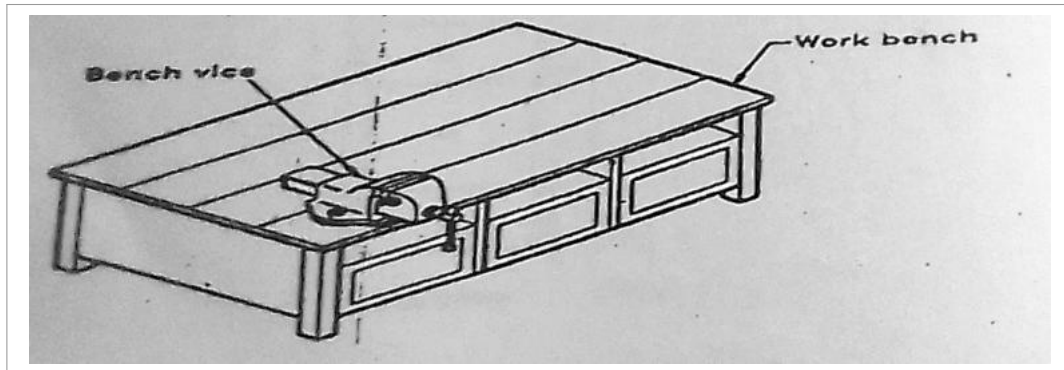


Figure 39: Fitter's work bench with a bench vice (Lab Manual, p. 44).

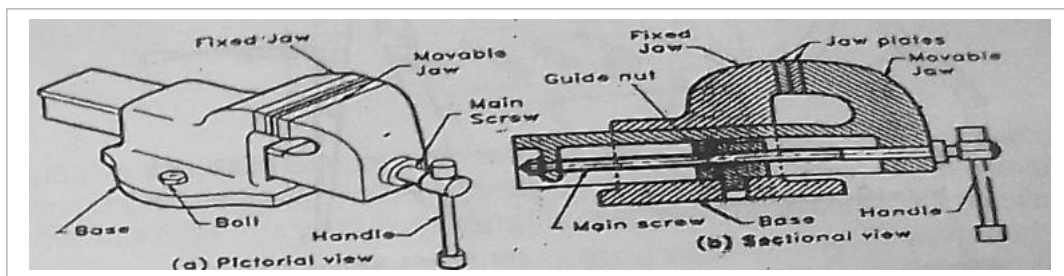


Figure 40: Bench vice: pictorial and sectional view (Lab Manual, p. 45).

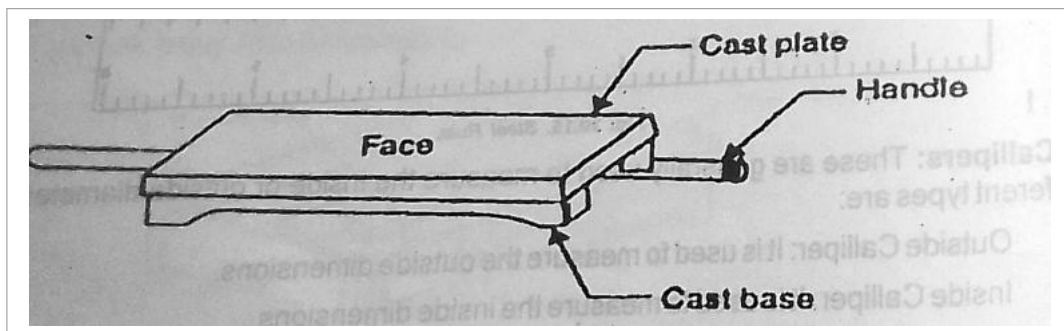


Figure 41: Surface plate (Lab Manual, p. 46).

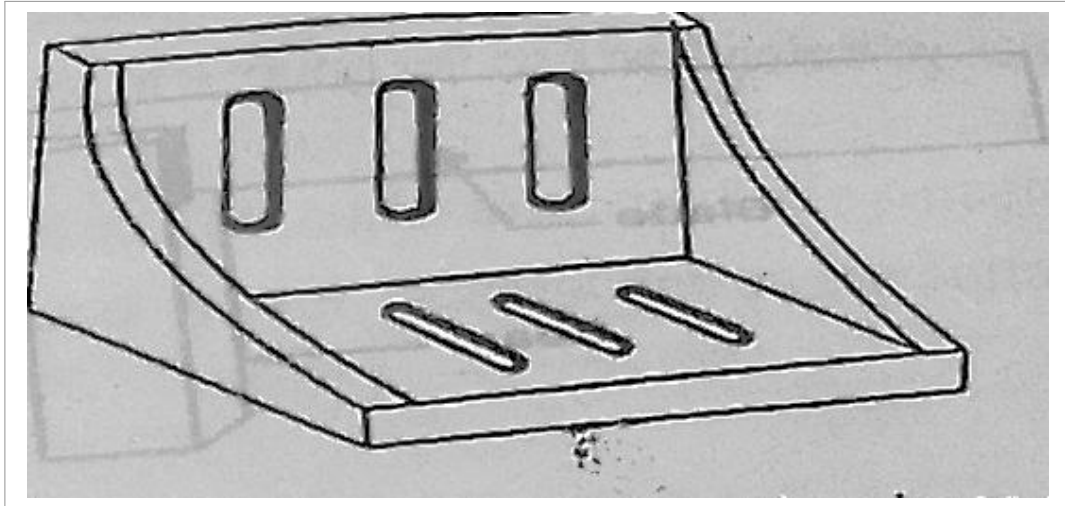


Figure 42: Angle plate (Lab Manual, p. 47).

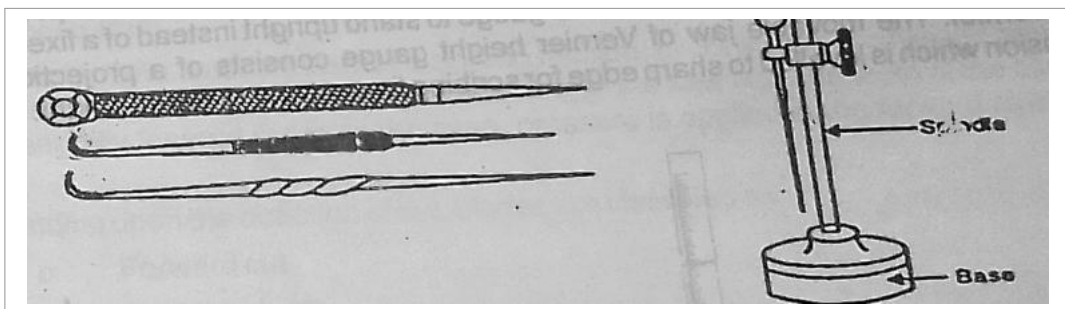


Figure 43: Scribes and surface gauge (Lab Manual, p. 47).

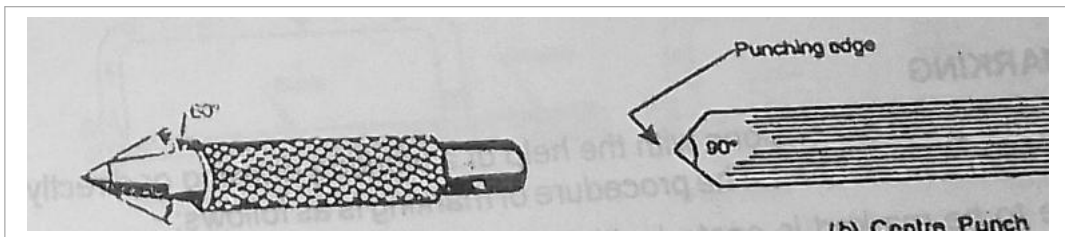


Figure 44: Dot punch (60 degrees) and centre punch (90 degrees) (Lab Manual, p. 47).

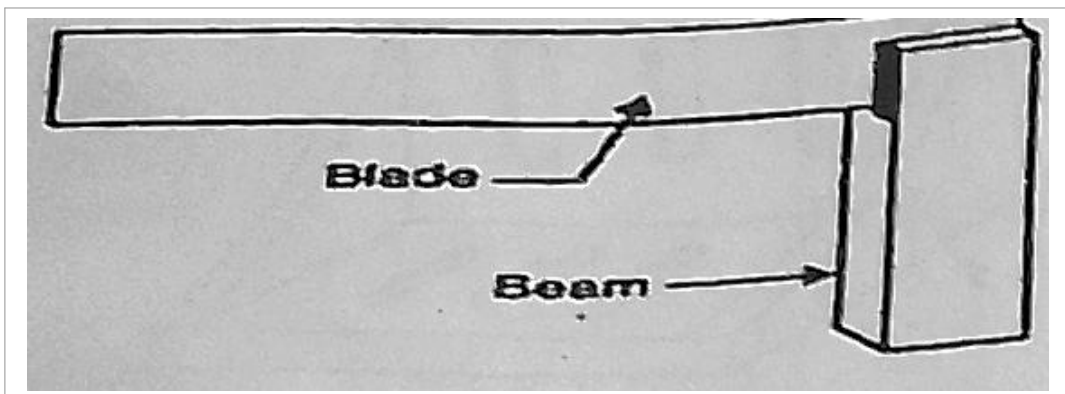


Figure 45: Try square (Lab Manual, p. 48).

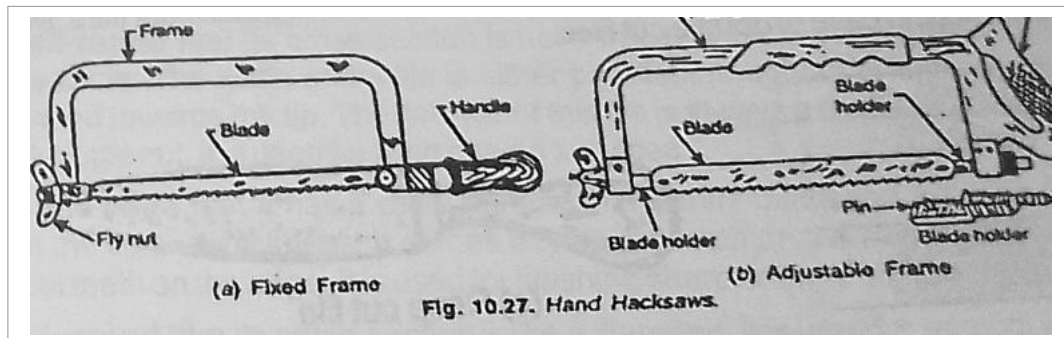


Figure 46: Hand hacksaws: fixed frame and adjustable frame (Lab Manual, p. 49).

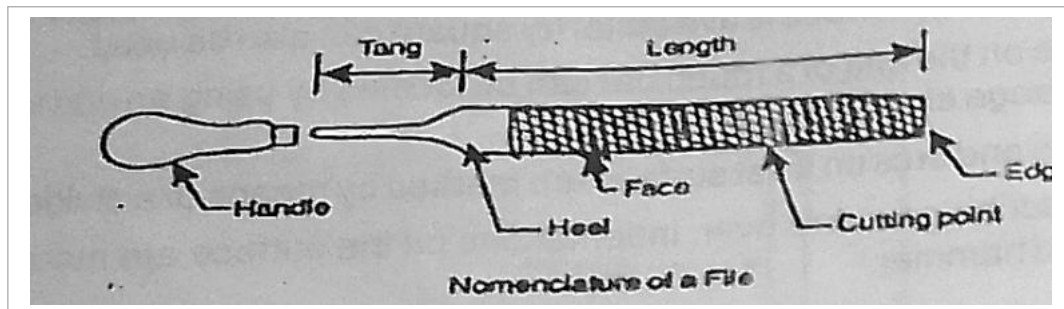


Figure 47: Nomenclature of a file (Lab Manual, p. 50).

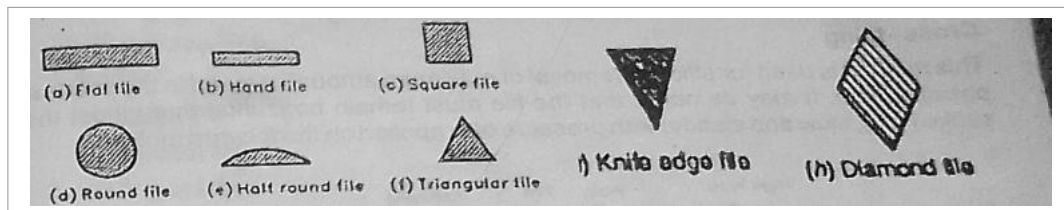


Figure 48: Cross-sectional shapes of files (Lab Manual, p. 51).

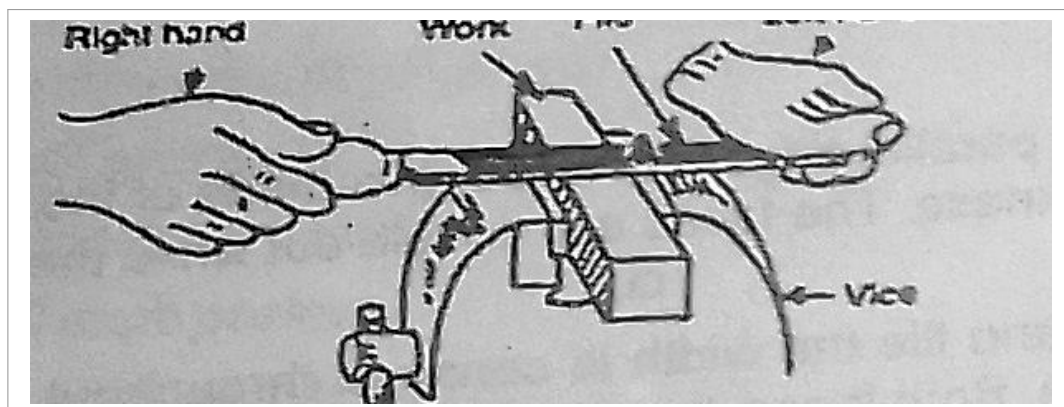


Figure 49: Cross filing (Lab Manual, p. 52).

5. Procedure

1. The tools of the fitting shop were introduced and their uses explained.
2. The bench vice, surface plate, angle plate, V-block and surface gauge were examined.
3. The marking procedure was demonstrated on a flat piece: chalk coating, setting up, scribing, dotting with the punch.
4. The hacksaw blade was fitted with the teeth pointing forward, and a cut was made on a flat piece.
5. Cross filing and draw filing were demonstrated, and the different files were identified.

6. Precautions

- Fix the work firmly in the vice with the line to be cut close to the jaws.
- Never use a file without a handle, and never use a file as a hammer.
- Use the correct blade pitch and make sure the teeth point forward.
- Keep the try square, surface plate and gauges clean and do not hammer on them.
- Wear safety glasses and keep the bench free of filings.

7. Discussion

The fitting shop is a place where accuracy is built by hand. We observed that each tool has a specific role: marking tools define the job, the vice holds it, the saw removes bulk metal, and files produce the final shape and finish. The progression from rough to smooth files demonstrated that metal can be removed quickly with a coarse file but a good surface requires a fine file and draw filing.

We also noted that marking depends strongly on the preparation of the surface. A chalk-coated surface shows a scribed line clearly, and a dotted line survives while filing and cutting. Parallax and wrong readings of the rule or gauge were identified as common sources of error.

8. Conclusion

The tools, marking methods, hacksawing and filing methods of the fitting shop were studied. We now understand how to prepare and hold a work piece, how to mark it out, and how to remove metal by sawing and filing to obtain the required fit.

9. Questions Asked in the Material

What are the classifications of fitting tools?

Work-holding tools, measuring and marking tools, cutting tools, striking tools, drilling tools and threading tools.

Name the grades of files.

Rough, bastard, second cut, smooth and dead smooth.

State the two methods of filing.

Cross filing (for fast removal of metal) and draw filing (for finishing).

10. What I Have Learned

- The names and functions of fitting shop tools.
- The correct method for marking out a work piece and for dotting the lines.
- The types, grades and shapes of files, and how to select a hacksaw blade.

BF 201 (Practical 1): Square Fit

1. Aims

To make a square fit from the given mild steel piece.

2. Objectives

- To file two edges square and straight using rough and smooth files and the try square.
- To mark out, punch, saw and file the work to form a square fit.
- To finish the fitted job to the dimensions given.

3. Apparatus / Materials

- Mild steel flat, 40 x 40 x 3 mm.
- 6 in try square, 6 in scribe, odd-leg calliper, steel rule.
- 12 in hacksaw frame and blades.
- 10 in rough file, 10 in smooth file, 10 in square file.
- Dot punch, ball pein hammer, chalk, bench vice.

4. Theory / Diagram

A square fit consists of two mating parts: a male part with a square projection and a female part with a square recess, which fit into each other with no gap and no rock. The accuracy of a square fit depends on the squareness of the edges, the straightness of the sides and the sizes of the mating dimensions. The manual specifies a raw material of 40 x 40 x 3 mm mild steel flat, to be filed to 38 mm square, and then divided to produce the square projection and recess.

The sequence of operations is filing, marking, punching, sawing, filing and finishing. Filing is done first to produce two edges square to each other, which serve as reference edges (datum). Marking is done by chalking the surface, then scribing lines with the odd-leg calliper and steel rule from the datum edges. The lines are punched with the dot punch so that they remain visible. Sawing removes the bulk of the waste metal, leaving a small allowance. The remaining metal is removed by filing to the line, checking with the try square against the datum edges, and the parts are finished with the smooth file.

Filing to a line by eye is not enough; the part must be checked with the try square, a steel rule and, by looking against light, for gaps. A rough file is used first for fast removal and a smooth file is used last for finish. The recess is finished with the square file, which has a safe edge for corners.

Diagrams

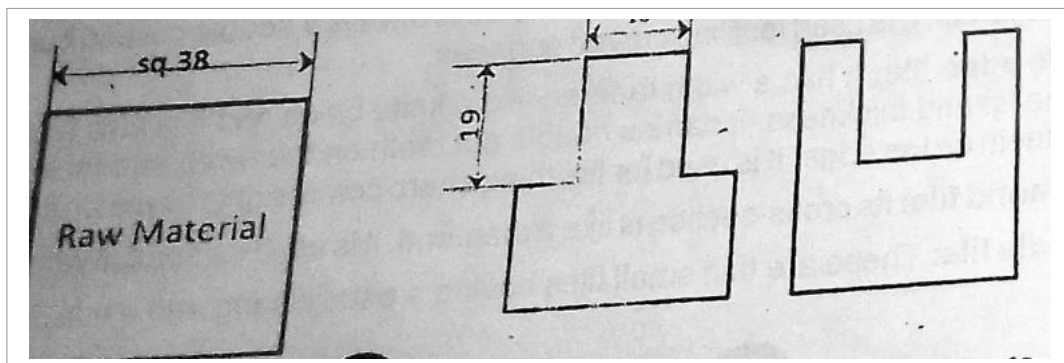


Figure 50: Raw material (38 mm square) and the dimensioned male and female square-fit parts (Lab Manual, p. 52).

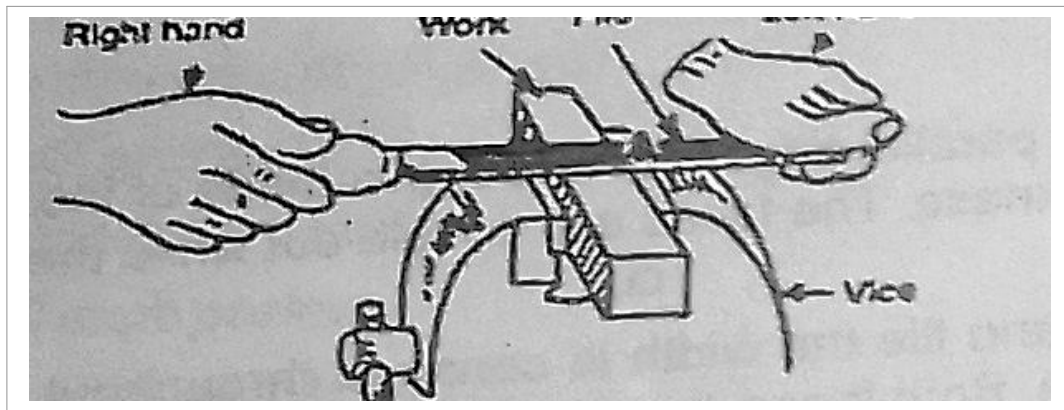


Figure 51: Cross filing (Lab Manual, p. 52).

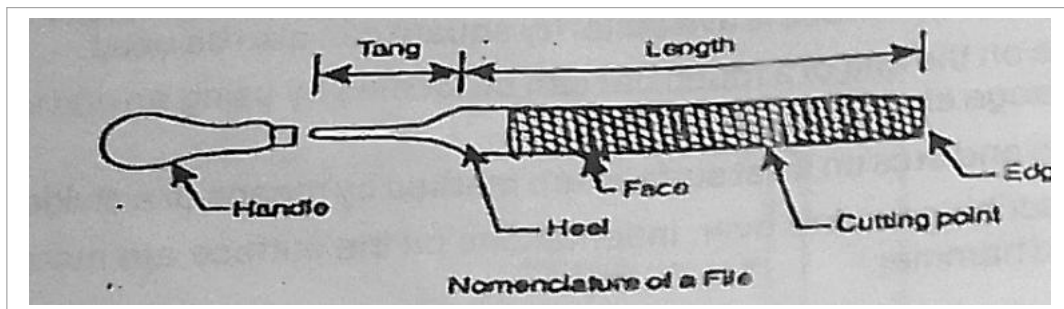


Figure 52: File nomenclature (Lab Manual, p. 50).

5. Procedure

1. The given mild steel flat piece was checked for the given dimensions.
2. One edge of the piece was filed to straightness with the rough and smooth files, and checked with the try square.
3. An adjacent edge was filed such that it was square to the first edge, and checked with the try square.
4. Wet chalk was applied on one side of the flat and dried for marking.
5. Lines were marked according to the given figure, using the odd-leg calliper and the steel rule.
6. Using the dot punch, punch marks were made along the marked lines.
7. The excess material was removed from the remaining two edges with the try square, levelling up to half of the marked dots.
8. Finally, the burrs were removed by filing the surface of the fitted job.

6. Precautions

- The perpendicularity of the faces and edges was checked carefully using the try square.
- Finishing was done using only smooth files.
- Marking was done without parallax error.
- The work was held rigidly in the vice, close to the jaws.
- File strokes were kept straight and level, not rocking.

7. Discussion

The most important result was that accurate marking and accurate filing of the datum edges made the rest of the job much easier. When the first two edges were not exactly square, every dimension marked from them was affected. The use of the try square after each stage helped to find and correct errors early.

Filing exactly to half of the punch marks allowed us to leave enough material to finish the job without overshooting the line. A common problem was a rocking motion of the file, which rounded the edges instead of keeping them flat. Maintaining a steady horizontal stroke, using the long, slow and steady forward stroke described for cross filing, produced flatter surfaces.

8. Conclusion

The square cutting and fitting were done successfully. The two parts of the square fit were made from the mild steel piece with accurate marking, sawing and filing, and the parts fitted into each other as required.

9. Questions Asked in the Material

Produce the working drawing of the square fit.

The drawing shows the 38 mm square raw material and the male and female parts with the dimensions in the figure above.

Using simple sketches enumerate sequentially the operational steps.

Filing, marking, punching, sawing, filing, finishing.

State the functions of each of the tools used.

Try square: checking squareness. Scriber and odd-leg calliper: marking lines. Steel rule: measuring. Dot punch and hammer: making marks permanent. Hacksaw: cutting away waste. Rough file: fast removal. Smooth file: finishing. Square file: filing the corners of the recess.

State the safety precautions that you observed.

See the precautions above.

10. What I Have Learned

- How to prepare datum edges by filing and testing with a try square.
- How to mark, punch, saw and file in sequence.
- That a good fit depends on checking the work repeatedly.

BF 201 (Practical 2): Dovetail Joint (Fitting)

1. Aims

To prepare two smooth rectangular blocks of mild steel and fit them together as a dovetail joint.

2. Objectives

- To file two mild steel blocks to the given rectangular dimensions.
- To mark and cut the dovetail form on the blocks.
- To fit the two blocks together so that the joint is accurate and flush.

3. Apparatus / Materials

- Mild steel pieces to produce two rectangular blocks of 5.5 x 5 x 0.4 cm.
- Bench vice, hacksaw, try square, hand file, surface plate, scale and scribe.

4. Theory / Diagram

A dovetail joint in metal is a sliding joint in which one part has a projection (the male part) with sloping sides that is wider at its end than at its base. The other part has a matching recess (the female part). Because the sides are sloping, the two parts can slide into each other along their length but cannot be separated at right angles to the sliding direction. The dovetail form is widely used in machine slides, such as the cross slide of a lathe, where it provides guidance as well as holding the two parts together.

Making the joint demands greater accuracy than the square fit because the angles must match as well as the sizes. The blocks are first made rectangular and flat. A pair of parallel lines is scribed 55 mm apart on the work with the scribe and the surface plate. Then the dovetail outline is marked, using the angle of the dovetail. The waste metal is sawn out with a hacksaw cutting inside the line, and the sloping surfaces are filed to the line with a file that has a safe edge, checking against the mating part.

The surface plate is used to test the flatness of the work: the work is placed on one of its faces, moved along the surface, and any high spots, where strain accumulates, are filed. A try square verifies that every edge is at right angles to its adjacent edge.

Diagrams

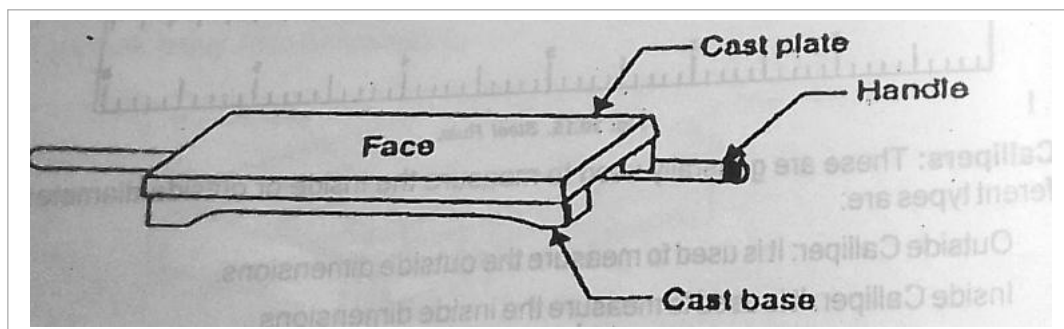


Figure 53: Surface plate used to test flatness (Lab Manual, p. 46).

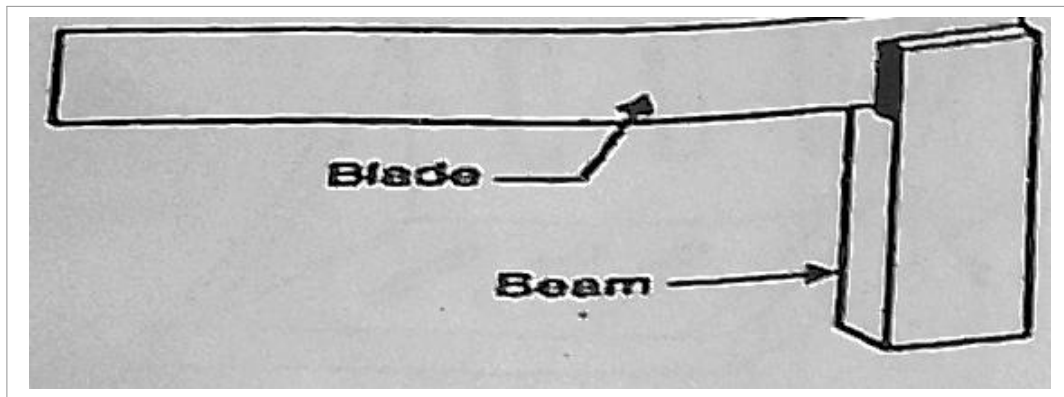


Figure 54: Try square used to check squareness (Lab Manual, p. 48).

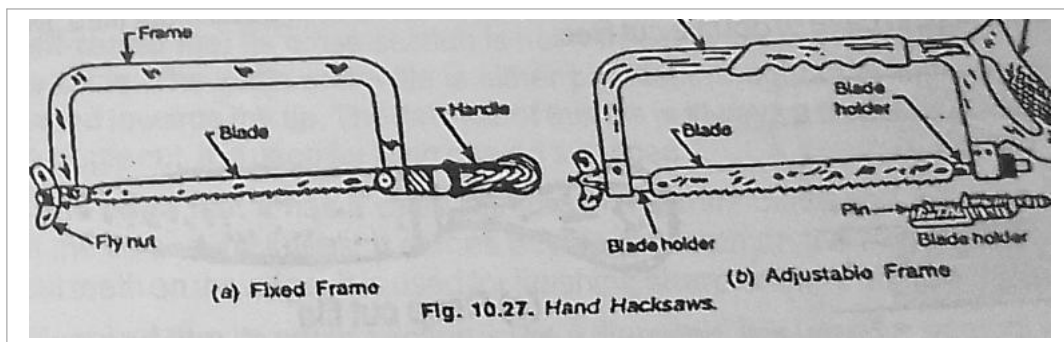


Figure 55: Hand hacksaw used to remove waste metal (Lab Manual, p. 49).

5. Procedure

1. The work pieces were placed on the table, and straight parallel lines were drawn on them 55 mm apart using the scribe.
2. The work pieces were fixed firmly in the jaws of the vice.
3. The pieces were cut along the marked line using the hacksaw.
4. The cut work pieces were placed firmly between the jaws of the vice such that a small portion of one of the edges was above the level of the jaw.
5. The edges were filed using a hand file.
6. A try square was used to ensure that all edges of the work pieces were at right angles to adjacent edges.
7. The work was placed on one of its faces on the surface plate and moved along it to check the areas where strain had accumulated, and these were filed properly.

6. Precautions

- While filing, the file should move in straight strokes and not slant.
- Dimensions should be checked again and again, and a single hand file used so that the piece of proper dimensions is obtained.
- The piece should be fixed straight in the jaws of the bench vice, otherwise an error in shape might occur.
- Wear safety glasses and keep fingers away from the saw.

7. Discussion

This exercise required patience and accuracy. When filing the sides of the blocks, the biggest challenge was to keep the surfaces flat and the edges square. We observed that checking the work again and again with the try square and the surface plate was the only reliable method

of avoiding errors. Using a single hand file near the end helped to prevent removing too much metal.

The surface plate was valuable because it exposed high spots that cannot be seen by the eye. By moving the work along the plate, the places where the metal touched the plate were easily seen, and we filed them away until the surface was flat.

8. Conclusion

Two smooth rectangular mild steel blocks were prepared and fitted as a dovetail-type joint. The exercise demonstrated how marking, sawing and filing are combined to achieve an accurate fit and how the surface plate and try square are used to check the result.

9. Questions Asked in the Material

Produce the working drawing of the dovetail joint.

The drawing shows two rectangular blocks with the dovetail projection and recess (see the manual, p. 54).

Using simple sketches enumerate the operational steps.

Mark parallel lines, hold in the vice, saw, file, check squareness, check flatness on the surface plate, fit the parts.

State the functions of each tool used.

Scriber: marking. Hacksaw: cutting. Hand file: filing flat surfaces. Try square: checking squareness. Surface plate: checking flatness. Scale: measuring.

10. What I Have Learned

- What a dovetail joint is, and why it is used in machine slides.
- How to check flatness with a surface plate.
- That a fit depends on accuracy of both size and angle.

BF 201 (Practical 3): V-Fitting

1. Aims

To make a V-fit from the given mild steel piece.

2. Objectives

- To produce two edges square and straight by filing.
- To mark out and cut a V-shaped notch and a matching projection.
- To fit the two V-parts together accurately.

3. Apparatus / Materials

- Mild steel flat, 40 x 40 x 3 mm.
- 6 in try square, 6 in scribe, odd-leg calliper, steel rule.
- 12 in hacksaw frame with 12 TPI blades.
- 10 in rough, smooth and triangular files, knife-edge file.
- Dot punch, ball pein hammer (0.5 lb).

4. Theory / Diagram

A V-fit is a fitting exercise in which a V-shaped notch is cut into one part and the matching V-shaped projection is made on the other. It tests the ability to mark and file angular surfaces. The V-fit is similar to the square fit in its sequence, but the sloping sides require a different method of filing and of checking. The manual gives a raw material of 38 mm square, a depth of 19 mm and a 10 mm opening at the sides, as shown in the figure.

The sequence of operations is the same as for the square fit: filing, marking, punching, sawing, filing and finishing. After making two adjacent edges square, the V is marked with a scribe and steel rule, using the odd-leg calliper to scribe lines parallel to the edges. The V is then punched and a hacksaw with a 12 TPI blade (a medium blade) is used to cut the waste out of the V. The triangular file, whose three faces are double cut and whose edges are single cut, is suitable for filing into the corners of the V. The knife-edge file reaches the sharp inside corners, and the smooth file finishes the surfaces.

The V is checked against the mating part by trying the two together; the places where the part sticks or shows light are marked and filed. This process of trial and adjustment is called fitting and is what gives the operation its name.

Diagrams

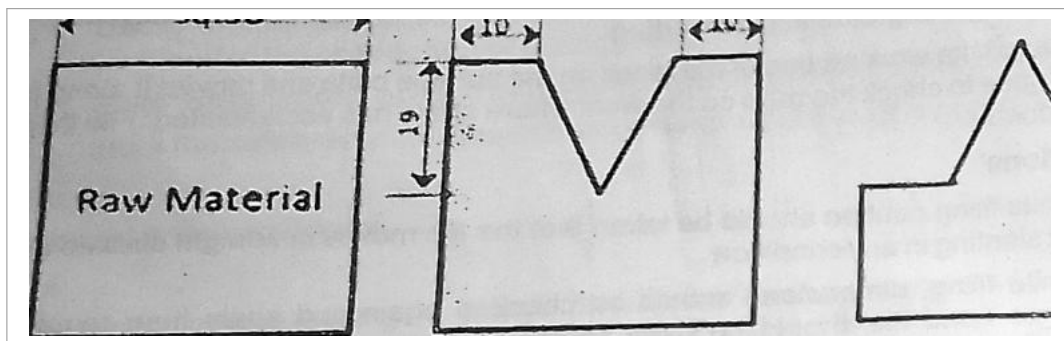


Figure 56: Raw material and the V-fit parts with the main dimensions (Lab Manual, p. 56).

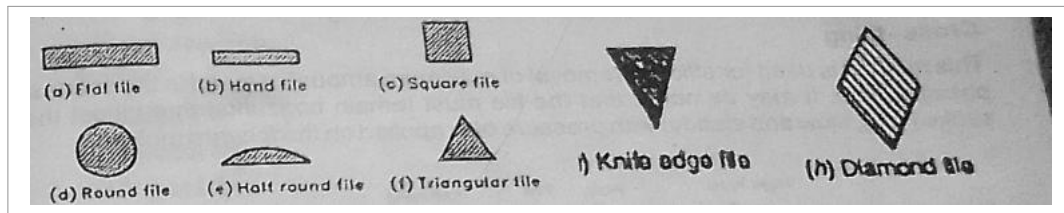


Figure 57: Types of file, including triangular and knife-edge files used for the V (Lab Manual, p. 51).

5. Procedure

1. The given mild steel flat was checked for the given dimensions.
2. One edge was filed with rough and smooth files and checked for straightness using the try square.
3. An adjacent edge was filed square to the first edge and checked with the try square.
4. Wet chalk was applied on one side of the flat and dried for marking.
5. Lines were marked according to the given figure, using the odd-leg calliper and the steel rule.
6. Using the dot punch, punches were made along the marked lines.
7. The excess material was removed from the remaining two edges with the try square, levelling up to half of the marked dots.
8. Finally, burrs were removed by filing the surface of the fitted job.

6. Precautions

- The perpendicularity of the faces and edges was checked carefully using the try square.
- Finishing was done only with smooth files.
- Marking was done without parallax error.
- The work was held rigidly and the hacksaw was used with the teeth facing forward.

7. Discussion

The V-fit was more demanding than the square fit because the angled sides cannot be checked directly with the try square. The surfaces had to be filed evenly on both sides of the V so that the angle remained symmetrical; filing more on one side moved the apex off-centre and the male part would not seat. The triangular file was well suited to the job, while the knife-edge file reached the sharp corners.

We also observed that the final fit was achieved by trial and adjustment, with the two parts tried together and the bright spots (where they touched first) filed down. This careful process is what gives a close, gap-free fit.

8. Conclusion

A V-fit was successfully produced from the mild steel flat. The exercise improved our accuracy in marking, sawing and filing angular surfaces and demonstrated the use of trial fitting to get a close fit.

9. Questions Asked in the Material

Produce the working drawing of the V-fit.

See the figure above: the 38 mm square raw material, a 19 mm deep V and 10 mm side shoulders.

Using simple sketches enumerate the operational steps.

Filing, marking, punching, sawing, filing and finishing.

State the functions of each tool used.

Triangular file: filing the V surfaces. Knife-edge file: sharp corners. Rough and smooth files: bulk removal and finishing. Hacksaw: removing waste. Dot punch: making lines permanent. Try square: checking squareness.

State the safety precautions.

See above.

10. What I Have Learned

- How to mark and cut an angular notch.
- How to use triangular and knife-edge files.
- How to fit parts by trial and adjustment.

HS 201: Introduction to the Welding Shop (Hot Shop) Practice

1. Aims

To study the principles of welding, the types of welding joint, welding positions, arc welding equipment and processes, and the safety recommendations for arc welding.

2. Objectives

- To know how to prepare welding edges before welding.
- To know the different welding joints and welding positions.
- To know the various arc welding processes and equipment.
- To know the safety recommendations for arc welding.
- To produce a butt joint using arc welding.

3. Apparatus / Materials

- Arc welding machine (air-cooled transformer type) with electrode holder and earth clamp.
- Mild steel electrodes, 3.1 mm x 350 mm.
- Welding table (metallic), tongs, chipping hammer and wire brush.
- Face shield or hand shield, apron, asbestos or leather gloves, protective clothing.
- Mild steel plates, rough and smooth files.

4. Theory / Diagram

Welding is a process for joining two similar or dissimilar metals or alloys by fusion. It joins different metals with or without the application of pressure and with or without the use of filler metal. The fusion of the metal takes place through heat, which may be generated by the combustion of gases, an electric arc, electric resistance or chemical reaction. Welding provides a permanent joint, but it normally affects the metallurgy of the components, and so it is usually accompanied by post-weld heat treatment for most critical components. Welding is widely used as a fabrication and repair process in industry, for example in ships, pressure vessels, automobile bodies, offshore platforms, bridges and welded pipes. The term weldability is the property of a metal which indicates the ease with which it can be welded with other similar or dissimilar metals.

The terms used for describing a weld are shown in the figure: the base metal, fusion zone, weld face, root face, root opening, toe and root.

Edge preparation

For welding, the surfaces to be joined are prepared first. Different edge preparations are used for welding butt joints, depending on plate thickness: straight (square), single-V, double-V, single-U, double-U, single-J, double-J, single-bevel and double-bevel. The purpose is to allow the weld metal to reach the root of the joint and give full penetration.

Welding joints

Welding joints are generally of two kinds, the lap joint and the butt joint. A single-lap joint is made by overlapping the edges of the plates; it has very little resistance to bending and is not recommended for most work. A double-lap joint is stronger but requires twice as much welding. The tee fillet weld is widely used but should not be used if an alternative design is possible. The butt weld includes the single-vee butt weld, used for plates up to about 15.8 mm thick with the plates spaced about 3.2 mm apart, and the double-vee butt weld for plates over

13 mm thick where welding can be done on both sides.

Welding positions

There are four welding positions. In the flat or downhand position, the welding is done from the upper side of the joint and the face of the weld is approximately horizontal. In the horizontal position, the plane of the work is vertical and the deposited weld bead is horizontal. This is commonly used in welding vessels and reservoirs. In the vertical position, the plane of the work is vertical and the weld is deposited upon a vertical surface; it is difficult to produce satisfactory welds because gravity acts on the molten metal. The overhead position is even more difficult because the pull of gravity against the molten metal is much greater.

Arc welding process

In arc welding an electric arc is established between an electrode and the work piece to melt the base metals. The arc temperature is very high, which melts the electrode tip and the parent metal, forming a weld pool that solidifies into the weld bead. Some processes use a shielding gas, and others use flux coatings to protect the molten pool from the atmosphere, and to stabilise the arc.

The equipment includes a switch box, secondary terminals, the welding machine, a current reading scale and a current-regulating hand wheel. The operator's accessories are the leather apron, asbestos gloves, protective glasses or face shield, electrode holder, chipping hammer, wire brush, earth clamp and a welding table.

The power source may be alternating current (AC) or direct current (DC). DC welding is usually obtained from generators driven by electric motors or by internal combustion engines. Where mains supply is available, transformers are used for AC welding to step down the supply voltage (200 to 400 V) to the open-circuit welding voltage (50 to 90 V). The choice of power source depends on the type of electrode and metal, the available supply, the output needed, the duty cycle, efficiency, cost, floor space and versatility.

Welding cables carry the current from the power source through the electrode holder, the arc, the work piece and back to the machine; they are insulated copper or aluminium cables. The electrode holder holds the electrode and conducts current to it, with sizes from 150 to 500 A. Welding electrodes are consumable (bare or coated) or non-consumable (carbon or tungsten). In a consumable electrode the tip melts and supplies filler metal. The flux coating on a coated electrode melts to give a gas shield, removes impurities as slag and stabilises the arc. Non-consumable electrodes, made of carbon or tungsten, do not melt appreciably.

Safety recommendations

The welding machine frame must be earthed. The arc produces infrared and ultraviolet light, so the operator must use a helmet or hand shield fitted with a special filter glass. Excess ultraviolet light can cause an effect like sunburn on the skin. The body is protected from sparks and molten metal by gloves, a leather apron and, for overhead work, a leather skull cap and jackets. Equipment should be inspected periodically and kept in safe working condition.

Diagrams

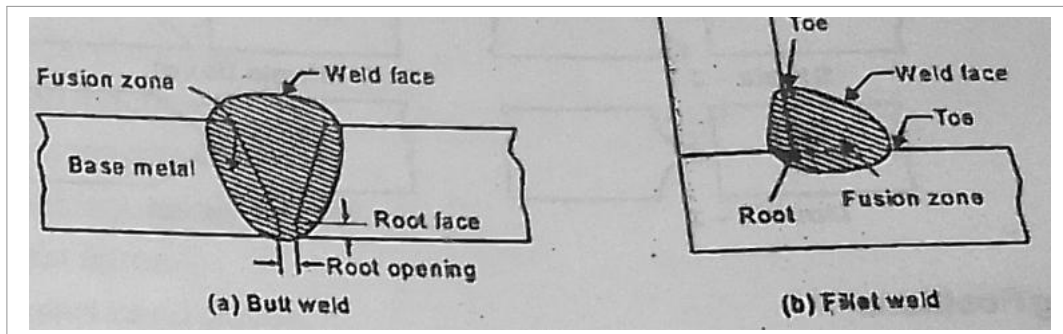


Figure 58: Terminology of the welding process: butt weld and fillet weld (Lab Manual, p. 59).

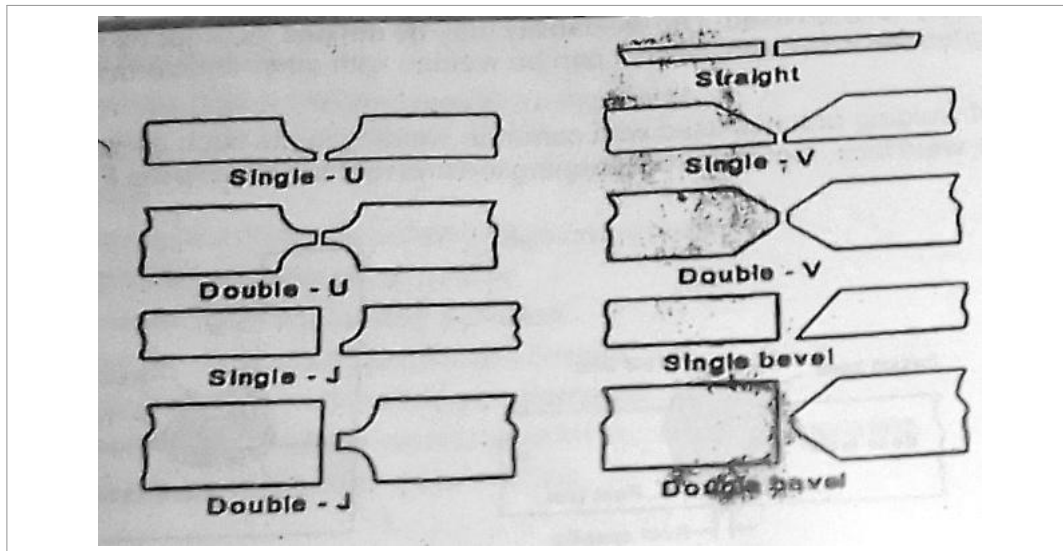


Figure 59: Edge preparations for butt joints (Lab Manual, p. 60).

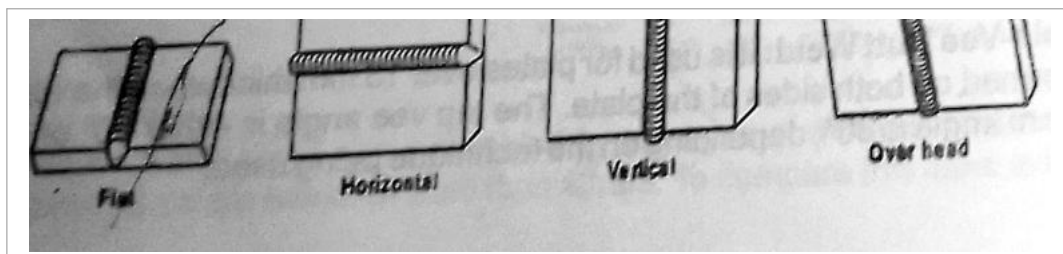


Figure 60: Welding positions: flat, horizontal, vertical and overhead (Lab Manual, p. 60).

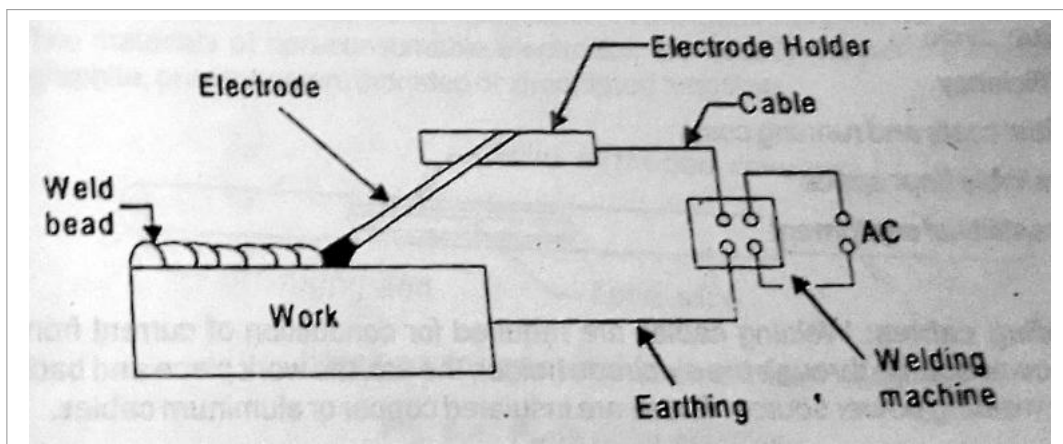


Figure 61: Basic arc welding circuit (Lab Manual, p. 61).

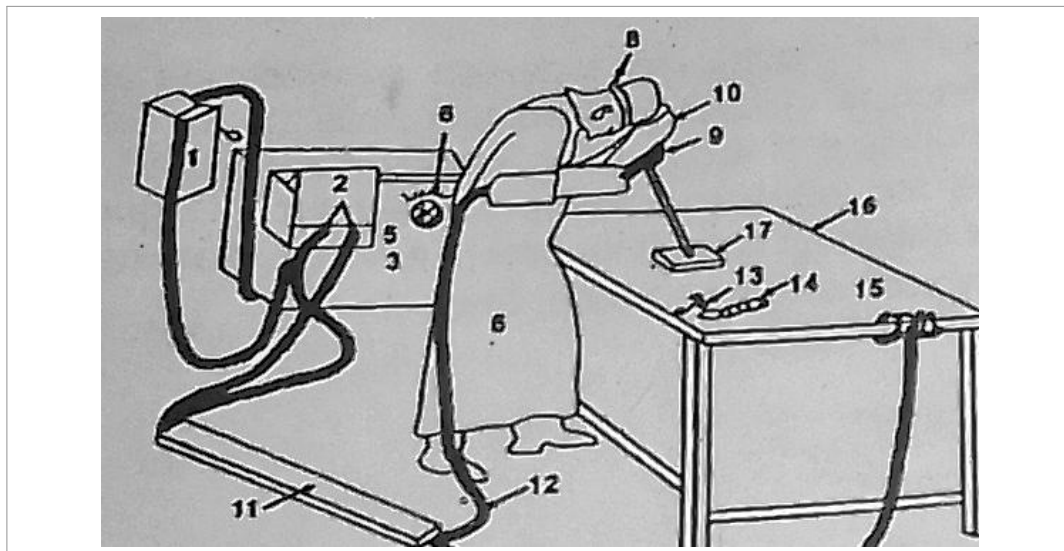


Figure 62: Arc welding equipment and accessories (Lab Manual, p. 62).

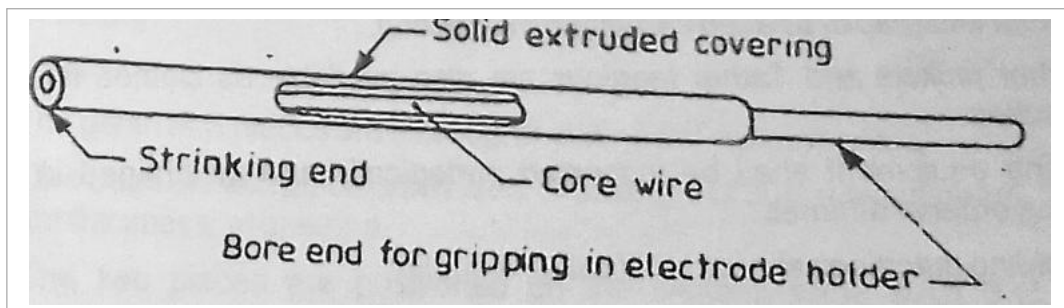


Figure 63: Parts of a coated electrode (Lab Manual, p. 63).

5. Procedure

1. We were introduced to the welding shop and given a safety briefing.
2. The welding terms, edge preparations, joints and positions were explained with examples.
3. The arc welding machine and its accessories were identified and their functions explained.
4. The correct use of the protective clothing, face shield and gloves was demonstrated.
5. A demonstration of striking an arc, running a bead, chipping the slag and cleaning was given by the instructor.

6. Precautions

- Always wear a face shield with a filter glass, gloves and an apron.
- Never look at the arc with the naked eye, and keep other people shielded from it.
- Make sure that the machine is earthed and the cables are not damaged.
- Do not weld on wet floors and keep flammable materials away from the area.
- Use tongs to handle hot work, and wear safety glasses when chipping the slag.

7. Discussion

The welding shop showed us how many different factors influence a sound weld: the edge preparation, the position, the electrode, the current and the skill of the welder. We understood why plates are bevelled for thick sections, since the weld metal has to reach the root of the joint for full penetration. The flat position is the easiest because gravity helps to keep the weld pool in place, whereas overhead welding requires much more skill.

The safety recommendations are particularly important because the dangers of arc welding are not always obvious. Ultraviolet light from the arc can burn the skin and eyes in a short time, and metal spatter and hot slag can cause burns, so full protective clothing is required.

8. Conclusion

The principles of welding, the types of joint, edge preparation, welding positions, equipment, electrodes and safety rules for arc welding were studied. This provides the necessary background for the practical butt and lap joints.

9. Questions Asked in the Material

What is welding?

A process for joining two similar or dissimilar metals by fusion, with or without pressure and with or without filler metal.

Name the four welding positions.

Flat (downhand), horizontal, vertical and overhead.

Why is the coating on an electrode important?

It protects the molten pool from the atmosphere, stabilises the arc and forms slag that protects the cooling weld.

10. What I Have Learned

- The meaning of welding, weldability and the terms used for a weld.
- The difference between lap and butt joints and the types of edge preparation.
- The equipment of an arc welding set and how the arc circuit works.
- The safety precautions that protect a welder from light, heat and electricity.

HS 201 (Practical 1): Butt Joint

1. Aims

To make a butt joint using the given two mild steel pieces by arc welding.

2. Objectives

- To prepare the edges of the plates for welding.
- To tack and weld two plates into a butt joint using an arc welding machine.
- To remove slag and clean the weld.

3. Apparatus / Materials

- Mild steel plates, 100 x 50 x 5 mm (2 pieces).
- Welding electrodes: M.S. electrodes, 3.1 mm x 350 mm.
- Air-cooled transformer welding machine (voltage 80 to 600 V, 3-phase supply, current up to 350 A).
- Rough and smooth files, protractor, tongs, ground clamp, electrode holder.
- Face shield, apron, gloves and chipping hammer.

4. Theory / Diagram

A butt joint is a joint in which two pieces are placed edge to edge in the same plane and welded along the line where they meet. Butt welds are widely used because they give an efficient transfer of load between the two plates and a smooth surface. For thin plates (about 5 mm, as here) the edges may be given a small bevel so that the weld penetrates to the root. The manual prescribes a bevel of about 30 degrees on one edge of each piece, leaving about a quarter of the plate thickness as the root face, which then forms a V-groove when the two plates are brought together.

The two plates are positioned on the welding table with a slight gap between them (the root opening), which allows the first run (the root pass) to melt through and fill the gap and give proper penetration. The plates are tack welded first at both ends (and in the centre) to hold the alignment, because heat from welding causes expansion and contraction which can pull the plates out of line.

The electrode is held at an angle of about 15 to 25 degrees from the vertical, tilted in the direction of welding. The arc is struck by tapping or scratching the electrode on the work, and then a short arc length (about the diameter of the electrode) is maintained as the electrode is moved along the joint at a uniform speed. A second run is deposited over the first one with a slight weave to fill the groove. Afterwards the slag, which is a glassy layer formed from the flux of the electrode, is removed with a chipping hammer and the weld is cleaned with a wire brush.

Diagrams

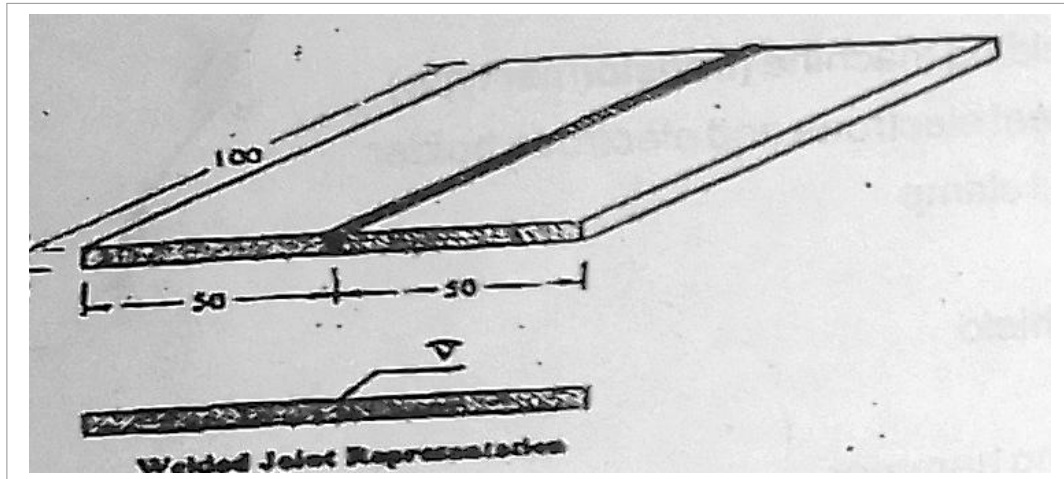


Figure 64: Butt joint: two 100 x 50 mm plates welded edge to edge, with the welded joint representation (Lab Manual, p. 66).

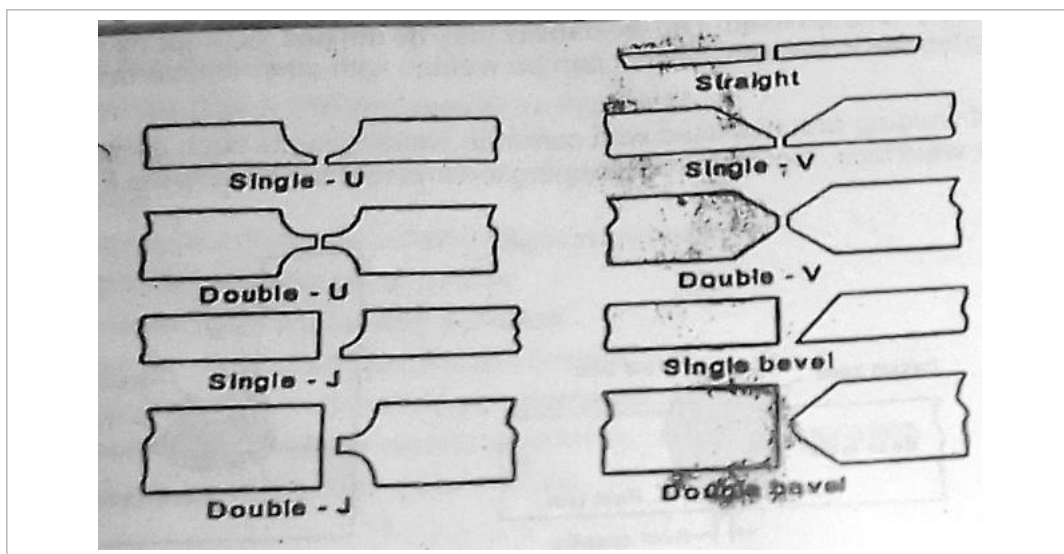


Figure 65: Edge preparation for butt joints (Lab Manual, p. 60).

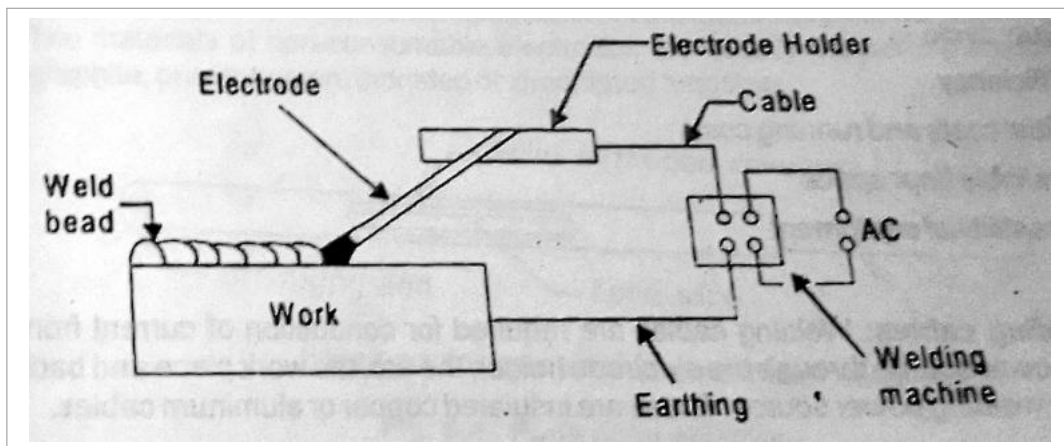


Figure 66: Arc welding circuit used for the practical (Lab Manual, p. 61).

5. Procedure

1. The given mild steel pieces were thoroughly cleaned of rust and scale.
2. One edge of each piece was bevelled to an angle of 30 degrees, leaving nearly a quarter of the thickness as a flat face, at one end.

3. The two pieces were positioned on the welding table such that they were slightly separated for better penetration of the weld.
4. The electrode was fitted into the electrode holder and the welding current was set to a proper value.
5. The ground clamp was fastened to the welding table.
6. Wearing the apron and using the face shield, the arc was struck and held to tack the two pieces together. The first run of the weld was then done to fill the root gap.
7. The second run of the weld was done with proper weaving and with uniform movement. During welding the electrode was kept at 15 to 25 degrees from the vertical and in the direction of welding.
8. The scale formed on the welds was removed using the chipping hammer.
9. Filing was done to remove any spatter around the weld.

6. Precautions

- Wear the apron, gloves and face shield at all times during welding.
- Do not strike the arc without covering the face with the shield.
- Ensure the ground clamp is firmly fixed to the table.
- Use tongs for handling hot pieces, and wear safety glasses while chipping slag.
- Keep flammable materials away from the welding area.

7. Discussion

The quality of the butt joint depended on four things: edge preparation, the gap, the current and the manner in which the electrode was moved. When the gap was too narrow the weld did not penetrate to the root, and when it was too wide the weld metal fell through. Tack welding at both ends and at the centre was very helpful, since it held the alignment while the weld cooled and contracted.

Striking the arc was initially difficult, as the electrode tended to stick to the plate. A smooth scratching movement and withdrawal to a short arc length gave a stable arc. The slag layer protected the weld as it cooled and was removed afterwards to reveal the bead. Defects that could be observed in a poor weld include undercut, lack of fusion, porosity and spatter, which are mostly caused by wrong current, speed or arc length.

8. Conclusion

A butt joint was successfully produced from the two mild steel pieces using the arc welding process. The exercise taught edge preparation, tacking, running the weld, chipping the slag and cleaning, and it emphasised the safety requirements for arc welding.

9. Questions Asked in the Material

Produce the working drawing of the butt joint.

The drawing shows two plates 100 x 50 x 5 mm joined edge to edge with a weld along the 100 mm length (see the figure).

Using simple sketches, state the operational sequence of how the butt joint was produced.

Marking, cutting, edge preparation, try square levelling, tacking, welding, cooling, chipping, cleaning.

State the safety precautions you observed.

See the precautions above.

10. What I Have Learned

- How to prepare a plate edge and set the root gap for a butt joint.
- How to strike and hold an arc and run a weld bead.
- Why tack welding is used and how slag is removed.

HS 201 (Practical 2): Lap Joint

1. Aims

To make a lap joint using the given two mild steel pieces and arc welding.

2. Objectives

- To position two plates in overlap and tack weld them.
- To weld the lap joint with a fillet weld along the edges.
- To check alignment and clean the weld.

3. Apparatus / Materials

- Mild steel plates, 100 x 50 x 5 mm (2 pieces).
- Welding electrodes: M.S. electrodes, 3.1 mm x 350 mm.
- Air-cooled transformer welding machine, up to 350 A.
- Rough and smooth files, protractor, tongs, ground clamp, electrode holder.
- Face shield, apron, gloves and chipping hammer.

4. Theory / Diagram

In a lap joint the two pieces overlap each other and are joined by fillet welds at the edge of the overlap. Overlapping increases the area over which the load is carried. A single-lap joint, made by overlapping the edges of two plates, is easy to prepare because no edge preparation is required, but it has little resistance to bending, since the load path is offset. For this reason a double-lap joint, welded on both sides, is stronger, but requires twice as much welding.

In a fillet weld the weld metal is deposited in the corner formed by the surface of one plate and the edge of the other. The fillet has a roughly triangular cross-section, and the strength depends on the throat thickness (the shortest distance from the root to the face). The terms toe and root, used in the butt weld, apply here too. A fillet weld is usually deposited in the flat or horizontal position.

The plates are tack welded at both ends and in the middle to hold them in position. The alignment is checked before the main weld is run, because distortion due to heating and cooling can pull the plates out of position. After welding, the slag is removed with the chipping hammer and the weld is cleaned.

Diagrams

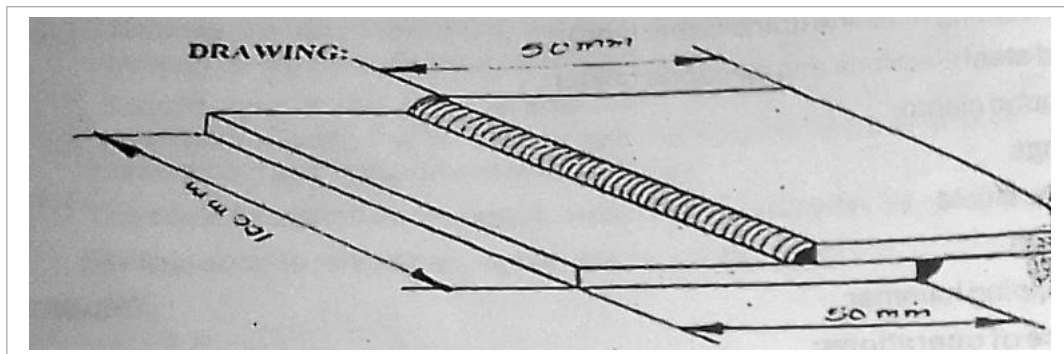


Figure 67: Lap joint of two 100 x 50 mm plates with an overlap of 50 mm and a weld along the edge (Lab Manual, p. 68).

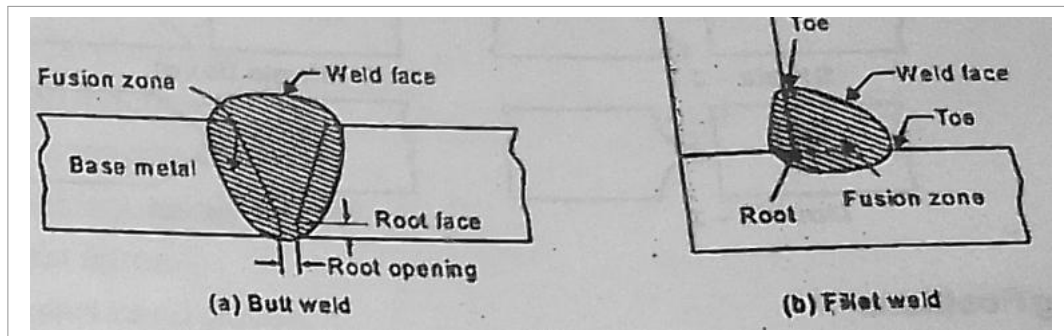


Figure 68: Terms used in a weld, including the fillet weld (Lab Manual, p. 59).

5. Procedure

1. The given mild steel pieces were thoroughly cleaned of rust and scale.
2. The two pieces were positioned on the welding table such that they overlapped one over the other as shown in the drawing.
3. The electrode was fitted into the electrode holder and the welding current was set to a proper value.
4. The ground clamp was fastened to the welding table.
5. Wearing the apron and using the face shield, the arc was struck, and the work pieces were tack welded at both the ends and at the centre of the joint.
6. The alignment of the lap joint was checked and the tack-welded pieces were welded as required.
7. The scale formed on the welds was removed using the chipping hammer.
8. Filing was done to remove any spatter around the weld.

6. Precautions

- Wear the face shield, apron and gloves at all times.
- Check the alignment before the final weld is made.
- Keep a short arc and uniform travel speed.
- Make sure the ground clamp is secured and that there are no damaged cables.
- Allow the job to cool, and handle it with tongs.

7. Discussion

The lap joint was easier to prepare than the butt joint since no edge preparation was needed, but obtaining a uniform fillet weld required a steady hand. When the electrode was held at the correct angle between the two plates, the fillet had an even, convex face. When the electrode was held too close to one plate, the weld metal fell on one side only and an undercut appeared on the other.

Distortion was visible after welding because the plates had been heated only along one edge, and it showed why tack welding and checking of the alignment are necessary. The lap joint is acceptable for light work, but because the load path is offset, it bends easily and a double-lap joint would be better for heavier loads.

8. Conclusion

A lap joint was successfully made from two mild steel plates by arc welding. The exercise demonstrated how to position, tack and weld a fillet, and how to clean the weld after welding.

9. Questions Asked in the Material

Produce the working drawing of the lap joint.

The drawing shows two 100 x 50 mm plates overlapping by 50 mm, joined by a weld along the overlap (see the figure).

Using simple sketches, state the operational sequence.

Marking, cutting, edge cleaning, try square levelling, tacking, welding, cooling, chipping, cleaning.

State the safety precautions you observed.

See above.

10. What I Have Learned

- How a lap joint and a fillet weld are made.
- The strength limitations of a single-lap joint.
- The importance of alignment checks and slag removal.

MS 201: Introduction to the Lathe and Machining Operations

1. Aims

To study the characteristic features of the lathe, its main parts, accessories and tools, and the types of operation that can be performed on it.

2. Objectives

- To study the characteristic features of the lathe.
- To study the types of lathe operations.
- To prepare for the production of a shaft by straight turning.

3. Apparatus / Materials

- Centre lathe machine with its accessories: three-jaw chuck, four-jaw chuck, face plate, centres, steady rest and follower rest.
- High-speed steel (H.S.S.) single point cutting tools, knurling tool and parting tool.
- Vernier calliper, steel rule, chuck key and tool post spanner.

4. Theory / Diagram

The lathe is considered one of the oldest machine tools and is widely used in industry. It is called the mother of machine tools. The first screw-cutting lathe is said to have been developed by an Englishman, Henry Maudslay, in 1797, and modern high-speed, heavy-duty lathes are based on that machine. The primary task of a lathe is to generate cylindrical work pieces. The process of machining a work piece to the required shape and size by moving the cutting tool either parallel or perpendicular to the axis of rotation of the work piece is known as turning, in which the excess unwanted metal is removed. The machine tool is capable of plain turning, taper turning, thread cutting, chamfering and knurling.

Main parts of a lathe

The main parts are the bed, headstock, spindle, tailstock, carriage (with saddle, apron, cross-slide, compound rest, compound slide and tool post), feed mechanism, lead screw, feed rod and the thread-cutting mechanism.

Bed and ways. The bed is mounted on the legs of the lathe, which are bolted to the floor. It forms the base of the machine and is made of cast iron, with its top surface machined accurately. The headstock is at the extreme left, the tailstock at the extreme right and the carriage between them. The top of the bed has flat or V-shaped guideways. The tailstock and carriage slide on these guideways. Inverted V-shaped guideways give better guidance and accurate alignment of the saddle and tailstock, and also allow the metal chips to fall through.

Headstock. The headstock is mounted permanently on the left side of the bed. It houses the hollow spindle and the mechanism for driving the spindle at multiple speeds. The drive may be a stepped cone pulley drive, a back gear drive or an all-gear drive.

Spindle. The spindle rotates on two large bearings in the headstock. A hole through the spindle allows long bars to pass through. The front end is threaded to take chucks, face plate, driving plate and catch plate, and has a taper hole to take the live centre. A gear at the other end transmits power through the tumbler gears to the lead screw.

Tailstock. The tailstock is located on the guideways at the right side of the bed. It has a body, a base and a spindle (or ram) which moves inside a bore. The spindle has a Morse taper hole

to receive the dead centre or the shanks of tools such as drills and reamers. When the hand wheel is turned clockwise, the spindle advances, and when turned anti-clockwise, it is withdrawn. The tailstock can be clamped at any position on the bed. Its uses are: supporting the other end of long work held between centres, holding drills, reamers and taps, turning tapers by the set-over method, and setting the cutting tool at the correct height.

Carriage. The carriage lies between the headstock and tailstock and moves along the bed. It consists of the saddle (an H-shaped casting which bridges the guideways), the apron (which carries the mechanism controlling the motion of the carriage and cross-slide), the cross-slide (which moves at right angles to the bed and carries the compound rest), the compound rest (which can be swivelled to any angle, used for taper turning), the compound slide, and the tool post. The cross-slide hand wheel is graduated so that the depth of cut can be set to about 0.05 mm.

Tool post. The tool post holds the tools. Types include the single-screw tool post, the four-bolt tool post, the four-way tool post and the open-side tool post. The four-way tool post carries four tools at a time, so machining time is reduced.

Feed rod and lead screw. The feed rod is a long shaft with a keyway, which transmits power from the spindle through gears to the apron to give automatic feed to the carriage and cross-slide. The lead screw is a long threaded shaft (usually Acme thread) used as a master screw in thread cutting. A split nut (half-nut) in the apron engages the lead screw to move the carriage by a calculated distance for each spindle revolution.

Types of lathe

Lathes are classified as speed lathes (woodworking, centring, polishing and metal spinning lathes), engine or centre lathes, bench lathes, tool-room lathes, semi-automatic lathes (capstan and turret), automatic lathes and special-purpose lathes (wheel lathe, gap-bed lathe, T-lathe and duplicating lathe). A speed lathe has a spindle speed of about 1200 to 3600 rpm and consists only of a headstock, a bed, a tailstock and a tool slide. The engine lathe is the most widely used, and it is so called because early lathes were driven by steam engines. A tool-room lathe is similar to an engine lathe but built more accurately with a wider range of speeds and feeds, for tool and die work.

Work-holding devices

These include chucks, face plate, driving plate, catch plate, carriers, mandrels, centres and rests. The three-jaw self-centring chuck has three jaws that move together by rotating one pinion with a chuck key; it holds round or hexagonal work quickly and centres it automatically. The four-jaw independent chuck has four jaws, each moved by its own screw, so it can hold irregular shapes and be set to run accurately but takes more time. The magnetic chuck holds thin ferrous work by magnetic flux. The collet chuck has a spring-steel bush that grips bar stock evenly, and it is used on capstan and automatic lathes. The face plate is a circular disc with slots for bolts and clamps to hold large or irregular work. Centres (live and dead) hold the work between the headstock and tailstock, and they have a Morse taper shank. The steady rest has three jaws and supports long work near the free end, while the follower rest is attached to the carriage and moves with the tool to prevent slender work from bending.

Lathe tools and operations

By construction, lathe tools are solid tools, brazed-tipped tools and tool bit with tool holders. By the operation they perform, they are turning tools, thread-cutting tools, facing tools, forming tools, parting tools, grooving tools, boring tools, internal thread cutting tools and knurling tools. By feed direction there are right-hand, left-hand and round-nose tools.

Facing is machining the end of the work to produce a flat surface square to the axis; the tool is fed perpendicular to the axis. Straight turning removes material to produce a cylinder with the tool fed parallel to the axis. Step turning produces different diameters on the same work. Chamfering bevels the end of the work. Grooving cuts a narrow groove on the cylindrical surface. Knurling embosses a diamond-shaped pattern on the surface to provide a good grip. Undercutting enlarges a diameter internally or reduces it externally over a short length. Taper turning is a uniform increase or decrease of diameter along the length, done by the form tool method, the compound rest method, tailstock set-over, taper turning attachment or combined feed. Thread cutting produces a helical groove by feeding the tool longitudinally by one pitch per revolution. Drilling, reaming, boring and tapping are also performed on the lathe.

Diagrams

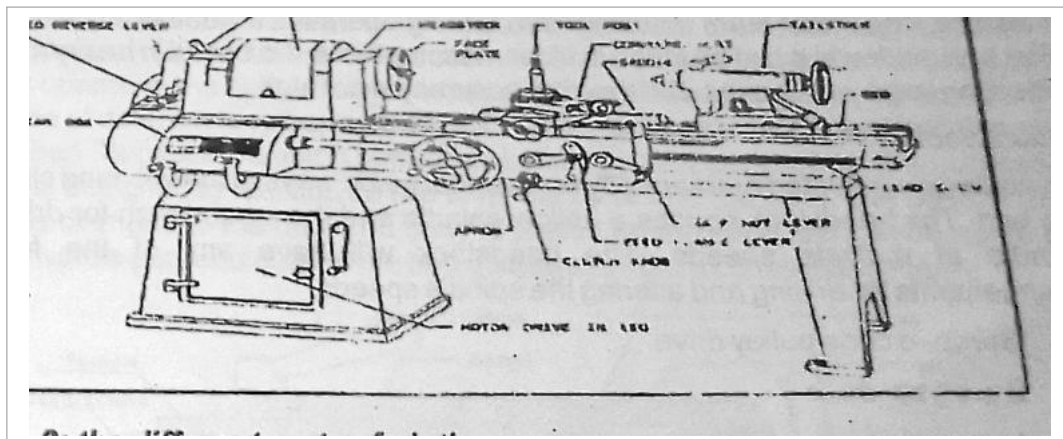


Figure 69: Parts of the lathe (Lab Manual, p. 69).

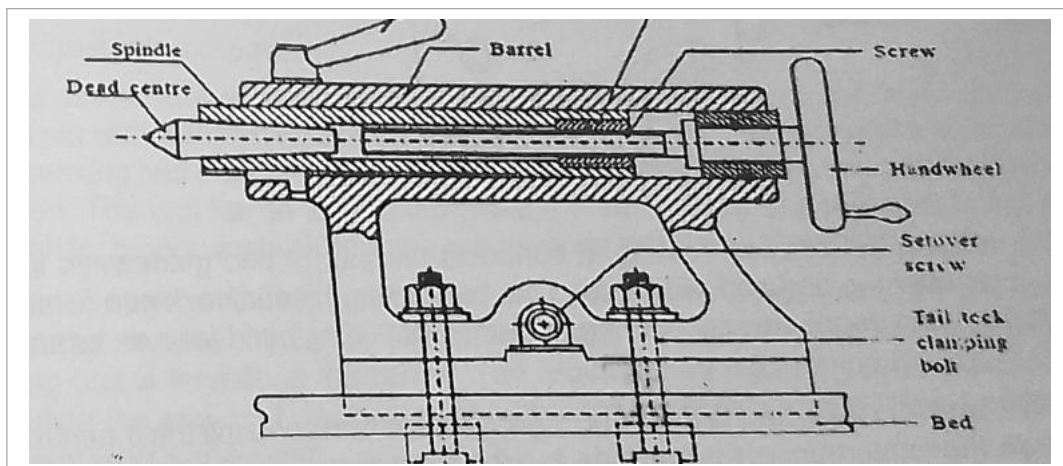


Figure 70: Tailstock of the lathe (Lab Manual, p. 71).

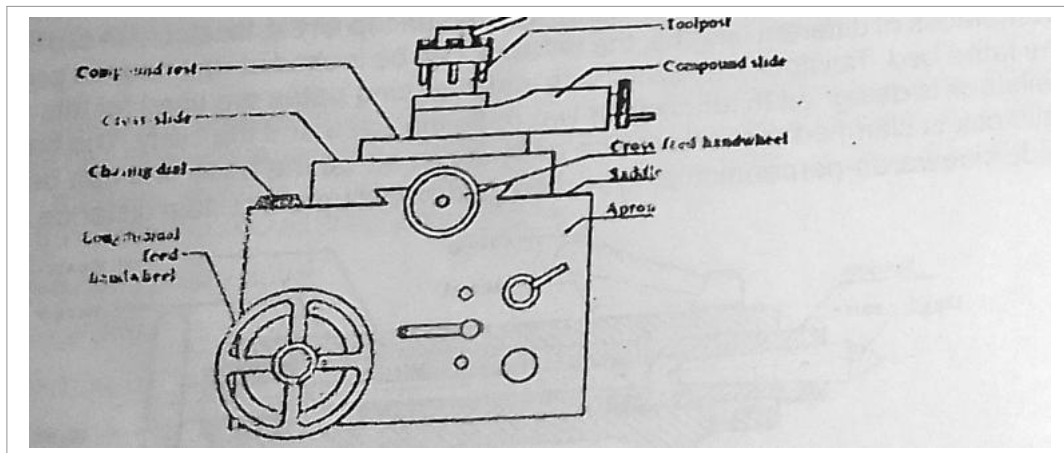


Figure 71: Carriage of the lathe with saddle, apron, cross-slide, compound rest and tool post (Lab Manual, p. 72).

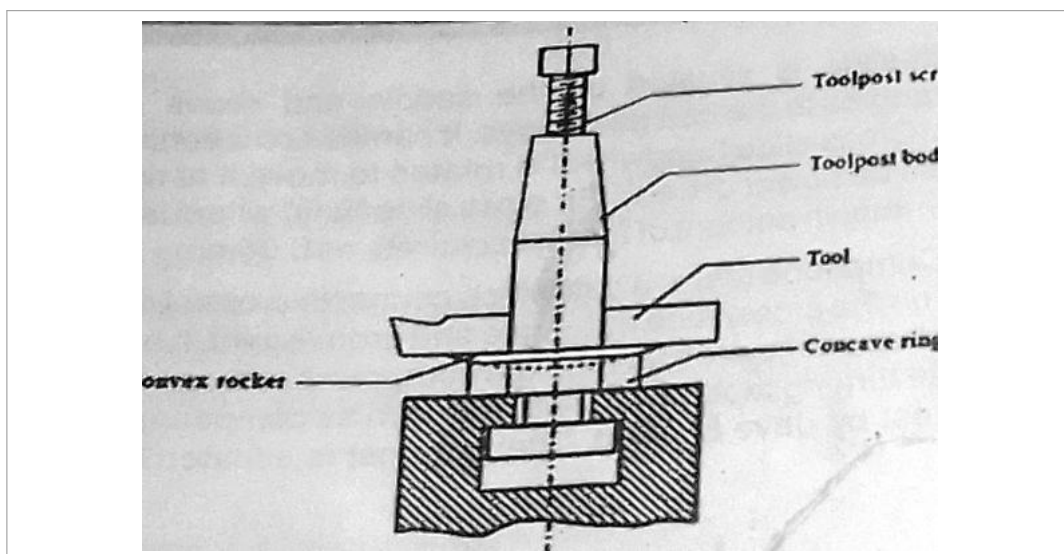


Figure 72: Single-screw tool post (Lab Manual, p. 74).

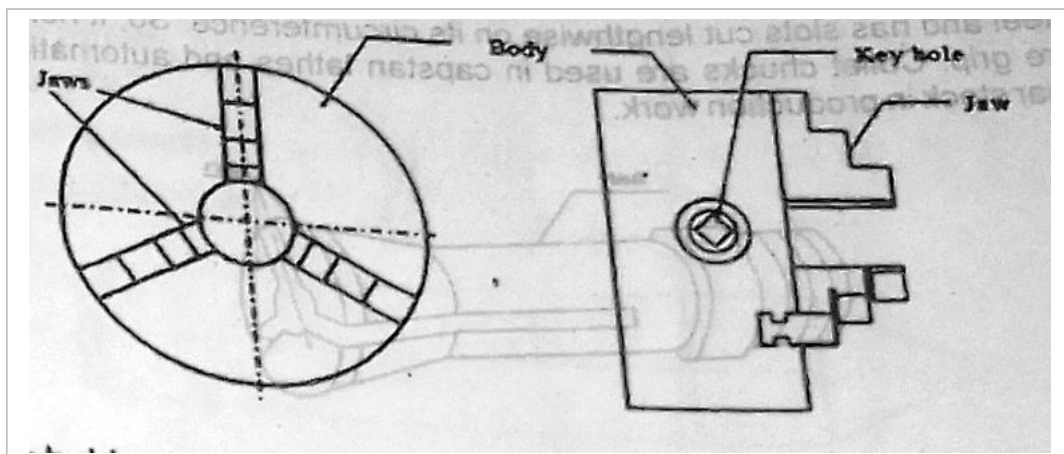


Figure 73: Three-jaw self-centring chuck (Lab Manual, p. 77).

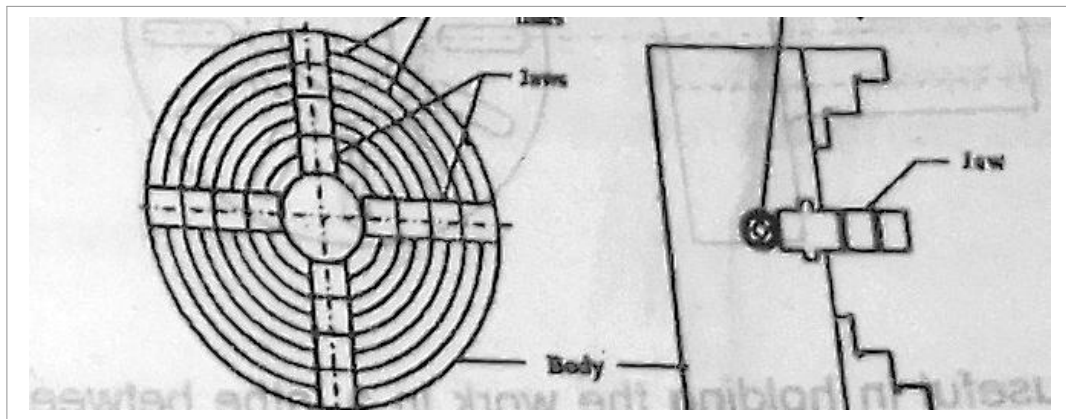


Figure 74: Four-jaw independent chuck (Lab Manual, p. 77).

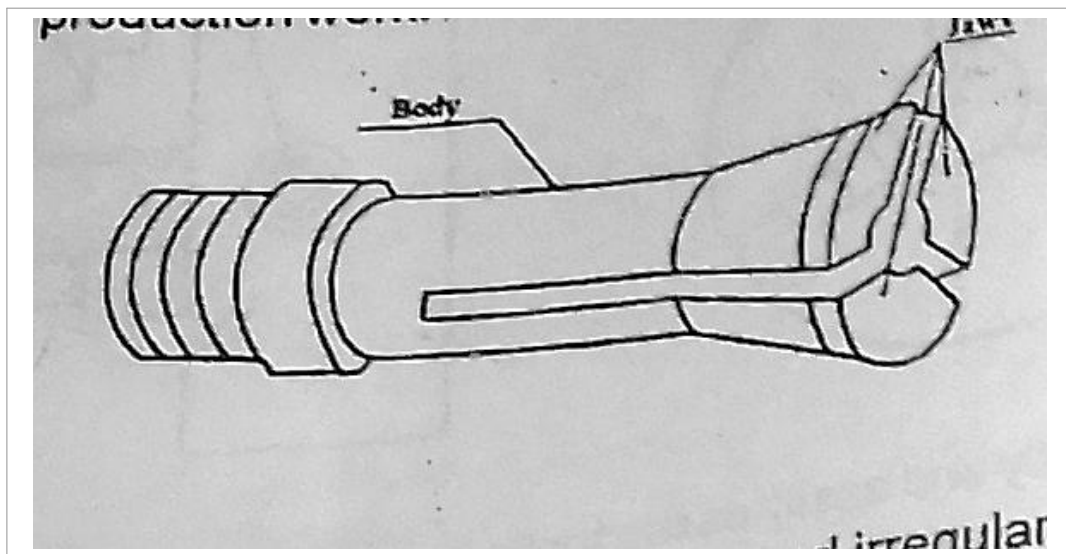


Figure 75: Collet chuck (Lab Manual, p. 78).

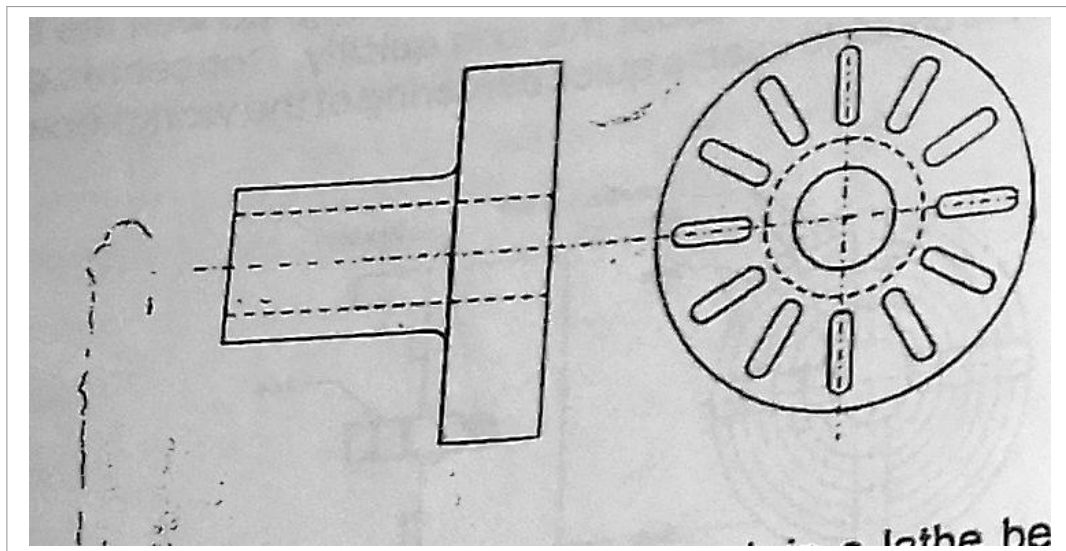


Figure 76: Face plate (Lab Manual, p. 78).

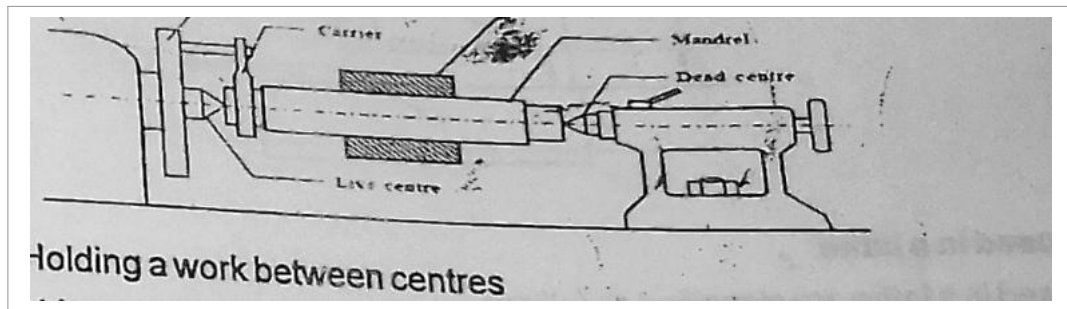


Figure 77: Holding work between centres (Lab Manual, p. 79).

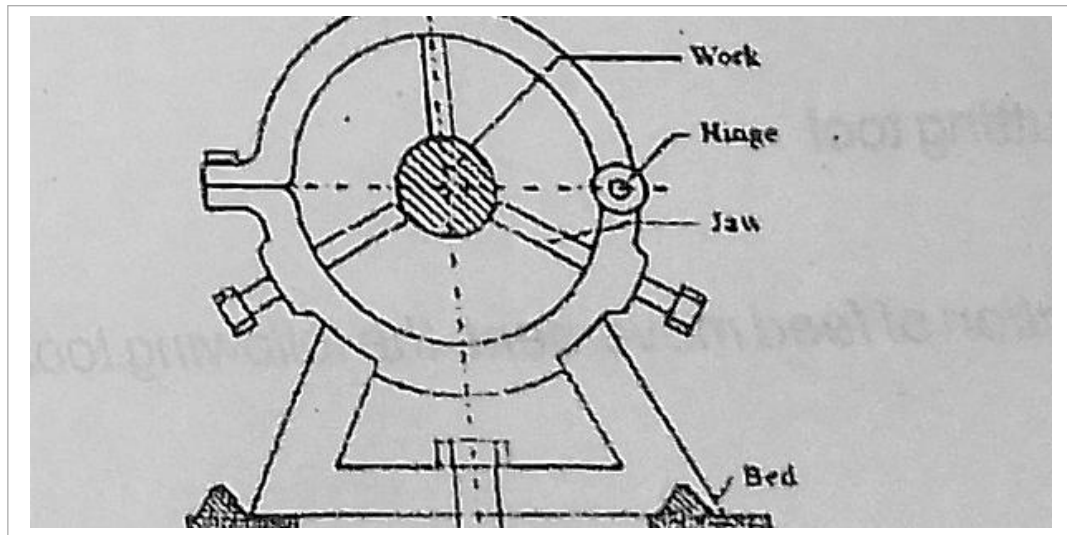


Figure 78: Steady rest (Lab Manual, p. 79).

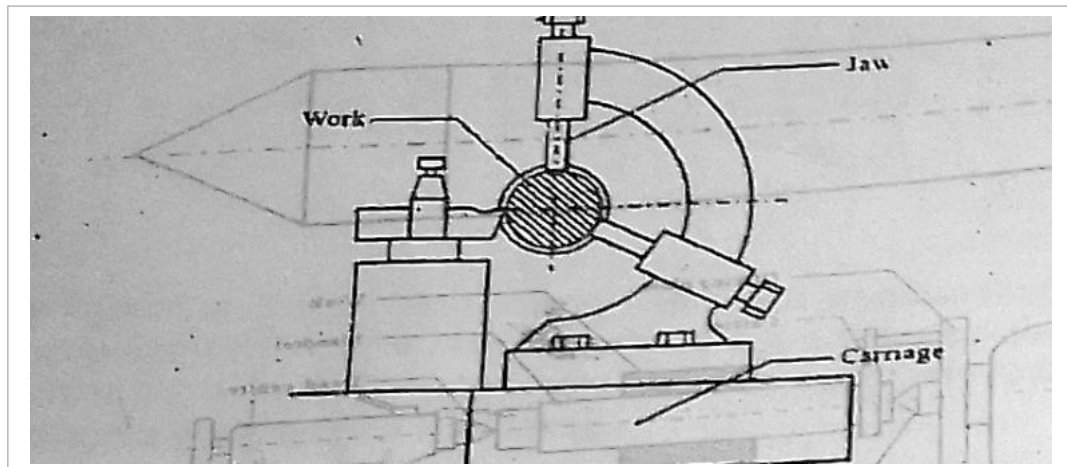


Figure 79: Follower rest (Lab Manual, p. 80).

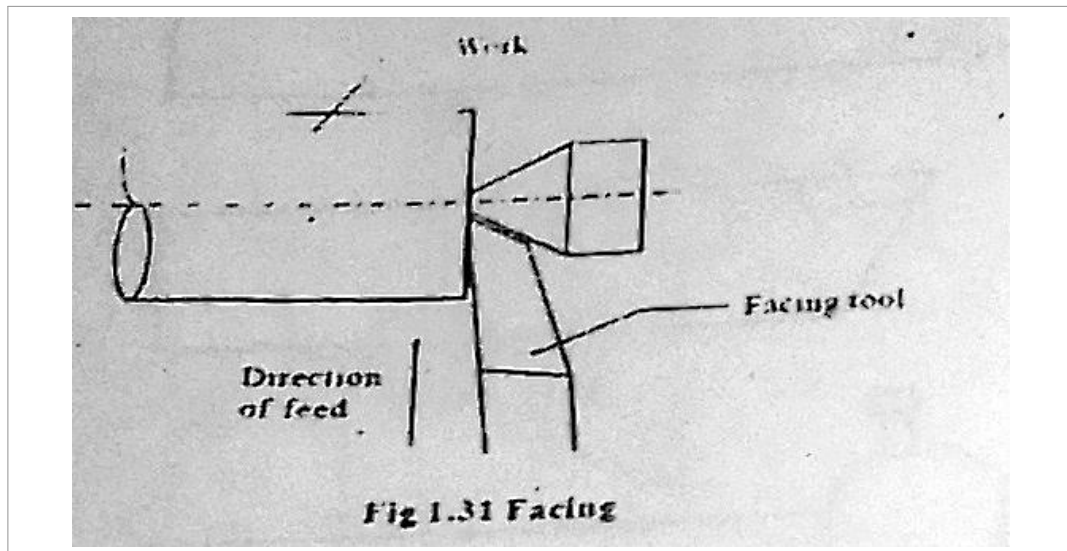


Figure 80: Facing operation (Lab Manual, p. 82).

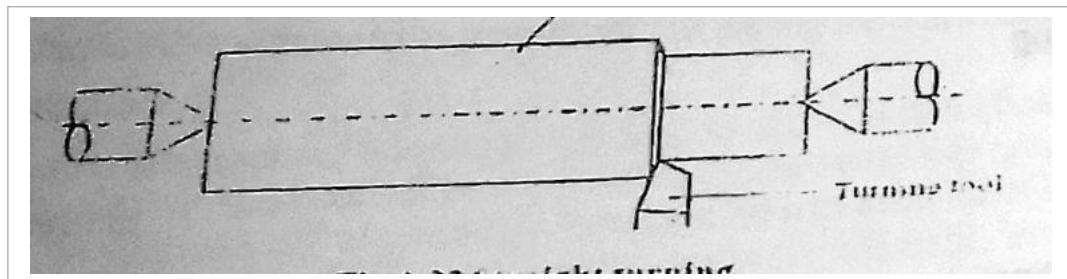


Figure 81: Straight turning (Lab Manual, p. 82).

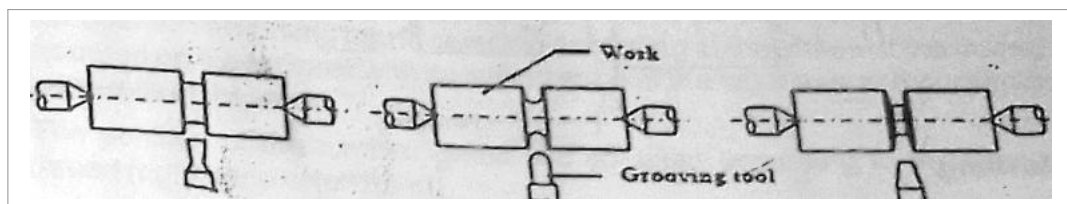


Figure 82: Types of groove (Lab Manual, p. 83).

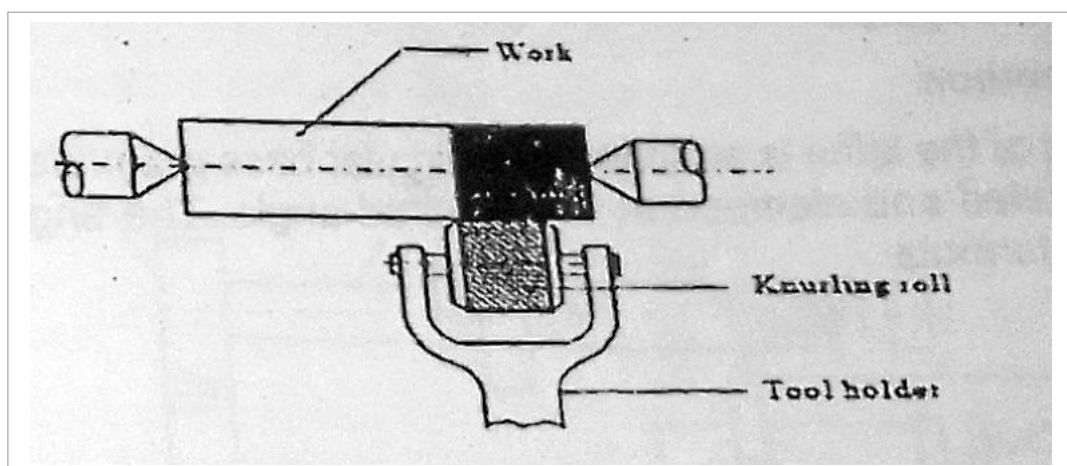


Figure 83: Knurling (Lab Manual, p. 83).

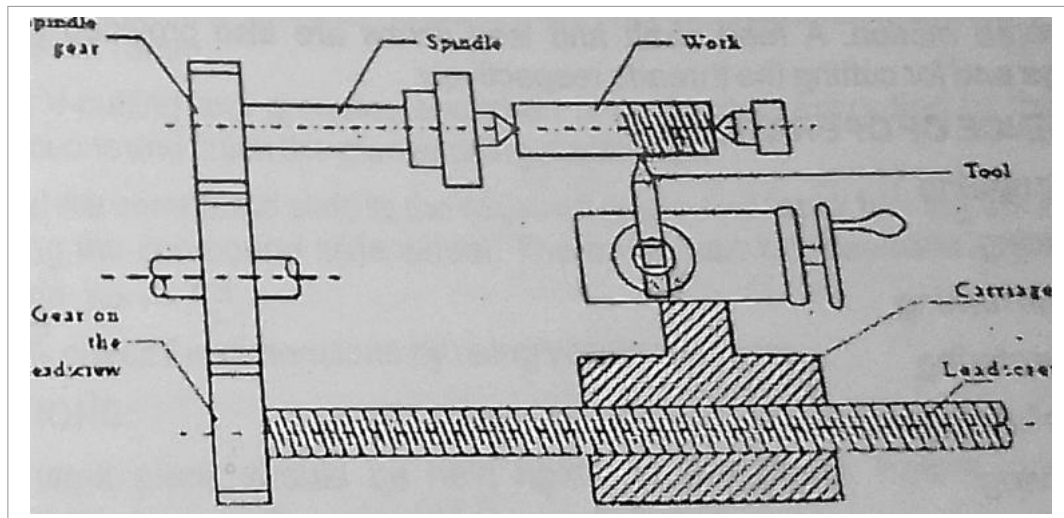


Figure 84: Thread cutting set-up on the lathe (Lab Manual, p. 85).

5. Procedure

1. The lathe machine was introduced and its main parts were identified with the labelled diagram.
2. The functions of the headstock, tailstock, carriage, lead screw and feed rod were explained.
3. The work-holding devices were examined and mounted, and the cutting tools were shown.
4. A demonstration of facing, straight turning, step turning, chamfering, grooving and knurling was given by the instructor.
5. The method of taper turning by compound rest and of thread cutting was explained.

6. Precautions

- Wear safety glasses and close-fitting clothing; remove rings, watches and ties and tie up long hair.
- Never leave the chuck key in the chuck.
- Never touch a rotating work piece or chuck, and never try to stop the chuck by hand.
- Do not remove chips by hand; use a brush or hook after stopping the machine.
- Stop the machine before measuring, changing tools or adjusting the work.
- Ensure the work and tool are firmly secured before starting.

7. Discussion

The lathe is described as the mother of machine tools because it can be used for many operations and also for making parts for other machines. Studying the different parts of the lathe showed that each has a specific function, and that the accuracy of the finished work depends on how rigidly the bed, headstock, carriage and tailstock are aligned. The lead screw and feed rod showed us two different drives: the feed rod provides smooth power feed for turning, and the lead screw with its half-nut gives a precise pitch for thread cutting.

Work holding is an important lesson. The three-jaw chuck is convenient for round bars, but when the work is irregular, it must be held in the four-jaw chuck or on the face plate. Long slender work needs a steady rest or follower rest, since otherwise it bends away from the tool and finishes oversize in the middle. Choice of speed, feed and depth of cut also determines the finish and tool life.

8. Conclusion

The characteristic features of the lathe, its parts, accessories, tools and operations were studied. The lathe can perform turning, facing, chamfering, grooving, knurling, taper turning, thread cutting, drilling, boring and other operations, and its correct use depends on good work holding, proper tool setting and strict observance of safety rules.

9. Questions Asked in the Material

List the functions of the lathe machine.

Plain (straight) turning, facing, step turning, chamfering, grooving, forming, knurling, undercutting, eccentric turning, taper turning, thread cutting, drilling, reaming, boring and tapping.

What are the main parts of a lathe?

Bed, headstock, spindle, tailstock, carriage (saddle, apron, cross-slide, compound rest, compound slide, tool post), feed mechanism, lead screw and feed rod.

What is the difference between a live centre and a dead centre?

The live centre fits in the headstock spindle and rotates with the work; the dead centre fits in the tailstock spindle and does not rotate, so it is hardened and needs lubrication.

What is the purpose of knurling?

To provide an effective gripping surface, to give a better appearance, and to slightly increase the diameter of the work.

10. What I Have Learned

- The names and functions of the main parts of the lathe.
- How different chucks, face plate, centres and rests hold the work.
- The types of lathe operations and the tools used for them.
- Safety rules specific to rotating machinery.

MS 201 (Practical 1): Step Turning and Taper Turning on a Lathe

1. Aims

To perform step turning and taper turning operations on a given work piece.

2. Objectives

- To perform centring, facing, plain turning and chamfering.
- To produce steps of different diameters on a bar, and a groove.
- To produce a taper using the compound rest method and to calculate the swivel angle.

3. Apparatus / Materials

- Mild steel rod, 25 mm diameter and 100 mm long.
- Vernier calliper, steel rule, spanner, chuck spanner and H.S.S. single-point cutting tool.
- Lathe specification: length of bed 1390 mm, width of bed 200 mm, height of centres 165 mm, admit between centres 700 mm, lead screw pitch 4 TPI, motor 1 h.p.

4. Theory / Diagram

A lathe removes undesired material from a rotating work piece in the form of chips with the help of a tool which is traversed across the work and can be fed to a depth in the work. The tool material should be harder than the work piece, and the tool is held securely and rigidly on the machine. The tool may be given linear motion in any direction. The lathe is used principally to produce cylindrical surfaces and plane surfaces at right angles to the axis of rotation, and it can also produce tapers and bellows.

A lathe, basically, consists of a bed to provide support, a headstock, a cross-slide to traverse the tool, and a tool post mounted on the cross-slide. The spindle is driven by a motor through a gear box to obtain a range of speeds. The carriage moves over the bed guideways parallel to the work piece and the cross-slide provides transverse motion. A feed shaft and a lead screw are also provided to power the carriage and for cutting threads respectively.

Sequence of operations: centring, facing, plain turning, chamfering, step turning, grooving and taper turning. Facing is done from the centre outwards, producing a flat end. Plain turning reduces the diameter. Step turning produces shoulders of different diameters, with the tool moved parallel to the axis. Chamfering bevels the corner. Grooving cuts a narrow V-shaped groove using a V-cutting tool and parting tool.

The taper is produced by the compound rest method. The compound rest is graduated in degrees and may be swivelled to the half angle of the taper. The angle of taper is calculated from $\tan(\alpha) = (D - d) / 2l$, where D is the larger diameter, d is the smaller diameter, l is the length of the taper and α is the half taper angle. The tool is then fed by hand using the compound slide handwheel, so the length of the taper is limited by the travel of the compound slide.

Example (from the dimensions in the manual): for a taper that reduces from 18 mm to 12 mm over 25 mm, $\tan(\alpha) = (18 - 12) / (2 \times 25) = 0.12$, giving α of about 6.8 degrees. The compound slide is therefore set at about 6.8 degrees from the lathe axis.

Diagrams

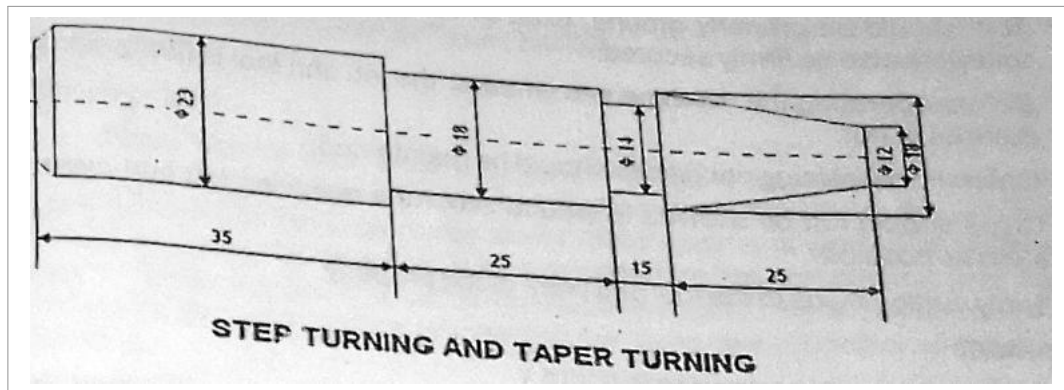


Figure 85: Step turning and taper turning drawing with dimensions in mm (Lab Manual, p. 87).

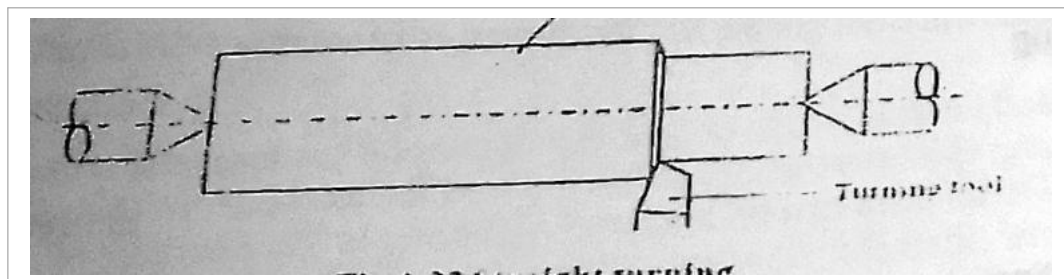


Figure 86: Straight turning (Lab Manual, p. 82).

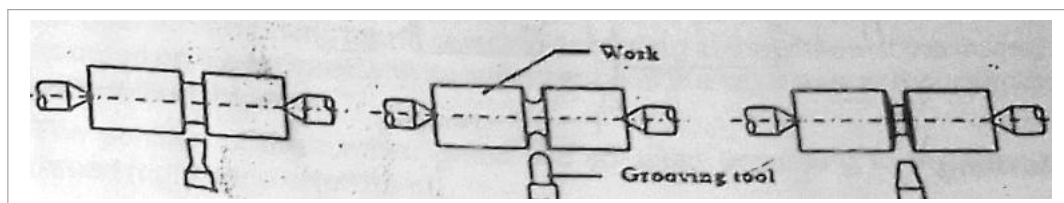


Figure 87: Grooving (Lab Manual, p. 83).

5. Procedure

1. The work piece was fixed in a three-jaw chuck with sufficient overhang.
2. The machine was adjusted to run the job at the required cutting speed.
3. The cutting tool was fixed in the tool post such that the centring operation was performed so that the axis of the job coincides with the lathe axis.
4. The feed and depth of cut were given to the cutting tool.
5. Facing was performed from the centre of the job towards the outside, or from the circumference towards the centre.
6. Plain turning was performed until the diameter of the work piece reduced to 23 mm.
7. The dimensions were checked using vernier callipers.
8. Chamfering was done on the 23 mm diameter surface.
9. The work piece was reversed in the chuck and the facing operation was performed to reduce the length of the work piece to the required dimension.
10. Plain turning was again performed until the diameter of the work piece reduced to 18 mm.
11. Using a V-cutting tool, the grooving operation was performed according to the given dimensions and the groove was finished using the parting tool.
12. The compound slide was swivelled to the required angle and the taper turning operation was performed by rotating the compound slide wheel. The angle can be calculated using $\tan(\alpha) = (D - d)/2l$.

13. Finally, the dimensions were checked using vernier callipers.

6. Precautions

- The work piece should be held rigidly in the chuck before operating the machine.
- The tool should be properly ground, fixed at the correct height and properly secured, and the work should also be firmly secured.
- Before operating the machine, check whether the job and tool are firmly secured in devices or not.
- Optimum machining conditions should be maintained.
- Chips should not be allowed to wind around a revolving job and should be cleared as often as possible.
- Apply cutting fluids to the tool and work piece properly.

7. Discussion

The practical showed that a good turned part is the result of preparation as much as cutting. Setting the tool exactly on centre height was necessary for a clean facing operation, because a tool above or below centre leaves a small pip at the centre of the face. Checking the diameters with the vernier calliper after each cut prevented us from turning undersize, since metal removed cannot be restored.

During step turning, the shoulders were kept square by feeding the tool only to the marked length and then withdrawing it along the face. The chamfers on the shoulders improved the appearance and removed sharp edges. For the taper, the compound rest had to be set accurately to the calculated half angle, and the tool had to be fed slowly and steadily by hand to give a smooth surface. Cutting fluid improved the surface finish and reduced tool wear. Continuous chips were a hazard, and they were cleared with a hook when the machine had stopped.

8. Conclusion

Step turning and taper turning operations were performed on the mild steel rod using a centre lathe. The required steps, chamfer, groove and taper were produced and checked with a vernier calliper, and the objective of the practical was achieved.

9. Questions Asked in the Material

Produce the working drawing in the figure.

See the drawing above: a 25 mm bar stepped down to 23 mm and 18 mm diameters and tapered at the end, with lengths of 35, 25, 15 and 25 mm.

Using simple sketches, enumerate the operational steps.

Centring, facing, plain turning, chamfering, step turning, grooving, taper turning, then checking the dimensions (see procedure).

List the functions of the lathe machine.

Turning, facing, chamfering, grooving, knurling, taper turning, thread cutting, drilling, reaming and boring.

List the functions of five accessories and attachments used with the lathe.

Three-jaw chuck: holds round work and centres it automatically. Four-jaw chuck: holds irregular work. Face plate: holds large or irregular work with bolts and clamps. Centres: hold work between the headstock and tailstock. Steady rest: supports long slender work.

Using simple sketches, describe step turning and taper turning.

Step turning: the tool is moved parallel to the lathe axis to reduce several diameters on the same bar and form shoulders. Taper turning: the compound rest is swivelled to the half-taper angle and the tool is fed along the angled slide.

List the safety precautions observed during the work.

See the precautions above.

10. What I Have Learned

- How to set a lathe tool on centre height and to take measured cuts.
- How to calculate the compound rest angle from $\tan(\alpha) = (D - d)/2l$.
- How to produce steps, a chamfer and a groove on a bar.
- The need for checking dimensions at every stage.

MS 201 (Practical 2): Production of a Metal Door Hinge Using a Centre Lathe

1. Aims

To produce a metal door hinge shaft (pin) using a centre lathe machine.

2. Objectives

- To perform centring, facing, plain turning, chamfering and drilling on the lathe.
- To produce the hinge shaft to the required dimensions.
- To apply the safety precautions of lathe operation.

3. Apparatus / Materials

- Mild steel bar of suitable diameter.
- Centre lathe, three-jaw chuck, H.S.S. turning tool, centre drill and drill bit in the tailstock.
- Vernier calliper, steel rule, chuck key.

4. Theory / Diagram

A door hinge consists of two leaves with knuckles joined by a pin (shaft) that passes through the knuckles, on which the leaves rotate. The pin must be round, straight and of uniform diameter so that the hinge turns smoothly, and it often has a chamfered end so that it enters the knuckles easily. The pin or shaft is a simple but typical turned component, because it requires only straight turning, facing, chamfering and, if required, drilling.

The sequence of operations is centring, facing, plain turning, chamfering and drilling. Centring is done by marking and drilling a small centre hole, using a centre drill held in the tailstock chuck, so that the end of the work is supported by the dead centre and the work does not whip. Facing squares the ends and sets the length. Plain turning, with the tool fed parallel to the axis, reduces the bar to the required diameter, and a finishing cut with a lower feed gives a smooth finish. Chamfering bevels the ends and removes sharp edges. Drilling, when needed, is done with the drill held in the tailstock spindle and fed by the tailstock hand wheel.

Turning speed is determined by the formula $N = 1000V/(\pi \times D)$, where V is the cutting speed for the material and tool (m/min) and D is the diameter in mm, so that a smaller diameter needs a higher rpm for the same cutting speed.

Diagrams

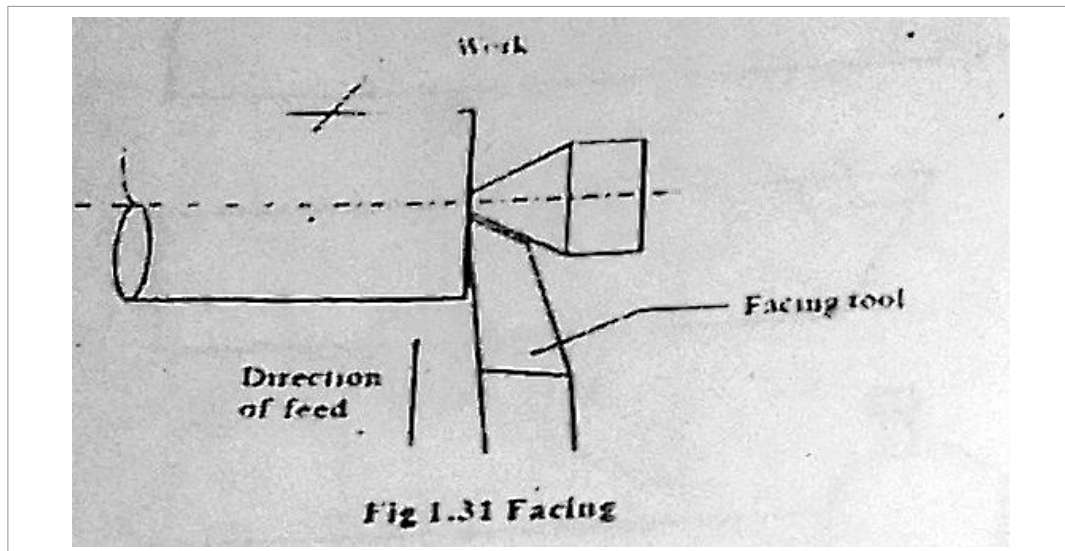


Figure 88: Facing operation (Lab Manual, p. 82).

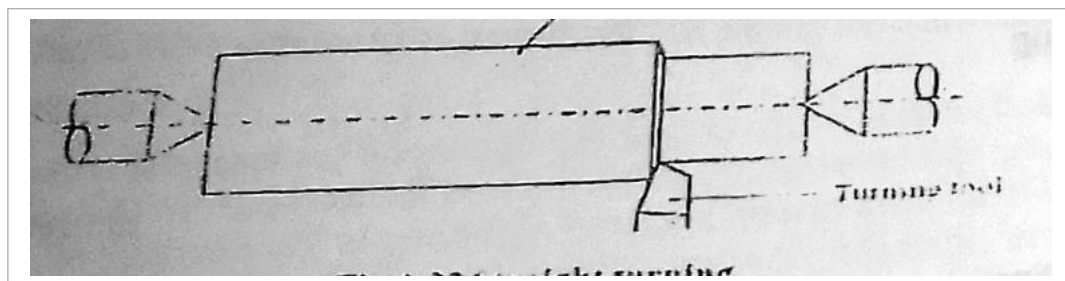


Figure 89: Straight turning (Lab Manual, p. 82).

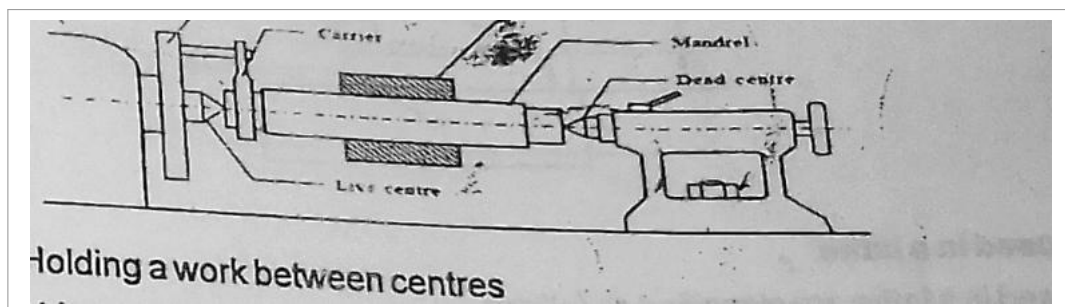


Figure 90: Holding work between centres (Lab Manual, p. 79).

5. Procedure

1. The mild steel bar was held in the three-jaw chuck with enough overhang.
2. The tool was set at centre height and a facing cut was made at the free end.
3. A centre hole was drilled in the end using the centre drill held in the tailstock.
4. The work was turned along its length to the required diameter by plain turning, taking roughing and then finishing cuts.
5. The ends of the shaft were chamfered.
6. The work was reversed in the chuck, faced to the required length and chamfered.
7. Drilling was done where required, using the tailstock.
8. Dimensions were checked with the vernier calliper.

6. Precautions

- Hold the work rigidly in the chuck and do not leave the chuck key in it.
- Set the tool at centre height and secure it firmly.

- Do not touch the rotating work or remove chips by hand.
- Use cutting fluid and clear chips often.
- Wear safety glasses and keep sleeves and hair away from the machine.

7. Discussion

The door hinge pin is a simple part, but it brought together several basic lathe operations in one job. We found that a sharp tool, correct speed and light finishing cuts produced a smooth, round surface, while taking too heavy a cut caused chatter and a rough finish. Centring the end of the bar and supporting it with the tailstock prevented the bar from bending away from the tool.

Checking the diameter at several points along the length was important, because taper can arise if the tailstock or tool post is out of alignment. Chamfering the ends was a small step, but it made the finished pin easy to fit into the hinge knuckle and safer to handle.

8. Conclusion

The hinge shaft was produced on a centre lathe by centring, facing, plain turning, chamfering and drilling. The practical consolidated our ability to do basic lathe operations to specified dimensions while following safety precautions.

9. Questions Asked in the Material

Produce the working drawing of the shaft.

The shaft is a plain cylinder of the specified diameter and length with chamfered ends (see the straight-turning figure).

Using simple sketches, enumerate the operational steps involved in the production of the shaft.

Centring, facing, plain turning, chamfering, drilling.

List the functions of the lathe machine.

Turning, facing, grooving, knurling, thread cutting, taper turning, drilling, boring.

Describe face and straight turning.

Facing: the tool is fed perpendicular to the axis to produce a flat end face. Straight turning: the tool is fed parallel to the axis to produce a cylindrical surface.

List the safety precautions observed during the work.

See the precautions above.

10. What I Have Learned

- How the basic lathe operations combine to make a shaft.
- The relationship between spindle speed, cutting speed and diameter.
- Why work should be supported by a centre in the tailstock.

AS 201: The Internal Combustion Engine

1. Aims

To study the internal combustion engine, its major parts, classification and four-stroke working cycle, and the routine checks and maintenance carried out on it.

2. Objectives

- To know the major parts of an internal combustion engine and their functions.
- To know the different classifications of the internal combustion engine.
- To describe the working cycle of a four-stroke spark ignition engine and a compression ignition engine.
- To be able to diagnose simple internal combustion engine faults.
- To know how to change the engine oil of an internal combustion engine.
- To identify the tools used to carry out repairs on an engine.
- To know the routine checks and routine maintenance carried out on an engine.

3. Apparatus / Materials

- A sectioned or complete internal combustion engine (petrol and/or diesel).
- Hand tools: spanner set, socket set, screwdrivers, pliers, plug spanner, torque wrench, feeler gauge.
- Oil drain pan, new engine oil, oil filter, funnel, dip stick, tyre pressure gauge, hydrometer.
- Personal protective equipment: gloves, overalls and safety glasses.

4. Theory / Diagram

The old steam engine was powerful, quiet and readily reversible, but it had low efficiency because the combustion took place externally, in a large and bulky boiler. Many engineers worked on replacing the boiler with an engine in which combustion takes place inside the cylinder. Several types of internal combustion engine were developed, of which only two, the spark ignition engine and the compression ignition engine, are still in general use.

An internal combustion engine is any engine that operates by burning its fuel in a confined space called the combustion chamber, which is usually an integral part of the engine. In contrast, a steam engine burns its fuel outside the engine. Examples are all vehicle engines, motor bikes, tricycle engines, tractors, earth-moving equipment, lawnmowers, motor saws, generators and some electric power plants.

Major parts

The cylinder block houses the cylinders in which the pistons move. The cylinder head closes the top of the cylinders and houses the valves, spark plugs (or injectors) and the combustion chambers. The piston moves up and down in the cylinder and transmits the force of combustion to the connecting rod through the gudgeon pin; piston rings seal the gap between the piston and cylinder wall. The connecting rod links the piston to the crankshaft, which converts the reciprocating motion of the pistons into rotary motion. The flywheel stores energy and smooths out the power impulses. The camshaft, driven by the crankshaft at half speed, operates the inlet and exhaust valves through followers, push rods and rocker arms. The sump (oil pan) stores lubricating oil, which is circulated by the oil pump. The cooling system (radiator, water pump, thermostat) removes excess heat, the fuel system supplies fuel (a carburettor or injection system), and the electrical system (battery, ignition coil, contact breaker or electronic ignition, spark plugs) supplies the spark.

Terms

Top dead centre (TDC) is the position of the piston when it is at the top of its travel (farthest from the crankshaft). Bottom dead centre (BDC) is the lowest position of travel. The stroke is the distance the piston travels between TDC and BDC. The combustion chamber is the space above the piston at TDC in which combustion takes place. Engine capacity (swept volume) is the volume swept by the piston in moving from TDC to BDC, multiplied by the number of cylinders. Ignition temperature is the temperature at which the fuel-air mixture ignites. Priming is the pouring of a small amount of fuel into the system, or the operation of the primer pump, to help starting, especially in a diesel engine. A spark ignition engine ignites the mixture with an electric spark, and a compression ignition engine ignites the fuel by the heat of compression.

Classification

Internal combustion engines can be classified by the type of ignition (spark ignition and compression ignition), by the number of strokes per cycle (two-stroke and four-stroke), by fuel used (petrol, diesel, gas), by cylinder arrangement (inline, V, flat or radial), by cooling system (air or water cooled), and by valve arrangement.

The four-stroke cycle

In a four-stroke engine the cycle is completed in four strokes of the piston, which is two revolutions of the crankshaft. (1) Suction stroke: the inlet valve opens and the piston moves down from TDC to BDC, drawing in an air-fuel mixture in a petrol engine, or only air in a diesel engine. (2) Compression stroke: both valves are closed and the piston moves up, compressing the charge. (3) Power stroke: near the end of compression, in a spark ignition engine the spark plug ignites the mixture; in a compression ignition engine, fuel is injected into the very hot compressed air and ignites. The expanding gases force the piston down, which produces power. (4) Exhaust stroke: the exhaust valve opens and the piston moves up, expelling the burnt gases. The cycle then repeats. The compression ratio in a diesel engine (about 15:1 to 22:1) is much higher than in a petrol engine (about 8:1 to 10:1).

Routine checks and maintenance

Routine checks include water level in the radiator, oil level using the dip stick, brake fluid level, tyre air pressure using a pressure gauge, and acid level in the battery. Routine maintenance includes changing the oil, oil filter and fuel filter, changing the spark plug and the contact breaker, checking tyre alignment, wheel balancing, and lubricating (greasing) the necessary parts.

Changing engine oil

The engine is warmed so that the oil flows easily and then switched off. The drain plug under the sump is removed and the old oil is drained into a pan. The oil filter is removed and replaced with a new one with its seal lightly oiled. The drain plug is refitted with its washer. New oil of the recommended grade is poured through the filler, the dipstick is used to check the level, and the engine is run briefly to circulate oil and checked for leaks, then the level is rechecked.

Diagrams

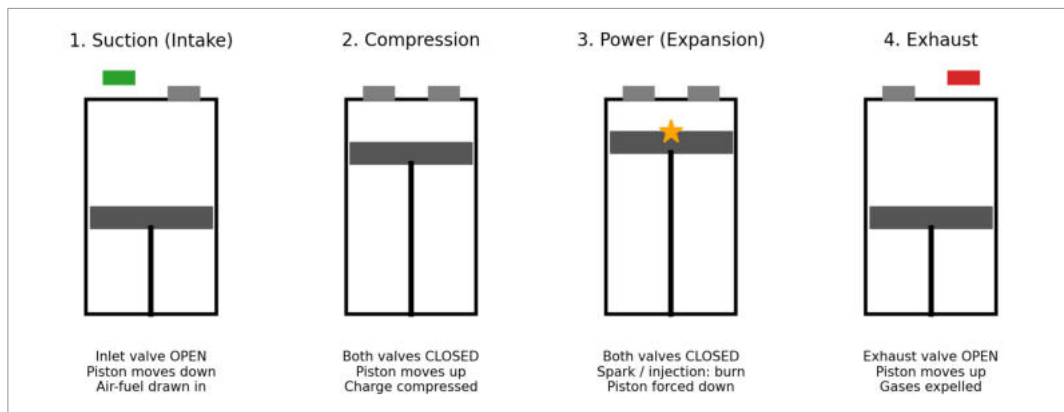


Figure 91: The four strokes of the four-stroke engine cycle: suction, compression, power and exhaust (schematic drawn for this report).

5. Procedure

1. The engine was examined externally and its major components were identified.
2. The functions of the parts were explained, and the valve and piston movements were observed by turning the engine over by hand.
3. The working of the four-stroke cycle was explained, and the differences between the spark ignition and compression ignition cycles were discussed.
4. Routine checks were carried out: the radiator water level, engine oil level with the dip stick, brake fluid level, tyre pressure and battery acid level.
5. The procedure for changing the engine oil and filters was demonstrated.
6. The tools used for repair and maintenance were identified.

6. Precautions

- Switch off the engine and remove the key before working on it.
- Do not open a radiator cap when the engine is hot.
- Allow the engine and exhaust to cool before touching them, and wear gloves.
- Dispose of used oil responsibly and clean spills immediately.
- Use the correct size of spanner, and keep the work area free from fuel and open flames.
- Run the engine only in a well-ventilated area because exhaust gas contains carbon monoxide.

7. Discussion

The experiment showed that the engine works through a repeating sequence in which every part has a precise role: the valves control the flow of gases, the piston transmits the force, and the crankshaft turns the reciprocating movement into rotation. The cycle needs two complete revolutions for one power stroke, and in a multi-cylinder engine the power strokes are arranged to overlap, giving smooth running.

The difference between the petrol and diesel engines is mainly in how the fuel is ignited. The spark ignition engine uses a spark, while the diesel engine depends on the high temperature produced by compressing air to a high ratio, so it needs a stronger construction. The routine checks highlighted how simple practices such as checking the oil, the water and the tyres prevent most breakdowns and prolong the life of the engine.

8. Conclusion

The major parts, classification, working cycle, routine checks and maintenance of an internal combustion engine were studied. The four-stroke cycle of suction, compression, power and exhaust was understood, and regular checks and maintenance were shown to be essential for reliable operation.

9. Questions Asked in the Material

Describe and illustrate with simple sketches the working cycle of a four-stroke spark ignition engine and compression ignition engine.

See the figure. In the spark ignition engine, a fuel-air mixture is drawn in, compressed, ignited by a spark and then exhausted. In the compression ignition engine, only air is drawn in and compressed to a high temperature, and the fuel is injected near the end of compression and ignites by itself.

What are the basic tools, spare parts and safety devices that a vehicle should have?

Tools: spanner set, screwdrivers, pliers, jack and wheel spanner, tyre pressure gauge. Spare parts: spare wheel, fan belt, fuses, bulbs, spark plugs. Safety devices: fire extinguisher, warning triangle, first aid box, seat belts.

Enumerate sequentially the steps taken when changing oil in an internal combustion engine.

Warm the engine and switch off; place a pan; remove the drain plug and drain; replace the filter; refit the plug; add new oil; check the level with the dip stick; run and check for leaks.

List and state two functions each of twenty components of an internal combustion engine (selected).

Piston: transmits the gas force; seals the cylinder with rings. Connecting rod: links piston to crankshaft; transmits force. Crankshaft: changes linear to rotary motion; delivers output. Camshaft: operates the valves; times their opening. Valves: admit mixture and release exhaust. Spark plug: ignites the mixture; seals the chamber. Flywheel: stores energy; smooths rotation. Radiator: cools the water; dissipates heat. Oil pump: supplies oil; maintains pressure. Water pump: circulates coolant; prevents overheating.

How often should one change the engine and gear oil of his vehicle?

Engine oil is generally changed every 5,000 km or as recommended by the manufacturer (about every 3 to 6 months), and gear oil about every 40,000 to 60,000 km, or according to the vehicle manual.

Assuming an engine refuses to start from rest, what are the likely problems and possible solutions?

Flat battery (charge or replace); faulty starter (repair); no fuel (refill, check pump); fouled spark plugs (clean or replace); wet or faulty ignition (dry, replace coil or contact breaker); blocked fuel filter (replace).

Assuming an engine went off suddenly while driving on the highway, what is the likely problem and a possible solution?

Fuel starvation (blocked filter or pump failure: replace); ignition failure (loose or damaged wiring: check connections); overheating (allow to cool and refill coolant).

List the different classifications of an internal combustion engine.

By ignition (spark or compression), strokes (two or four), fuel, cylinder arrangement, cooling system and valve arrangement.

List ten tools used to effect repair on an internal combustion engine.

Spanner set, socket set, screwdriver, pliers, plug spanner, torque wrench, feeler gauge, hammer, oil filter wrench and a jack.

Define the following terms.

Spark ignition engine: an engine in which the mixture is ignited by a spark. Compression ignition engine: one in which fuel ignites by the heat of compressed air. Stroke: the piston movement between TDC and BDC. TDC: top dead centre, the highest piston position. BDC: bottom dead centre, the lowest piston position. Combustion chamber: the space where the fuel burns. Ignition temperature: the temperature at which the mixture ignites. Two-stroke engine: one that completes the cycle in two strokes (one revolution). Four-stroke engine: one that completes the cycle in four strokes (two revolutions). Engine capacity: the total swept volume of all cylinders. Priming: supplying a small amount of fuel to help starting.

10. What I Have Learned

- The main components of the internal combustion engine and what each does.
- How the four strokes combine to produce power.
- Routine checks and maintenance that prolong engine life.
- The basic procedure for changing engine oil and diagnosing simple faults.

FS 201: Introduction to the Foundry Shop

1. Aims

To study the processes and materials of the foundry shop, including patterns, moulding sand and the hand tools used in mould making.

2. Objectives

- To understand what patterns are.
- To know what moulding sands are.
- To become familiar with the tools and materials used in the foundry shop.
- To produce a simple cast.

3. Apparatus / Materials

- Patterns (solid or single-piece), moulding sand, parting sand, facing sand, backing sand.
- Moulding boxes (cope and drag), moulding board and bottom board.
- Hand tools: riddle, shovel, rammers, strike-off bar, sprue pin, trowels, lifter, vent wire, slick, swab, gate cutter, bellows and draw spike.

4. Theory / Diagram

A foundry is a factory that produces metal castings from ferrous or non-ferrous alloys. Metals are turned into parts by melting the metal into a liquid, pouring it into a mould, and removing the metal from the mould after it has solidified. The most common metal alloys processed are aluminium and cast iron, but other metals such as steel, magnesium, copper, tin and zinc can also be processed.

Patterns

A pattern is a model or replica of the object to be cast. It is embedded in moulding sand and suitable ramming of moulding sand is done around the pattern. The pattern is then withdrawn, leaving a cavity (the mould) in the sand. The common pattern materials are wood, metal, plastic, plaster and wax or mercury. Wood is the most common material because it is cheap and easy to shape. Patterns are made larger than the casting to allow for shrinkage of the metal on cooling, and have tapered sides (draft) so that they can be withdrawn without damaging the mould.

Types of pattern include: single-piece (solid) pattern, two-piece (split) pattern, cope and drag pattern, three-piece or multi-piece pattern, loose piece pattern, match plate pattern, follow board pattern, gated pattern, sweep pattern, skeleton pattern and segmental pattern. The single-piece pattern is made of one piece without joints, partings or loose pieces and is the simplest form. A two-piece or split pattern is split into two parts, joined by dowel pins, when a solid pattern is difficult to withdraw from the mould cavity.

Moulding sand

Moulding sands come from the beds of seas, rivers, lakes, granular elements of rocks and deserts. They may be natural or synthetic. Natural sands contain enough binder. Synthetic sand is made artificially using basic sand constituents (silica sand, 88 to 92 %), binder (6 to 12 %), water or moisture (3 to 6 %) and other additives in proper proportions.

Binders may be inorganic (clays such as kaolinite, ball clay, fire clay, limonite, fuller's earth and bentonite, as well as sodium silicate and Portland cement) or organic (dextrin, molasses, cereal binders, linseed oil, resins, phenol formaldehyde, urea formaldehyde). The bentonite

variety of clay is the most common, though clay alone cannot develop bonds without the presence of moisture.

Additives improve properties of the sand. Coal dust produces a reducing atmosphere during casting; corn flour increases collapsibility; dextrin increases dry strength; sea coal fills the pores among sand grains; wood flour prevents grains from making contact with one another; and silica flour increases hot strength and finish.

Kinds of moulding sand are: green sand (a freshly prepared mixture of silica sand with 18 to 30 % clay and 6 to 8 % water), dry sand (green sand dried or baked in an oven), loam sand (a mixture of sand and clay with water to a thin plastic paste), facing sand (next to the pattern surface), backing sand (floor sand that fills the flask), parting sand (without binder, to prevent sticking), and core sand (silica sand mixed with oil binders).

Properties required in moulding sand are: refractoriness (the ability to withstand high temperature without fusing), permeability (porosity that allows gases to escape), cohesiveness (the ability of sand grains to stick to each other), green strength, dry strength, flowability or plasticity, adhesiveness (the ability to stick to the moulding box) and collapsibility (the ability to collapse after the metal solidifies so that it can contract freely).

Hand tools

The riddle is a screen with a standard wire mesh in a circular wooden frame, used to clean the sand of foreign material and to break up lumps. The shovel is used to mix, temper and move sand. Rammers pack the sand in the moulding box uniformly around the pattern. The strike-off bar is a flat bar used to scrape the excess sand from the top of the box after ramming. The sprue pin is a tapered rod of wood or iron pushed into the cope to form the passage through which molten metal is poured. Trowels finish flat surfaces and corners in a mould. The lifter repairs the mould and removes loose sand from it. The vent wire is a thin steel rod with a pointed end used to make small vent holes in the sand so that gases and steam can escape. The slick repairs and finishes the mould surfaces and edges after the pattern is withdrawn. The swab is a small hemp-fibre brush used to moisten the edges of the sand mould around the pattern before it is withdrawn. The gate cutter is used to cut runners and gates linking the sprue to the mould cavity. The bellows blow loose sand away from the mould cavity. The draw spike is a tapered steel rod with a loop and a sharp end screwed into the pattern for withdrawing it.

The moulding box

The moulding box or flask is a wood or metal frame made of two parts, the cope (upper half) and the drag (lower half), held in alignment by pins and lugs.

Diagrams

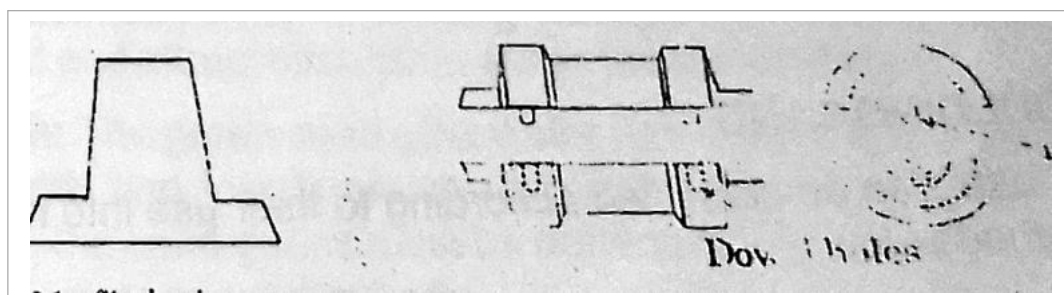


Figure 92: Single-piece (solid) pattern and two-piece (split) pattern with dowel holes (Lab Manual, p. 91).

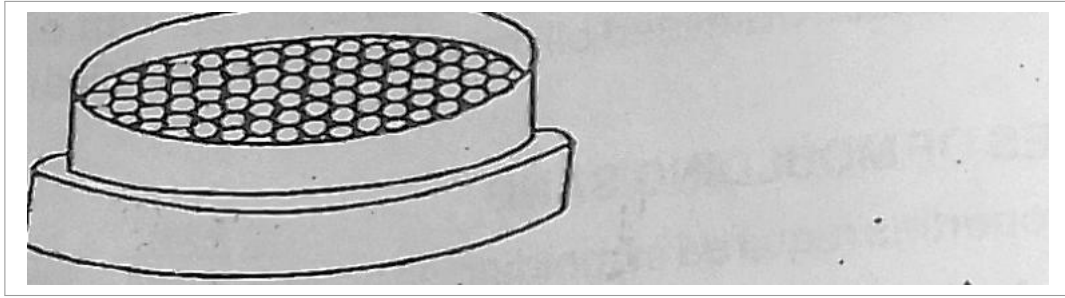


Figure 93: Hand riddle (Lab Manual, p. 94).

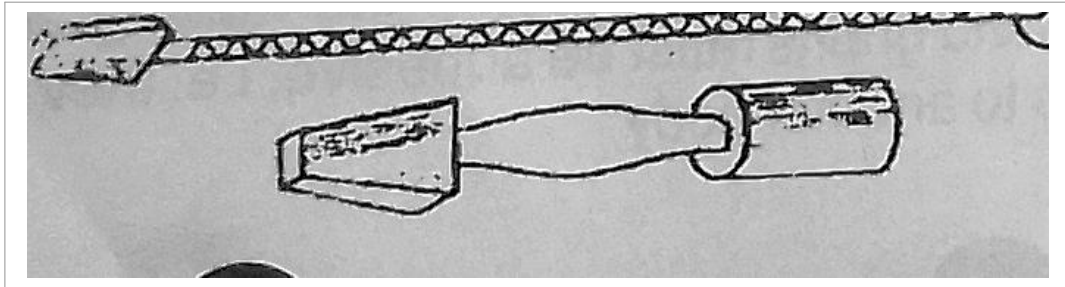


Figure 94: Rammers (Lab Manual, p. 94).

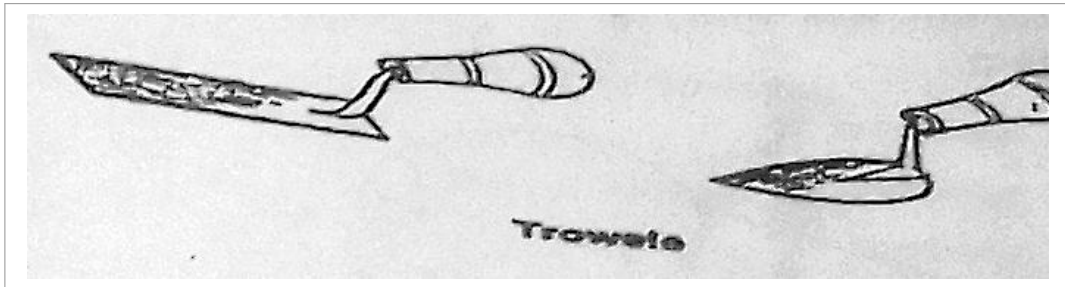


Figure 95: Trowels (Lab Manual, p. 95).

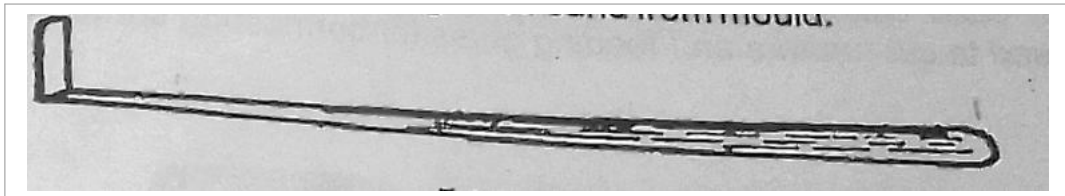


Figure 96: Lifter (Lab Manual, p. 95).

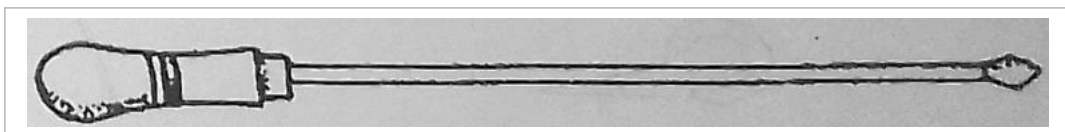


Figure 97: Vent wire (Lab Manual, p. 95).

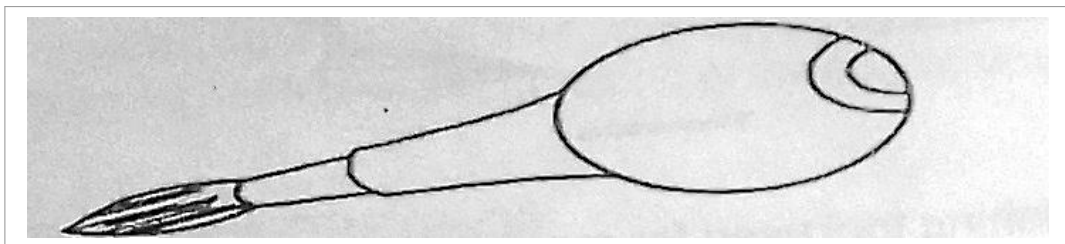


Figure 98: Swab (Lab Manual, p. 96).

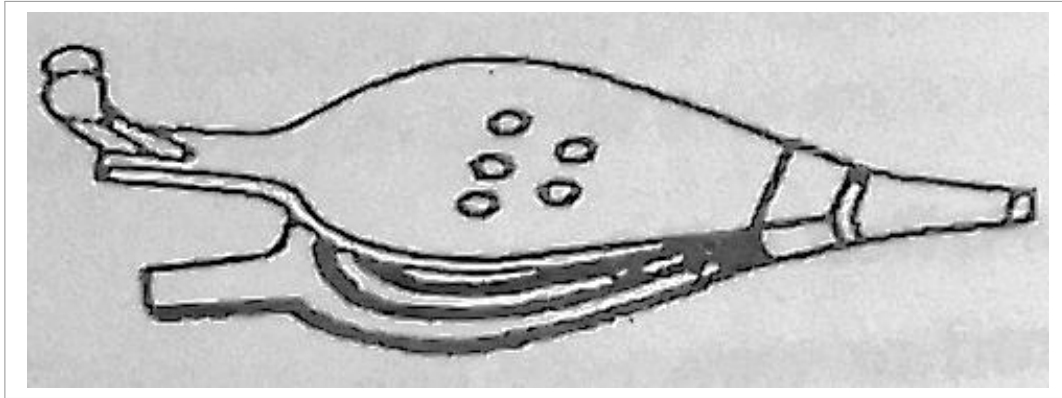


Figure 99: Bellows (Lab Manual, p. 96).

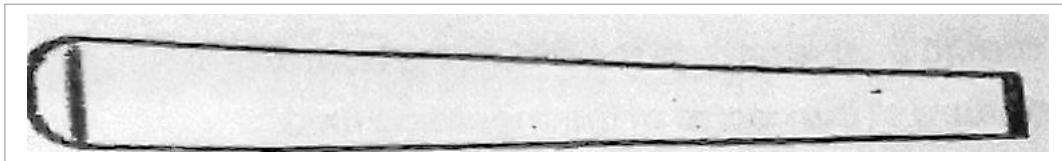


Figure 100: Sprue pin (Lab Manual, p. 97).

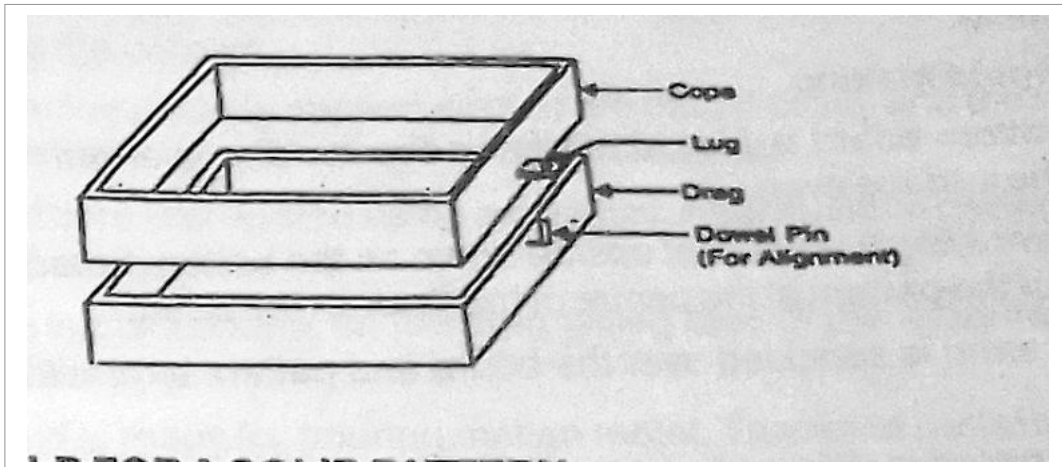


Figure 101: Moulding box: cope and drag with dowel pins for alignment (Lab Manual, p. 97).

5. Procedure

1. The foundry shop was introduced and the process of casting was explained.
2. Different patterns and pattern materials were examined, and the types of pattern were discussed.
3. The types and properties of moulding sand were explained, and the sand was prepared by riddling and mixing.
4. The hand tools of the foundry were identified and their uses described.
5. A moulding box was examined, and the cope and drag were identified.

6. Precautions

- Wear safety glasses, gloves, closed shoes and overalls.
- Keep the floor free of sand and tools to prevent tripping.
- Never pour molten metal into a damp mould, because steam can cause an explosion.
- Keep the area around the pouring point clear of people and combustible materials.
- Handle patterns and tools carefully, and return them to their places after use.

7. Discussion

The foundry shop shows that casting begins long before any metal is melted. The quality of the casting depends on the pattern, the mould and especially the sand. A sand with poor permeability traps gas and produces blow holes, and one with poor refractoriness fuses with the metal and gives a rough surface. The sand also needs enough strength to retain its shape, but must collapse after solidification so that the metal can contract without cracking.

The hand tools may look simple, but each one has a particular purpose that affects the final casting: the rammer controls the density of the sand, the vent wire opens escape paths for gases, and the swab and slick prepare and repair the mould surface. Correct use of these tools gives a clean mould cavity with sharp edges.

8. Conclusion

The patterns, moulding sand and hand tools of the foundry shop were studied. We learned the types of pattern, the types and properties of moulding sand, and the use of the foundry hand tools, which provides the basis for making a sand mould and a casting.

9. Questions Asked in the Material

What is a pattern?

A model or replica of the object to be cast, used to form the cavity in the mould.

List the materials used for making patterns.

Wood, metal, plastic, plaster, wax and mercury.

List the properties of moulding sand.

Refractoriness, permeability, cohesiveness, green strength, dry strength, flowability, adhesiveness and collapsibility.

10. What I Have Learned

- What a pattern is and the types of pattern in use.
- The composition, types and properties of moulding sand.
- The names and uses of foundry hand tools.

FS 201 (Practical 1): Mould for a Solid Pattern

1. Aims

To prepare a sand mould using the given single-piece (solid) pattern.

2. Objectives

- To prepare moulding sand and to ram a drag and cope around a solid pattern.
- To make the sprue, riser, runner and vent holes in the mould.
- To withdraw the pattern and prepare the mould cavity ready for pouring.

3. Apparatus / Materials

- Raw materials: moulding sand, parting sand, facing sand, backing sand, single-piece solid pattern, bottom board, moulding boxes.
- Tools: moulding board, drag and cope boxes, rammer, strike-off bar, bellows, riser and sprue pins, gate cutter, vent rod, draw spike and wire brush.

4. Theory / Diagram

Making a sand mould is the process of building a cavity in sand, in the shape of the casting wanted. The solid pattern is placed in the drag (lower half of the moulding box) and sand is rammed round it. The cope (upper half) is then placed on top with sprue and riser pins, and rammed. The two halves are separated, the pattern is removed, and the halves are put back together with the cavity between them. Molten metal is poured through the sprue, flows down through the runner and gate into the cavity, and the riser feeds metal to make up for shrinkage as the casting solidifies.

Parting sand, which has no binder or moisture, is sprinkled on the pattern and the joint surface so that the cope and drag do not stick, and so the pattern can be withdrawn cleanly. Facing sand, which has fine grains and a high refractoriness, is placed against the pattern to give a good surface finish. Backing sand is used to fill the remainder of the box. The sand must be rammed enough to hold its shape when the pattern is withdrawn, but not so hard that gases cannot escape. For this reason, vent holes are made with the vent wire so that steam and gases can leave the mould during casting.

The gating system consists of the pouring basin, sprue, runner and gates, and its purpose is to lead the metal smoothly into the cavity with minimum turbulence. A good mould also allows for shrinkage, and the pattern has a taper so that it can be withdrawn without damaging the sand.

Diagrams

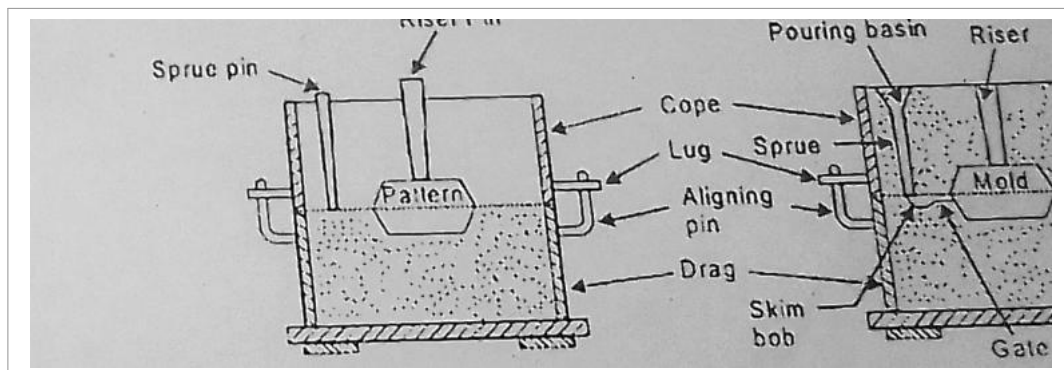


Figure 102: Cross-section of a sand mould showing the cope, drag, sprue, riser, pattern, lug and aligning pin (Lab Manual, p. 99).

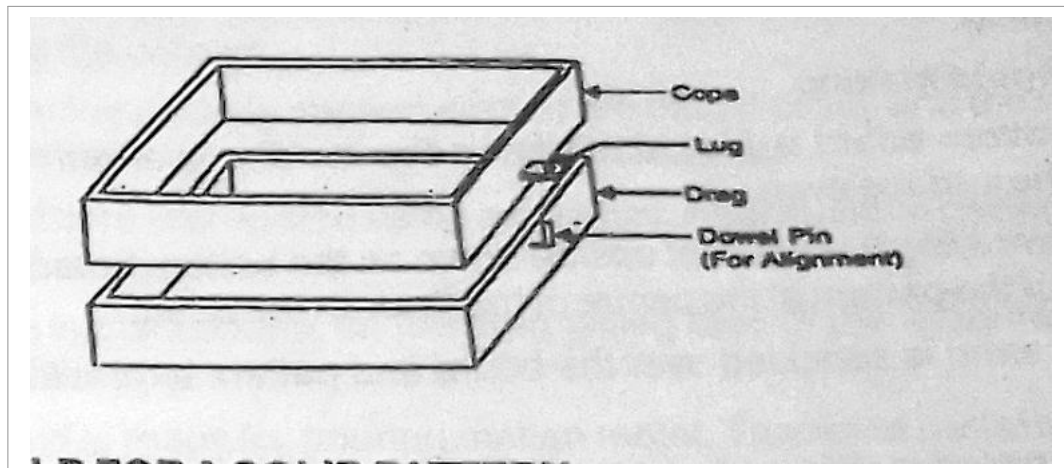


Figure 103: Moulding box (cope and drag) with dowel pins (Lab Manual, p. 97).



Figure 104: Sprue pin used to make the pouring passage (Lab Manual, p. 97).

5. Procedure

Mould making

1. A bottom board was placed on the moulding platform or floor, making the surface even.
2. The drag moulding flask was kept upside down on the bottom board along with the drag part of the pattern at the centre of the flask on the board.
3. Dry facing sand was sprinkled over the board and pattern to provide a non-sticky layer.
4. Freshly prepared moulding sand of requisite quality was poured into the drag and on the pattern to a thickness of 30 to 50 mm.
5. The rest of the drag flask was completely filled with the backup sand and uniformly rammed to compact the sand.
6. After the ramming was over, the excess sand in the flask was scraped using a flat bar (strike-off bar) to the level of the flask edges.
7. With a vent wire of 1 to 2 mm diameter with a pointed end, vent holes were made in the drag to the full depth of the flask as well as to the pattern, to facilitate the removal of gases during casting solidification. This completes the preparation of the drag.
8. The finished drag flask was rolled over to the bottom board, exposing the pattern.
9. Using a slick, the edges of sand around the pattern were repaired.
10. The cope flask on top of the drag was located, aligning it with the help of the drag box pins.
11. The sprue of the gating system for making the sprue passage was located at a small distance of about 50 mm from the pattern. The sprue base, runners and in-gates were also located as shown, and risers were also placed. Freshly prepared facing sand was poured around the pattern.
12. The moulding sand was then poured into the cope box. The sand was adequately rammed, excess sand scraped and vent holes made all over in the cope.
13. The sprue and the riser were carefully withdrawn from the flask.
14. Later the pouring basin was cut near the top of the sprue.

15. The cope was separated from the drag and any loose sand on the cope and drag interface was blown off with the help of the bellows.
16. Now the cope and the drag pattern halves were withdrawn by using the draw spikes and rapping the pattern all around to slightly enlarge the mould cavity so that the walls are not spoiled by the withdrawing pattern.
17. The runners and gates were then removed or cut in the mould carefully without spoiling the mould.
18. Any excess or loose sand in the runners and mould cavity was blown away using the bellows.
19. Now the facing paste was applied all over the mould cavity and the runners to give the finished casting a good surface finish.
20. A dry sand core was prepared using a core box. After suitable baking it was placed in the mould cavity (where the casting required one).
21. The cope was placed back on the drag taking care of the alignment of the two by means of the pins.
22. The mould was then ready for pouring molten metal. The liquid metal was allowed to cool and become solid, giving the casting desired.

6. Precautions

- Wear safety glasses, gloves and closed shoes.
- Do not ram the sand too hard or too soft; uniform ramming is required.
- Use parting sand properly so the halves do not stick.
- Withdraw the pattern carefully, rapping lightly before lifting it, to avoid damage to the mould.
- Make sure the mould and the ladle are dry before pouring, because moisture can cause steam explosions.
- Keep spectators away from the pouring area and wear protective clothing when pouring.

7. Discussion

The mould was successfully prepared, but the exercise showed us that good moulding is largely about control of the sand. When the sand was too wet it stuck to the pattern and made a ragged cavity, and when it was too dry it crumbled when the pattern was withdrawn. Ramming density had to be even: loose areas collapsed, while over-rammed areas were impermeable and would trap gas.

Venting with the vent wire was a small step with a large effect, since gases that cannot leave the mould cause blow holes in the casting. Rapping the pattern slightly before withdrawing it enlarged the cavity just enough that the pattern came out clean without breaking the edges. The riser and gating system were arranged to deliver metal smoothly and to supply extra metal during solidification shrinkage.

8. Conclusion

The sand mould for a solid pattern was successfully prepared. The cavity was formed in the drag and cope with the gating system and vents, and the mould was ready for pouring. The exercise taught the sequence of moulding operations and the importance of sand properties and mould preparation.

9. Questions Asked in the Material

Produce the working drawing of the mould for a solid pattern.

The drawing is the cross-section in the figure above, showing cope, drag, pattern, sprue, riser, vents and parting line.

Produce the mould for a solid pattern.

See the procedure above.

Using simple sketches, outline sequentially the operational steps in the production of the mould for a solid pattern and the cast.

Sand preparation; place the drag on the board; place the pattern; ram the drag; place the runner and riser; ram the cope; remove the pattern, runner and riser; cut the gates; reassemble; pour.

List the functions of at least five tools you used.

Rammer: compacts the sand uniformly. Strike-off bar: removes excess sand level with the flask. Vent wire: makes vent holes for gases. Sprue pin: forms the pouring passage. Draw spike: withdraws the pattern. Gate cutter: cuts runners and gates. Bellows: blows loose sand out of the cavity.

List the safety precautions observed.

See the precautions above.

10. What I Have Learned

- The complete sequence of making a sand mould.
- The roles of facing, backing and parting sand in the mould.
- Why vent holes, a gating system and a riser are necessary.
- That a good casting begins with a well-made mould.